PMBOK® Guide – Eighth Edition & PMP® Exam — Complete Study Guide

A self-study companion for the *PMBOK® Guide – Eighth Edition* (PMI, 2025) and the PMP® Examination Content Outline that took effect in July 2026

> **Read this first.** This is an independent study aid written to help you understand, remember and *apply* the content you will be tested on. It explains the concepts of the *PMBOK® Guide – Eighth Edition* in plain language and connects them to the way PMP® questions are written. It does not reproduce the PMBOK® Guide, so keep your copy of the Guide next to you and use this book to make sense of it. Exam logistics (number of questions, time, fees, eligibility) can change, so confirm them in PMI's current *PMP Certification Handbook* before you book your exam. PMI, PMP, CAPM and PMBOK are registered marks of the Project Management Institute, Inc.

How this study package is organised

File	What it is	When to use it
01 – Study Guide (this file)	21 chapters + glossary. Each chapter ends with review questions; answers with explanations are in Appendix B.	Weeks 1–6: read one chapter, then answer its questions *before* looking at the key.
02 – Comprehensive Question Bank	Questions covering the whole body of knowledge, grouped in parts, each part followed by its answers and explanations.	After you finish the guide, to find weak areas.
03 – Demo Exams 01–10	The ten demo exams as documents (180 questions, 240 minutes each) with answer keys, explanations and a score-analysis sheet.	If you prefer to practise on paper.
04 – Exam Simulator	The exam platform: the ten demo exams plus mixed exams drawn from all of them, with a timer, breaks, flagging, scoring by domain and task, and a PDF of your corrections after every attempt.	Weeks 7–8 (and beyond): one exam per sitting, under real conditions.

How to study with this guide

1. **Read actively.** Each chapter starts with what you should be able to do afterwards. Keep a notebook of terms you could not explain to a colleague.
2. **Test yourself honestly.** Answer the review questions without looking back. Mark any question you guessed as "wrong" even if you guessed right.
3. **Close the gaps.** If you score below 80% on a chapter, re-read the parts you missed before moving on.
4. **Integrate.** After the last chapter, take the Comprehensive Question Bank. It mixes topics the way the exam does.

5. **Simulate.** Take the demo exams under exam conditions: 240 minutes, no notes, breaks only after questions 60 and 120. Review *every* explanation — including those for questions you got right.
6. **Think like PMI.** Chapter 21 explains the mindset behind the "best" answer. Re-read it before every demo exam.

Suggested 8-week plan (about 10–12 hours per week)

Week	Read	Practise
1	Ch. 1–4 (exam, PMBOK 8 structure, value delivery, principles)	Chapter questions
2	Ch. 5–7 (life cycles, focus areas & processes, Governance)	Chapter questions
3	Ch. 8–11 (Scope, Quality, Schedule, Finance)	Chapter questions + formula drills (Ch. 20)
4	Ch. 12–14 (Stakeholders, Resources & procurement, Risk)	Chapter questions
5	Ch. 15–17 (Agile & hybrid, People, Business environment)	Chapter questions
6	Ch. 18–21 (Tailoring, toolbox, formulas, exam strategy)	Comprehensive Question Bank
7	Revise weak chapters	Demo Exams 01–04
8	Revise weak domains; re-read Ch. 21	Demo Exams 05–10 (retake any below 75%)

Conventions used in this guide

- **Exam tip** boxes tell you how a concept tends to appear in questions.
- **Trap** boxes describe a wrong answer that looks attractive.
- **What changed** boxes compare the Eighth Edition with the Sixth and Seventh Editions — useful if you have older materials.
- Process names follow the Eighth Edition; when an older name is still common in prep materials it is given in brackets, e.g. *Develop Scope Structure (formerly Create WBS)*.

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- Appendix B — Answer Keys for the Chapter Review Questions

Chapter 1 — The PMP® Exam (2026) and How the PMBOK® Guide Fits

After this chapter you should be able to:

- describe the format of the PMP® exam that has been delivered since July 2026;
- list the three exam domains, their weights and the 26 tasks;
- explain the relationship between the exam content outline and the *PMBOK® Guide – Eighth Edition*;
- plan your preparation and know what "good enough" looks like on practice exams.

1.1 Which certification does this guide target?

PMI offers several certifications. The two that rely most heavily on the PMBOK® Guide are:

	PMP® (Project Management Professional)	CAPM® (Certified Associate in Project Management)
Who it is for	Practitioners who have led projects	Entry-level and aspiring practitioners
Experience required	Yes (see 1.6)	No
Question style	Mostly situational ("What should the project manager do next?")	More knowledge-based ("Which artifact…?")
Based on	PMP Examination Content Outline (ECO)	CAPM Examination Content Outline

This guide is written for the **PMP**, because it is the certification most people mean when they study the PMBOK® Guide. Everything you need for the CAPM is also here: the CAPM tests the same vocabulary, processes, life cycles, agile frameworks and business-analysis basics, only at a lower level of difficulty.

1.2 Exam format (from 9 July 2026)

PMI introduced a new PMP Examination Content Outline (ECO) in July 2026. The most widely reported format is:

Element	Detail
Number of questions	**180** — of which **170 are scored** and **10 are unscored pretest items** (you cannot tell which ones)
Time	**240 minutes**
Breaks	Two optional 10-minute breaks. Once you start a break you cannot go back to the previous block of questions
Delivery	Pearson VUE test centre or online proctored

Element	Detail
Item types	Multiple choice (one answer), multiple response (choose two or three), matching / drag-and-drop, hotspot (click on part of a graphic), limited fill-in-the-blank (usually a calculation), and **case-study sets** in which several questions share one scenario that may include tables or charts
Approaches	About **40% predictive**, about **60% agile or hybrid**

> **Exam tip.** Pace yourself at roughly **80 seconds per question**. With 240 minutes for 180 questions you have 1.33 minutes per question. A good checkpoint plan is: question 60 at about 80 minutes, question 120 at about 160 minutes, leaving a few minutes at the end to revisit flagged items.

> **Trap.** Some websites still show the pre-2026 format (230 minutes, 175 scored + 5 unscored questions). Always check the current PMP Certification Handbook on pmi.org.

1.3 The three domains

The exam is organised into three **domains** (areas of practice). The 2026 outline moved a lot of weight from People and Process into the Business Environment:

Domain	2021 outline	2026 outline	What it covers
People	42%	**33%**	Leading and developing the team, conflict, stakeholder engagement and expectations, communication, knowledge transfer
Process	50%	**41%**	Planning and managing scope, schedule, finance, resources, procurement, quality, value delivery, status and closure
Business Environment	8%	**26%**	Governance, compliance, change control, impediments and issues, risk, continuous improvement, organisational change, external environment

On 180 questions this is roughly **59 People, 74 Process and 47 Business Environment** questions.

1.4 The 26 tasks

Each domain contains tasks. Each task has **enablers** — examples of the work involved. You do not need to memorise task numbers, but you should know *where* a topic lives, because it tells you what the question is really testing.

Domain I — People (33%)

#	Task	What questions typically test
1	Develop a common vision	Helping the team and stakeholders share one understanding of the goal: vision statements, project/product vision, team charter, alignment workshops
2	Manage conflicts	Recognising the source and stage of conflict, choosing an appropriate resolution approach, facilitating rather than imposing

#	Task	What questions typically test
3	Lead the project team	Servant leadership, empowerment, motivation, coaching, team development, psychological safety, performance
4	Engage stakeholders	Identifying and analysing stakeholders, building relationships, choosing engagement strategies
5	Align stakeholder expectations	Surfacing expectations early, facilitating agreement among stakeholders with different interests
6	Manage stakeholder expectations	Monitoring satisfaction, adjusting engagement, keeping expectations consistent with what can be delivered
7	Help ensure knowledge transfer	Lessons learned, documentation, mentoring, handovers, transfer of knowledge to operations and to future projects
8	Plan and manage communication	Communication requirements, methods, frequency, channels, feedback, information radiators, reporting

Domain II — Process (41%)

#	Task	What questions typically test
1	Develop an integrated project management plan and plan delivery	Choosing the approach, integrating subsidiary plans, planning releases and iterations
2	Develop and manage project scope	Requirements, scope statement, WBS/backlog, validating deliverables, preventing scope creep
3	Help ensure value-based delivery	Identifying value with stakeholders, prioritising by value, incremental delivery, measuring benefits
4	Plan and manage resources	Team and physical resources, acquisition, allocation, resource constraints
5	Plan and manage procurement	Make-or-buy, contract types, selecting sellers, administering contracts
6	Plan and manage finance	Estimates, budgets, reserves, earned value, forecasting, tracking spend
7	Plan and optimise quality of products/deliverables	Quality planning, acceptance criteria, definition of done, quality assurance and control
8	Plan and manage schedule	Estimating, sequencing, critical path, compression, release planning, velocity
9	Evaluate project status	Performance measurement, metrics, dashboards, reporting, forecasts
10	Manage project closure	Formal acceptance, transition, contract closure, lessons learned, archiving, releasing resources

Domain III — Business Environment (26%)

#	Task	What questions typically test
1	Define and establish project governance	Decision rights, escalation paths, phase gates, roles (sponsor, steering committee, PMO, CCB)
2	Plan and manage project compliance	Regulatory, legal, contractual and organisational requirements; audits; consequences of non-compliance
3	Manage and control changes	Integrated change control, change requests, the change control board, backlog-based change in agile
4	Remove impediments and manage issues	Issue logs, escalation, servant-leader impediment removal
5	Plan and manage risk	Risk planning, identification, analysis, responses, monitoring, reserves
6	Continuous improvement	Retrospectives, lessons learned, process improvement, Kaizen, PDCA
7	Support organisational change	Readiness, change management models, adoption, resistance, transition to operations
8	Evaluate external business environment changes	Regulations, market, technology, economic and social changes and their effect on the project

> **What changed.** Risk and change control now sit in the **Business Environment** domain, and the **Process** domain has separate tasks for *finance*, *resources* and *procurement*. Expect more questions on governance, compliance, sustainability, artificial intelligence and organisational change than in older practice tests.

1.5 How the PMBOK® Guide relates to the exam

The PMP exam is built from the **Examination Content Outline**, not from a single book. PMI uses several references when it writes questions, and the *PMBOK® Guide – Eighth Edition* is the core one. Questions rarely ask you to recite the Guide. Instead they describe a situation and ask what a competent project manager would do. To answer them you need three things:

1. **Vocabulary and concepts** from the PMBOK® Guide (principles, performance domains, processes, artifacts, techniques).
2. **Judgment** — knowing which action comes *first*, which is *best* and which is merely possible.
3. **Context awareness** — recognising whether the scenario is predictive, agile or hybrid, because the right action can differ.

1.6 Eligibility and scoring (summary)

To apply for the PMP you need **35 contact hours** of project management education (or an active CAPM) and experience leading projects within the last eight years:

- **36 months** with a four-year (bachelor's) degree or global equivalent;
- **60 months** with a high-school diploma or associate's degree;

- **24 months** with a bachelor's or postgraduate degree from a GAC-accredited programme.

Your application may be selected for **audit**, in which case you must provide evidence (diplomas, training certificates, supervisor signatures).

Scoring. PMI does not publish a fixed pass mark. A panel of experts determines the passing standard using psychometric methods, and your report shows **Pass/Fail** plus a rating per domain: **Above Target, Target, Below Target, Needs Improvement**. As a practical rule, if you consistently score **75% or more** on good-quality full-length practice exams, you are in a strong position.

1.7 What the exam rewards — a first look at the "PMI mindset"

You will see this in detail in Chapter 21. In short, the best answer usually:

- **analyses before acting** — understand the situation, check the plan, assess the impact;
- **addresses the root cause** rather than a symptom;
- **involves the right people** — talk directly to the person concerned, collaborate with the team, engage the sponsor or product owner when it is their decision;
- **follows the agreed process** (for example, a change request in predictive work, backlog prioritisation by the product owner in agile work);
- **serves the team** — remove impediments, coach, protect focus, empower decisions;
- **keeps value, ethics and sustainability in view.**

Answers that ignore a problem, escalate too early, blame people, bypass governance or let the project manager make decisions that belong to others are usually wrong.

Chapter 1 Review Questions

Answer all questions first, then check the answer key for Chapter 1 in Appendix B.

1.1 How many questions on the PMP exam delivered since July 2026 contribute to your score?

- **A.** 170
- **B.** 175
- **C.** 180
- **D.** 185

1.2 What is the weighting of the Business Environment domain in the 2026 PMP Examination Content Outline?

- **A.** 8%
- **B.** 26%
- **C.** 33%
- **D.** 41%

1.3 In the 2026 Examination Content Outline, in which domain does the task "Plan and manage risk" appear?

- **A.** Business Environment
- **B.** Process
- **C.** It is not an exam task; risk is tested only through the PMBOK Guide

- **D.** People

1.4 A candidate has a bachelor's degree from a non-GAC-accredited university. How many months of project leadership experience are required to apply for the PMP?

- **A.** 24 months
- **B.** 36 months
- **C.** 48 months
- **D.** 60 months

1.5 Approximately what share of PMP exam questions reflect predictive approaches?

- **A.** 20%
- **B.** 40%
- **C.** 50%
- **D.** 75%

1.6 Which two tasks belong to the People domain in the 2026 outline? (Choose two.)

- **A.** Evaluate project status
- **B.** Support organisational change
- **C.** Plan and manage procurement
- **D.** Align stakeholder expectations
- **E.** Help ensure knowledge transfer

1.7 Which statement best describes how the PMBOK Guide relates to the PMP exam?

- **A.** The exam tests the PMBOK Guide word for word, chapter by chapter, including all definitions
- **B.** The exam only tests the processes and ITTOs of the PMBOK Guide
- **C.** The exam follows the content outline; the PMBOK Guide is a core reference for questions
- **D.** The PMBOK Guide is not used at all; only agile references matter

1.8 What does the PMP score report show for each domain?

- **A.** The exact number of correct answers in the domain, out of the scored questions
- **B.** A percentile compared with other candidates
- **C.** A performance rating, such as Above Target or Needs Improvement
- **D.** A letter grade from A to F

1.9 During the exam, a candidate takes the first optional break after question 60. What is true about the questions in the first section after the break?

- **A.** They can be reviewed but not changed
- **B.** They can be reviewed and changed until the end of the exam
- **C.** They are automatically marked as incorrect if they were flagged
- **D.** They can no longer be reviewed or changed

1.10 A practice question describes a team that works in two-week sprints with a product owner. Why does identifying this context matter before choosing an answer?

- **A.** Agile questions are unscored pretest items
- **B.** Agile questions always have two correct answers

- **C.** It does not matter, because the PMI mindset and the correct actions are identical in every context and approach
- **D.** The best action may differ by context, e.g. backlog reprioritisation instead of a change request

1.11 Match each element of the 2026 PMP exam with its value.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Scored questions	**A.** 170
2. Exam duration	**B.** 240 minutes
3. Weight of the People domain	**C.** 41%
4. Weight of the Process domain	**D.** 33%

Chapter 2 — PMBOK® Guide 8 at a Glance: Structure, Evolution and Core Vocabulary

After this chapter you should be able to:

- describe how the Eighth Edition is organised and why it changed;
- explain the "why – what – how" logic of principles, performance domains and processes;
- define the core terms used everywhere in the exam (project, program, portfolio, operations, value, outcome, benefit…);
- translate older knowledge-area vocabulary into Eighth-Edition terms.

2.1 What the PMBOK® Guide is

The *PMBOK® Guide – Eighth Edition* (published by PMI in late 2025) is a single volume containing two parts:

1. **The Standard for Project Management** — a short, consensus-based (ANSI-accredited) description of *what* project management is: the system for value delivery, the project environment and the **six principles** that guide behaviour.
2. **A Guide to the Project Management Body of Knowledge** — the larger part, describing the **seven performance domains**, the **40 processes** grouped in **five focus areas**, **tailoring**, and the **models, methods and artifacts** practitioners use.

It closes with appendices. Those most relevant to the exam are **X2 – Project Management Office (PMO)**, **X3 – Artificial Intelligence (AI)** and **X4 – Procurement**, plus a glossary. (X1 lists contributors and X5 describes the evolution of the Guide.)

> **Exam tip.** The PMBOK® Guide is *descriptive and non-prescriptive*: it describes good practice that is applicable to most projects most of the time, and the project team **tailors** it. An answer that says "always follow every process in the Guide" is suspicious.

2.2 How we got here — the evolution of the Guide

Edition	Year	Organising idea
1st–5th	1996–2013	Process-based: process groups × knowledge areas, inputs/tools & techniques/outputs (ITTOs)
6th	2017	**5 process groups, 10 knowledge areas, 49 processes**; agile content added in each knowledge area; *Agile Practice Guide* bundled
7th	2021	Radical shift to **principles**: **12 principles**, **8 performance domains**, tailoring, **models, methods and artifacts**; processes moved to PMIstandards+
8th	2025	Synthesis: **6 principles, 7 performance domains, 5 focus areas, 40 processes**, tailoring, models/methods/artifacts, new appendices on PMO, AI and procurement

Why the change? Practitioners found the Sixth Edition too process-heavy for adaptive work, and the Seventh Edition too abstract to apply ("we know the principles, but *how* do we do it?"). The Eighth Edition keeps the principles-and-domains foundation of the Seventh Edition, simplifies it, and **reintroduces processes** — but as non-prescriptive guidance that applies to predictive, adaptive and hybrid work alike.

The three layers: why, what and how

Layer	Eighth-Edition element	Question it answers
Why	**6 principles**	What beliefs and behaviours should guide every decision?
What	**7 performance domains**	Which areas of activity must be managed effectively for the project to deliver its outcomes?
How	**40 processes in 5 focus areas** + tailoring + models, methods and artifacts	What do we actually do, in which order, with which tools?

The six principles (Chapter 4)

1. **Adopt a holistic view**
2. **Focus on value**
3. **Embed quality into processes and deliverables**
4. **Be an accountable leader**
5. **Integrate sustainability within all project areas**
6. **Build an empowered culture**

The seven performance domains (Chapters 7–14)

Domain	Main concern	Number of processes
Governance	Decision structures, oversight, integration of plans, execution, quality assurance, knowledge, change, closure	9
Scope	Requirements, scope definition and structure, validation and control (including quality control)	6
Schedule	Planning, developing and controlling the schedule	3
Finance	Financial planning, cost estimates, budget, financial control and funding	4
Stakeholders	Identification, engagement and communications	7
Resources	Team and physical/virtual resources, acquisition, leadership, resource control	5
Risk	Risk planning, identification, analysis, responses and monitoring	6
	Total	**40**

The five focus areas (Chapter 6)

Initiating → Planning → Executing → Monitoring and Controlling → Closing. These are the former *process groups*. They describe the *nature* of the work, not phases in time; they repeat in every phase, release or iteration.

2.3 What happened to the ten knowledge areas?

If you studied with Sixth-Edition material, this translation table will save you time:

Sixth Edition knowledge area	Where it lives in the Eighth Edition
Integration Management	**Governance** domain (charter, integrated plan, execution, knowledge, monitoring, change control, closure)
Scope Management	**Scope** domain
Schedule Management	**Schedule** domain (activity definition, sequencing and estimating are folded into *Develop Schedule*)
Cost Management	**Finance** domain (broadened to funding and financial value)
Quality Management	Spread out: quality **planning** inside *Define Scope*, quality **assurance** in Governance (*Manage Quality Assurance*), quality **control** inside *Monitor and Control Scope* — reflecting the principle *Embed quality*
Resource Management	**Resources** domain (*Develop Team* and *Manage Team* merge into *Lead the Team*)
Communications Management	**Stakeholders** domain
Risk Management	**Risk** domain (qualitative and quantitative analysis merge into *Perform Risk Analysis*)
Procurement Management	Sourcing strategy in **Governance** (*Plan Sourcing Strategy*); acquisition and control through the **Resources** domain; strategic detail in **Appendix X4**
Stakeholder Management	**Stakeholders** domain

And the Seventh Edition's eight domains map like this: *Stakeholders → Stakeholders; Team → Resources; Development Approach & Life Cycle → Governance and tailoring; Planning → spread across all domains; Project Work → Governance and Resources; Delivery → Scope (plus value and quality); Measurement → Governance (monitoring) and each domain's control process; Uncertainty → Risk.* Schedule and Finance became domains in their own right.

2.4 Core vocabulary

You must be able to use these terms precisely — many distractors depend on confusing them.

Term	Meaning
Project	A **temporary** endeavour undertaken to create a **unique** product, service or result. Temporary = definite beginning and end (not short); unique = something not done exactly before.
Progressive elaboration	Developing plans and details iteratively as more information becomes available.
Project management	The application of knowledge, skills, tools and techniques to project activities to meet project requirements and deliver value.
Operations	Ongoing, repetitive work that sustains the organisation (e.g., running payroll). Projects often *change* operations.
Program	Related projects, subsidiary programs and program activities managed in a **coordinated** way to obtain **benefits** not available from managing them individually.
Portfolio	Projects, programs, subsidiary portfolios and operations managed **as a group** to achieve **strategic objectives**. Components need not be related to each other.
Product	An artifact that is produced, quantifiable, and can be an end item or a component. Products often outlive the projects that create or change them.
Deliverable	Any unique and verifiable product, result or capability required to complete a process, phase or project.
Output	What the project produces (a deliverable).
Outcome	The end result or consequence of using the output (e.g., faster order processing).
Benefit	A gain realised by the organisation and beneficiaries from the outcome (e.g., lower costs, higher customer satisfaction).
Value	The worth, importance or usefulness of something — financial or non-financial. Projects exist to create value.
Organisational project management (OPM)	A framework that integrates portfolio, program and project management with organisational enablers to achieve strategic objectives.
Stakeholder	An individual, group or organisation that may affect, be affected by, or perceive itself to be affected by a decision, activity or outcome of the project.
Sponsor	The person or group that provides resources and support for the project and is accountable for enabling success (and usually for benefits).
Baseline	The approved version of a work product (scope, schedule, cost) that can be changed only through formal change control and is used for comparison.
Enterprise environmental factors (EEFs)	Conditions *not under the immediate control of the team* that influence, constrain or direct the project (culture, regulations, market, infrastructure).
Organisational process assets (OPAs)	Plans, processes, policies, procedures and knowledge bases *specific to and used by* the performing organisation (templates, lessons learned, historical data).
Tailoring	Deliberate adaptation of approach, governance and processes to suit the context.
Model / Method / Artifact	A model is a thinking strategy to explain a process or framework; a method is a means of achieving an outcome (a technique); an artifact is a template, document, output or deliverable.

Project vs operations vs program vs portfolio

Aspect	Project	Program	Portfolio	Operations
Objective	Deliver specific outputs and outcomes	Deliver benefits through coordination	Achieve strategic objectives; optimise investment	Sustain the business
Duration	Temporary	Temporary (often long)	Ongoing	Ongoing
Change	Controlled through the plan	Expected; managed to optimise benefits	Monitored in the broader environment	Minimised; continuous improvement
Success measured by	Product and project quality, timeliness, budget, satisfaction, value	Benefits realised	Aggregate investment performance and benefits	Efficiency, KPIs

Why projects are started

Projects are vehicles for change. Common drivers:

- **Regulatory, legal or social requirements** (new safety law, accessibility rules);
- **Stakeholder requests or needs** (a customer asks for a new feature, a community requests a facility);
- **Business or technological strategies** (digital transformation, entering a new market, adopting AI);
- **Creating, improving or fixing products, processes or services** (fixing defects, improving performance).

> **Exam tip.** Questions often hide a portfolio or program concept inside a project scenario. If two projects compete for the same scarce resources and the decision affects strategy, the right level for the decision is usually **portfolio** governance, not the project manager.

2.5 Reading the rest of this guide

Chapters 3–5 cover the Standard (value delivery, environment, principles) and life cycles. Chapter 6 introduces focus areas and all 40 processes. Chapters 7–14 go domain by domain. Chapters 15–19 cover agile/hybrid practice, people skills, the business environment, tailoring and the toolbox of models, methods and artifacts. Chapter 20 collects every formula, and Chapter 21 prepares you for the exam itself.

Chapter 2 Review Questions

Answer all questions first, then check the answer key for Chapter 2 in Appendix B.

2.1 Which combination correctly describes the structure of the PMBOK Guide – Eighth Edition?

- **A.** 5 process groups, 10 knowledge areas and 49 processes
- **B.** 6 principles, 8 performance domains, 5 process groups and 49 processes

- **C.** 6 principles, 7 performance domains, 5 focus areas and 40 processes
- **D.** 12 principles, 8 performance domains and models, methods and artifacts

2.2 Which of the following is NOT one of the seven performance domains of the Eighth Edition?

- **A.** Resources
- **B.** Finance
- **C.** Quality
- **D.** Governance

2.3 A new IT system is delivered (output). Customer orders are now processed in four hours instead of two days (outcome). What is the best example of the resulting benefit?

- **A.** The trained help-desk team
- **B.** The project closure report
- **C.** Higher customer retention and lower processing costs
- **D.** The installed and tested IT system, accepted by the customer

2.4 What distinguishes a program from a portfolio?

- **A.** A program is ongoing, while a portfolio is temporary
- **B.** A program coordinates related components for added benefits; a portfolio groups components to meet strategy
- **C.** A portfolio contains only projects, while a program contains only operations and maintenance work of the organisation
- **D.** There is no difference; the terms are interchangeable

2.5 Which of the following is an organisational process asset rather than an enterprise environmental factor?

- **A.** The lessons learned repository from previous projects
- **B.** New government safety regulations
- **C.** Current market exchange rates
- **D.** The organisation's culture and its appetite for risk in new projects

2.6 In the Eighth Edition, where does Communications Management (a Sixth Edition knowledge area) primarily reside?

- **A.** Resources performance domain
- **B.** Governance performance domain
- **C.** It was removed from the Eighth Edition
- **D.** Stakeholders performance domain

2.7 A colleague says, "A project is temporary, so it must be short." What is the best response?

- **A.** Temporary means a definite beginning and end, not a short duration
- **B.** Temporary means that the team is always disbanded after each phase of the project
- **C.** Correct; long-duration work is always operations
- **D.** Correct; projects longer than one year are programs

2.8 Why did PMI reintroduce processes in the Eighth Edition?

- **A.** Because the PMP exam is now based only on ITTOs

- **B.** To remove agile and hybrid approaches from the Guide and return to a purely predictive model
- **C.** To make the Guide mandatory for all projects
- **D.** Principles alone were too abstract, so processes returned as non-prescriptive guidance

2.9 Which two appendices of the Eighth Edition are newly added or significantly expanded and frequently referenced for exam preparation? (Choose two.)

- **A.** Business Analysis for Practitioners
- **B.** Artificial Intelligence
- **C.** Procurement
- **D.** Earned Value Management standard
- **E.** Agile Manifesto

2.10 A company replaces its payroll software, then continues to run payroll every month with the new system. Which statement is correct?

- **A.** Both activities are operations because payroll is recurring
- **B.** Replacing the software is a project; running monthly payroll is operations
- **C.** Replacing the software is a portfolio, and running the monthly payroll is a program
- **D.** Both activities are projects because they involve software

2.11 Match each layer of the Eighth Edition with the question it answers.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Principles	**A.** What — which areas of activity must be managed effectively?
2. Performance domains	**B.** How — which work is performed, in what order, with which tools?
3. Processes and focus areas	**C.** Why — which beliefs and behaviours should guide decisions?
4. Tailoring	**D.** Fit — how should the approach be adapted to this context?

2.12 A government ministry passes a new law on data protection that affects a software project. For the project, this law is best classified as:

- **A.** An organisational process asset of the ministry
- **B.** An enterprise environmental factor
- **C.** A work performance report
- **D.** A project deliverable

Chapter 3 — The Standard: A System for Value Delivery, Governance and the Project Environment

After this chapter you should be able to:

- explain how portfolios, programs, projects, products and operations work together to create value;
- distinguish organisational governance from project governance and describe the main governance roles;
- classify internal and external environmental factors and organisational process assets;
- compare organisational structures and their effect on the project manager's authority;
- describe the business documents that justify and authorise a project (needs assessment, business case, benefits management plan, charter);
- describe the project manager's role and the PMI Talent Triangle.

3.1 Value is the reason projects exist

A project is not successful just because it delivered on time and on budget. It is successful when it contributes to **value** — the worth, importance or usefulness that stakeholders obtain. Value can be:

- **tangible**: revenue, cost savings, market share, assets, equity;
- **intangible**: reputation, brand recognition, public benefit, strategic alignment, employee goodwill, sustainability.

The chain that connects work to value is:

Activities → Outputs (deliverables) → Outcomes (changes in behaviour or capability) → Benefits (gains for the organisation) → Value

> **Exam tip.** When a question asks how to measure success, prefer answers that refer to **outcomes, benefits and value** (customer adoption, business KPIs, benefit realisation) over answers that refer only to outputs or to the triple constraint.

3.2 The system for value delivery

Organisations do not deliver value through isolated projects. They use a **system for value delivery**: a collection of strategic business activities aimed at building, sustaining and advancing the organisation. Its components are:

Component	Role in value delivery
Portfolios	Select and prioritise the right work to execute strategy; balance investments and risks
Programs	Coordinate related projects and activities to realise benefits that would not be obtained individually
Projects	Create the outputs that enable outcomes and benefits

Component	Role in value delivery
Products	Deliver value continuously over their life cycle; may be created or changed by many projects
Operations	Run the business and *sustain* the value once deliverables are transitioned

Information flows in both directions. Strategy, priorities and constraints flow *down* from leadership through portfolios and programs to projects. Performance data, lessons, risks and benefit realisation information flow *up*, so leadership can adjust strategy and investments. A project manager who reports only "green" status is breaking this feedback loop.

Value is delivered continuously

In adaptive and product-oriented organisations, value is delivered in increments: each release is used, measured and fed back into the backlog. Value delivery does not wait for project closure. Predictive projects may deliver most value at the end, but benefits are still tracked after the transition to operations.

Products and product management

The Eighth Edition gives explicit attention to **product management considerations**:

- A **product life cycle** (introduction, growth, maturity, decline, retirement) is longer than any single project.
- Organisations increasingly fund **long-lived product teams** rather than temporary project teams ("project-to-product" shift).
- Projects may create a new product, add capabilities, or retire one.
- A product focus emphasises customer outcomes, continuous discovery, roadmaps and backlog management — vocabulary you will meet again in Chapter 15.

3.3 Governance: organisational and project

Governance is the framework of authority, accountability, policies, decision rights and oversight that directs and controls work. Two levels matter:

Level	Examples	Purpose
Organisational governance	Board, executive committees, portfolio governance, policies, compliance rules	Sets strategy, risk appetite and rules for the whole organisation
Project governance	Sponsor, steering committee, phase gates, change control board, escalation paths, PMO oversight	Guides and oversees the specific project within the organisation's framework

A good governance framework answers: *Who decides what? Within what limits (thresholds)? How are issues escalated? How is performance reviewed? How are changes approved? How are compliance and ethics ensured?*

Key governance roles

Role	Typical responsibilities
Sponsor	Champions the project, provides funding and resources, approves the charter, makes decisions above the PM's authority, removes organisational obstacles, is usually accountable for benefits
Steering committee / project board	Provides strategic direction, approves major changes and phase gates, resolves escalated issues
Portfolio manager / portfolio board	Prioritises investments, allocates scarce resources across projects, decides on start/stop
Program manager	Coordinates related projects, manages interdependencies and benefits
PMO	Standardises governance, provides methods, templates, training, reporting and sometimes direct management of projects
Change control board (CCB)	Reviews, approves, defers or rejects change requests according to authority thresholds
Product owner (adaptive)	Owns the product backlog, maximises value, decides priorities and accepts work
Project manager	Leads the team to achieve objectives, integrates the work, operates within delegated authority
Functional / resource managers	Provide people and expertise, often hold administrative authority over team members

3.4 Functions needed in every project

Regardless of job titles, every project needs certain **functions** to be fulfilled. One person may perform several functions and one function may be shared:

- **Provide oversight and coordination** — plan, monitor, integrate (often the PM, PMO or scrum master/team facilitator).
- **Present objectives and feedback** — articulate needs, priorities and acceptance (customer, product owner, users).
- **Facilitate and support** — encourage participation, consensus, removing impediments (PM, scrum master, coach).
- **Perform work and contribute insights** — create the deliverables (team members, contractors).
- **Apply expertise** — subject-matter experts, specialists.
- **Provide business direction and insight** — clarify value and priorities (sponsor, product owner, business analyst).
- **Provide resources and direction** — sponsor, functional managers, governance bodies.
- **Maintain governance** — approve, oversee and ensure compliance (steering committee, PMO, CCB).

3.5 The project environment

Projects exist inside an environment that influences them. The Guide distinguishes **internal** and **external** factors.

Internal environment	External environment
Organisational culture, structure and governance	Marketplace conditions (competitors, market share, brand)
Geographic distribution of facilities and people	Social and cultural influences, political climate, ethics
Infrastructure, IT systems and software	Regulatory and legal environment (laws, standards, data protection, safety)
Resource availability and employee capability	Commercial databases and benchmarking data
Process assets (templates, procedures, policies)	Academic research and industry standards
Data assets and knowledge assets (lessons learned, historical data)	Financial considerations (exchange rates, interest rates, inflation, tariffs)
Security and safety policies	Physical environment (weather, working conditions)

Two traditional labels are still widely used in exam questions:

- **Enterprise environmental factors (EEFs)**: conditions *not under the team's immediate control* (internal or external) that influence, constrain or direct the project.
- **Organisational process assets (OPAs)**: plans, processes, policies, procedures and knowledge bases *of the performing organisation*. They are of two kinds: **processes, policies and procedures** (usually not updated by the project) and **organisational knowledge repositories** (updated by the project — e.g., lessons learned, historical information).

> **Trap.** An organisation's *culture* is an EEF, not an OPA, even though it is internal. Templates and lessons learned are OPAs.

3.6 Organisational structures and the project manager's authority

Structure	PM authority	Resource availability	Who controls the budget	PM role
Organic / simple	Little or none	Little or none	Owner/operator	Part-time, coordinator
Functional (centralised)	Little or none	Little or none	Functional manager	Part-time; often called *project coordinator* or *expediter*
Multi-divisional	Little or none	Little or none	Functional manager	Part-time
Weak matrix	Low	Low	Functional manager	Part-time; coordinator/expediter
Balanced matrix	Low to moderate	Low to moderate	Mixed	Full-time
Strong matrix	Moderate to high	Moderate to high	Project manager	Full-time, with admin staff
Project-oriented (projectized)	High to almost total	High to almost total	Project manager	Full-time

Structure	PM authority	Resource availability	Who controls the budget	PM role
Virtual / networked	Low to moderate	Low to moderate	Mixed	Full- or part-time
Hybrid	Mixed	Mixed	Mixed	Mixed
PMO-led	High	High	Project manager	Full-time

- A **project expediter** acts as a staff assistant and communications coordinator with no decision authority.
- A **project coordinator** has some limited authority and usually reports to a higher-level manager.
- In a **matrix**, team members have two bosses: the functional manager (administrative, career, often technical) and the project manager (project work). Conflicts over priorities and resources are common — the PM must **negotiate** and use influence rather than formal power.

3.7 Business documents: justifying and authorising the project

Document	Owner	Content	Purpose
Needs assessment	Business / business analyst	Problem or opportunity, current state, desired future state, gaps	Understand the business need before investing
Business case	Sponsor (prepared with business analysts)	Business need, situation analysis, options (do nothing, do minimum, do more), recommended option, cost-benefit analysis, financial measures (NPV, IRR, payback), risks, success criteria	Justify the investment; reviewed at phase gates to confirm the project is still worthwhile
Benefits management plan	Sponsor / benefit owner	Target benefits, strategic alignment, timeframe for realisation, **benefit owner**, metrics, assumptions, risks	Describe how and when benefits will be delivered and measured
Project charter	Issued by the **sponsor** (or initiator); developed with the PM	Purpose, measurable objectives and success criteria, high-level requirements, boundaries and key deliverables, overall risk, summary milestones, pre-approved financial resources, key stakeholders, approval and exit criteria, **assigned PM and authority level**, sponsor's name and authority	**Formally authorises the project** and gives the PM authority to apply organisational resources

Exam tip. The **charter** authorises the project; the **business case** justifies it; the **benefits management plan** describes how value will be realised and measured. If a question asks "the project no longer seems worthwhile — what should be reviewed?", the answer is usually the **business case** (and the benefits plan), often at a phase gate.

Agreements and contracts

When the project is performed for an external customer, an **agreement** (contract, memorandum of understanding, letter of intent, service-level agreement) defines the initial intentions and is an input to initiating the project.

3.8 The project manager's role and competencies

The project manager is the person assigned by the performing organisation to lead the team responsible for achieving the project objectives. A common metaphor is the **conductor of an orchestra**: the PM does not play every instrument but integrates and directs. The PM's spheres of influence include the **project** (team, sponsor, stakeholders), the **organisation** (other PMs, PMO, executives), the **industry** (trends, standards, regulations) and the **profession** (knowledge transfer, continuing education, mentoring).

The PMI Talent Triangle®

Side	Examples
Ways of Working	Mastery of predictive, agile, hybrid, design thinking and other approaches; selecting and tailoring methods
Power Skills	Collaborative leadership, communication, innovative mindset, for-purpose orientation, empathy, negotiation, problem-solving
Business Acumen	Understanding of the organisation's strategy, finance, markets, competition, regulations; linking the project to value

> **What changed.** Earlier versions of the Talent Triangle used *Technical Project Management*, *Leadership* and *Strategic and Business Management*. You may still see these labels in older materials.

Integration — the PM's core job

Integration happens at several levels: **process** (making the pieces of the plan fit together), **cognitive** (applying knowledge and judgment), **context** (adapting to the environment and technology) and **complexity** (managing the system behaviour of the project). The PM, not the team, is ultimately accountable for integrating the project as a whole.

Chapter 3 Review Questions

Answer all questions first, then check the answer key for Chapter 3 in Appendix B.

3.1 Which document formally authorises the existence of a project and gives the project manager authority to apply organisational resources?

- **A.** Business case
- **B.** Benefits management plan
- **C.** Project charter
- **D.** Project management plan

3.2 Midway through a predictive project, a competitor launches a product that makes the planned solution far less attractive. At the next phase gate, which document should the steering committee primarily revisit to decide whether to continue?

- **A.** The business case
- **B.** The communications management plan
- **C.** The project schedule
- **D.** The resource calendar

3.3 Who is typically accountable for the realisation of benefits after a project's deliverables are transferred to operations?

- **A.** The project team, which builds the deliverables
- **B.** The project manager, who delivers the outputs
- **C.** The procurement department
- **D.** The sponsor or the designated benefit owner

3.4 In which organisational structure does the project manager have the least authority and usually acts as a coordinator or expediter?

- **A.** Strong matrix
- **B.** PMO-led
- **C.** Functional
- **D.** Projectized

3.5 A project manager in a balanced matrix organisation needs a senior engineer who reports to a functional manager. What is the best approach?

- **A.** Assign the engineer directly, because the project has an approved charter and budget
- **B.** Hire a contractor without consulting anyone
- **C.** Negotiate with the functional manager for the engineer's time
- **D.** Escalate immediately to the CEO

3.6 Which of the following is classified as an enterprise environmental factor?

- **A.** The lessons learned repository
- **B.** The organisation's documented change control procedure for projects
- **C.** The company's project schedule template
- **D.** The organisation's culture and appetite for risk

3.7 Which two items would you expect to find in a benefits management plan? (Choose two.)

- **A.** The detailed WBS dictionary
- **B.** The benefit owner responsible for realisation
- **C.** The timeframe for realising target benefits
- **D.** The critical path of the schedule
- **E.** The team's ground rules

3.8 Why is information expected to flow both up and down the system for value delivery?

- **A.** Strategy guides the work, and performance and benefit information lets leaders adjust strategy
- **B.** So that operations can approve project change requests

- **C.** So the project manager can bypass the sponsor when needed and report directly to the board of directors
- **D.** Because portfolios report to projects

3.9 An organisation funds a permanent cross-functional team to evolve its mobile banking app over several years. This reflects which concept?

- **A.** A temporary project with a fixed end date and a dedicated, cross-functional team
- **B.** A product-oriented approach that delivers value continuously
- **C.** A portfolio of unrelated projects
- **D.** A projectized organisational structure

3.10 Which side of the PMI Talent Triangle includes negotiation, communication and collaborative leadership?

- **A.** Technical Project Management
- **B.** Ways of Working
- **C.** Business Acumen
- **D.** Power Skills

3.11 A sponsor asks the project manager to prepare the charter. What is the most appropriate response?

- **A.** Develop it with the sponsor, who issues and signs it
- **B.** Write and sign the charter alone, then start planning
- **C.** Skip the charter because a business case already exists
- **D.** Refuse, because only the PMO may write charters

3.12 In the system for value delivery, which component sustains the value of deliverables after a project ends?

- **A.** The project team
- **B.** Operations
- **C.** The PMO
- **D.** The change control board

Chapter 4 — The Six Principles of Project Management

After this chapter you should be able to:

- state the six principles of the Eighth Edition and explain each in your own words;
- recognise behaviours that apply (or violate) each principle in a scenario;
- relate the six principles to the twelve principles of the Seventh Edition and to the PMI Code of Ethics.

4.1 What a principle is (and is not)

A **principle** is a fundamental norm, truth or value that guides behaviour and decisions. Principles are **not prescriptive**: they do not tell you which template to use. They apply to every project, every development approach, every industry. When a scenario seems to have no "process" answer, ask yourself which principle the best action would honour.

The principles are consistent with the **PMI Code of Ethics and Professional Conduct** (responsibility, respect, fairness and honesty — see Chapter 16). They also overlap: one good action usually honours several principles at once.

4.2 Principle 1 — Adopt a holistic view

Idea in one sentence: see the project as a system of interacting parts that lives inside larger systems (the organisation, the market, society), and make decisions with the whole in mind.

Key ideas

- **Systems thinking.** A change in one area ripples elsewhere: compressing the schedule may raise cost, risk and team stress; a supplier change may affect quality and compliance.
- **Integration across domains.** Scope, schedule, finance, resources, risk and stakeholders are interdependent — the seven domains are *interactive, interrelated and interdependent*.
- **Life-cycle perspective.** Consider the product's whole life — operation, maintenance, support, disposal — not only the project's delivery.
- **Complexity awareness.** Many sources of complexity (human behaviour, system behaviour, ambiguity, technological innovation) cannot be removed by planning harder; they require sensing, adapting and feedback.
- **Context.** The organisation's strategy, culture, governance and environment shape what "good" looks like.

In practice: impact analysis before approving changes; cross-functional reviews; considering downstream users and operations; tracking interdependencies with other projects and programs.

Violations to recognise: optimising one constraint at the expense of the whole ("we met the date, the help desk can deal with the defects"); ignoring effects on other projects that share resources.

4.3 Principle 2 — Focus on value

Idea in one sentence: continually evaluate and adjust the project so that it delivers the intended outcomes, benefits and value — not just outputs.

Key ideas

- **Value is the ultimate success indicator.** It may be financial or non-financial, and may be realised during or after the project.
- **Clarify value early** with stakeholders: what problem are we solving, for whom, and how will we know?
- **Prioritise by value.** In adaptive work the product owner orders the backlog by value; in predictive work the business case and benefits plan anchor decisions.
- **Deliver early and often** where possible, so value is realised sooner and assumptions are validated by real feedback.
- **Re-evaluate continuously.** If the business case is no longer valid, the right decision may be to change direction — or to stop the project.

In practice: minimum viable products (MVPs); value-based prioritisation (MoSCoW, cost of delay, weighted shortest job first); benefit KPIs; phase-gate reviews of the business case.

Violations: continuing a project that no longer makes sense because "we already spent so much" (sunk-cost thinking); measuring success purely by "on time, on budget".

4.4 Principle 3 — Embed quality into processes and deliverables

Idea in one sentence: quality is designed and built in from the beginning — in the way work is done and in what is produced — rather than inspected in at the end.

Key ideas

- **Two dimensions of quality:** the *process* (how we work) and the *product* (what we deliver). Both must meet the expectations of stakeholders and the acceptance criteria.
- **Prevention over inspection.** Preventing a defect is cheaper than finding it; finding it early is cheaper than finding it late.
- **Quality is defined by requirements and fitness for use**, including non-functional requirements (performance, security, usability, sustainability).
- **Everyone owns quality.** Teams use definitions of done, peer reviews, test automation and standards; leaders create conditions for doing it right.
- **Continuous improvement.** Retrospectives, lessons learned, root-cause analysis and PDCA cycles improve both process and product.

Why it matters for the Eighth Edition structure: quality is no longer a separate domain. **Quality planning** is part of *Define Scope*, **quality assurance** is the Governance process *Manage Quality Assurance*, and **quality control** is part of *Monitor and Control Scope*. The principle explains why: quality is everyone's responsibility in every domain.

Violations: skipping testing to meet a date; adding extra features nobody asked for (gold plating — more is not "higher quality"); treating quality as the QA department's problem.

4.5 Principle 4 — Be an accountable leader

Idea in one sentence: take ownership of decisions, actions and results; lead with integrity and transparency, whatever your formal role.

Key ideas

- **Leadership is not a title.** Anyone on the team can demonstrate leadership behaviours: setting direction, motivating, facilitating, taking ownership.
- **Accountability and ownership.** Leaders own outcomes, admit mistakes, report status honestly and follow through on commitments.
- **Stewardship.** Act as a responsible steward of the organisation's resources and of the trust of stakeholders: integrity, care, trustworthiness and compliance with laws and policies.
- **Ethics.** Follow the PMI Code of Ethics; act transparently; avoid conflicts of interest; protect confidential information.
- **Adapt the style.** Use servant leadership to empower teams, directive leadership in a crisis, coaching for less experienced members (situational leadership).
- **Decision-making.** Make timely, informed decisions at the right level, and know when a decision belongs to someone else (product owner, sponsor, CCB).

In practice: honest status reports (including bad news, early); escalating through the agreed path when a decision exceeds your authority; protecting the team from inappropriate pressure; modelling the behaviours you expect.

Violations: hiding problems ("watermelon" status — green outside, red inside); blaming team members publicly; accepting gifts that create conflicts of interest; making decisions that belong to the customer or product owner.

4.6 Principle 5 — Integrate sustainability within all project areas

Idea in one sentence: consider the long-term economic, environmental and social impacts of the project and its product in every decision.

Key ideas

- **The triple bottom line: people** (social), **planet** (environmental) and **prosperity/profit** (economic). Many organisations frame this as **ESG** — environmental, social and governance.
- **Across all domains:** sustainable sourcing (Resources/Governance), life-cycle costs and green finance (Finance), environmental and social risks (Risk), community and stakeholder impacts (Stakeholders), sustainable requirements and acceptance criteria (Scope), energy-efficient scheduling and logistics (Schedule).
- **Product life cycle:** design for reuse, recycling and minimal waste (circular economy); consider operation and disposal.
- **Social sustainability includes the team:** sustainable pace, well-being, fair labour practices, diversity and inclusion.
- **Compliance and beyond.** Meet legal requirements (environmental permits, reporting rules) and the organisation's sustainability commitments (e.g., net-zero targets, UN Sustainable Development Goals).

In practice: including sustainability criteria in supplier selection; measuring the carbon footprint of a solution; engaging affected communities; choosing options with lower life-cycle impact even when initial cost is slightly higher, when the business case supports it.

Violations: ignoring environmental regulations to save time; selecting a supplier with known unethical labour practices because it is cheapest; consistently demanding overtime ("death marches").

4.7 Principle 6 — Build an empowered culture

Idea in one sentence: create an environment of trust, psychological safety and shared ownership in which people and stakeholders can contribute their best, make decisions and adapt.

Key ideas

- **Empowerment.** Push decisions to the people closest to the work; give teams autonomy within clear boundaries; self-organising teams decide *how* to do their work.
- **Psychological safety.** People can raise problems, admit mistakes and challenge ideas without fear.
- **Collaboration and shared ownership.** Team agreements (team charter, ground rules), collective accountability for results.
- **Engaged stakeholders.** Stakeholders participate actively rather than receiving reports passively.
- **Diversity, equity and inclusion.** Diverse perspectives improve decisions; inclusive practices ensure all voices are heard.
- **Learning and adaptability.** A culture that experiments, learns from failure, and adapts to change is more resilient.

In practice: servant leadership; facilitating rather than dictating; recognising contributions; retrospectives with real follow-through; coaching; delegating decisions with clear guardrails.

Violations: micromanagement; punishing people who report bad news; assigning tasks to self-organising teams; excluding remote team members from decisions.

4.8 How the twelve principles of the Seventh Edition live on

This mapping is an interpretive study aid (the Eighth Edition does not require you to memorise it), but it helps if your other materials still use the Seventh-Edition list:

Seventh Edition principle	Mainly reflected in (Eighth Edition)
Be a diligent, respectful and caring steward	Be an accountable leader; Integrate sustainability
Create a collaborative project team environment	Build an empowered culture
Effectively engage with stakeholders	Build an empowered culture; Stakeholders domain
Focus on value	Focus on value
Recognise, evaluate and respond to system interactions	Adopt a holistic view
Demonstrate leadership behaviours	Be an accountable leader
Tailor based on context	Tailoring (Guide); Adopt a holistic view

Seventh Edition principle	Mainly reflected in (Eighth Edition)
Build quality into processes and deliverables	Embed quality into processes and deliverables
Navigate complexity	Adopt a holistic view
Optimise risk responses	Risk performance domain
Embrace adaptability and resiliency	Build an empowered culture; Adopt a holistic view
Enable change to achieve the envisioned future state	Focus on value; Build an empowered culture (and organisational change, Chapter 17)

4.9 Using principles to answer exam questions

When two answers both look reasonable, check which one better reflects the principles:

If the scenario is about...	Prefer the answer that...
A change with ripple effects	analyses the impact across scope, schedule, cost, risk, quality and other projects (holistic view)
Continuing or stopping work	revisits value, the business case and benefits (focus on value)
Defects found late	addresses process causes and prevention, not just rework (embed quality)
A mistake or bad news	is transparent, owns the problem and acts ethically (accountable leader)
Environmental or social impacts	considers long-term impacts and stakeholders affected (sustainability)
Team decisions or morale	empowers, facilitates and builds safety (empowered culture)

Chapter 4 Review Questions

Answer all questions first, then check the answer key for Chapter 4 in Appendix B.

4.1 A project manager proposes to compress the schedule by adding a night shift. Before approving, the PM asks the team to analyse effects on quality, safety, costs, team well-being and the neighbouring community. Which principle is most directly demonstrated?

- **A.** Build an empowered culture
- **B.** Adopt a holistic view
- **C.** Embed quality into processes and deliverables
- **D.** Focus on value

4.2 Market research shows that the product being built will no longer meet customers' needs. The project is 60% complete and on budget. What does the principle "Focus on value" suggest?

- **A.** Reduce testing to finish faster
- **B.** Ask the team to add more features to increase value
- **C.** Finish the project, because 60% of the budget has already been spent and must not be wasted
- **D.** Re-evaluate the business case with the sponsor and consider changing direction

4.3 Which practice best reflects the principle "Embed quality into processes and deliverables"?

- **A.** Inspecting all of the deliverables thoroughly at the end of the project, before handover
- **B.** Adding features beyond the requirements to delight the customer
- **C.** Agreeing on a definition of done and automated tests before development starts
- **D.** Letting the quality department own all quality decisions

4.4 A project manager discovers that a status report sent last week understated a cost overrun. What should the project manager do, consistent with being an accountable leader?

- **A.** Ask the finance analyst to take responsibility for the error
- **B.** Inform the sponsor promptly, correct the report and explain the planned actions
- **C.** Wait until the overrun has been recovered, so that the issue never needs to be reported to anyone
- **D.** Remove cost information from future reports

4.5 During supplier selection, the lowest bidder uses a manufacturing process with significant environmental damage, while the second-lowest bidder meets the organisation's environmental commitments at a 3% higher price. Which principle should most influence the recommendation?

- **A.** Build an empowered culture
- **B.** Adopt a holistic view, so the cheapest option is preferred
- **C.** Integrate sustainability within all project areas
- **D.** Focus on value, so price is the only criterion

4.6 An agile team asks the project manager to decide how to split a feature into technical tasks. What best reflects the principle "Build an empowered culture"?

- **A.** Encourage the team to decide the breakdown and offer support
- **B.** Ask the sponsor to decide the breakdown
- **C.** Postpone the feature until the project manager has time to analyse it in detail
- **D.** Prepare the task list and assign tasks to each developer

4.7 Which two situations violate the principle of being an accountable leader? (Choose two.)

- **A.** Escalating a decision that exceeds the PM's authority through the agreed path
- **B.** Reporting a project as green although critical milestones are slipping
- **C.** Accepting a valuable gift from a bidder during an open tender
- **D.** Asking the team for input before an important decision
- **E.** Admitting a planning error in the team retrospective

4.8 In the Eighth Edition, why is quality not a separate performance domain?

- **A.** Because quality is fully covered by the risk domain
- **B.** Because quality is only relevant to predictive projects
- **C.** Because quality is no longer considered important in modern, value-focused projects of any type
- **D.** Because embedding quality makes it part of every domain, especially Governance and Scope

4.9 A team has worked overtime for six consecutive sprints and morale is falling. Which principle is most at risk?

- **A.** Integrate sustainability, including the team's social sustainability and a sustainable pace
- **B.** Focus on value

- **C.** Adopt a holistic view only
- **D.** Embed quality into processes, because overtime always improves the quality of the work produced by teams

4.10 Which Seventh Edition principle is now primarily addressed through the Risk performance domain rather than as a separate principle?

- **A.** Focus on value
- **B.** Optimise risk responses
- **C.** Demonstrate leadership behaviours
- **D.** Create a collaborative team environment

4.11 A product owner wants to measure the success of a new self-service portal. Which metric best reflects the principle of focusing on value?

- **A.** Number of status meetings held
- **B.** Number of lines of code written
- **C.** Lower call-centre volume and higher customer satisfaction after launch
- **D.** Percentage of the approved budget that has been spent by the end of the project

4.12 Which statement about the six principles is correct?

- **A.** They replace the PMI Code of Ethics and Professional Conduct
- **B.** They are mandatory procedures that must be documented in the project management plan
- **C.** They apply only to adaptive projects
- **D.** They are non-prescriptive and guide behaviour in every type of project

Chapter 5 — Life Cycles and Development Approaches

After this chapter you should be able to:

- distinguish project life cycle, product life cycle, phases and phase gates;
- compare predictive, iterative, incremental, adaptive and hybrid approaches;
- select a development approach using product, project and organisational factors;
- recognise common hybrid patterns and delivery cadences in exam scenarios.

5.1 Key definitions

Term	Meaning
Project life cycle	The series of phases a project passes through from start to completion (e.g., feasibility → design → build → test → deploy → close).
Phase	A collection of logically related activities that culminates in the completion of one or more deliverables.
Phase gate (also stage gate, kill point, phase review, decision gate)	A review at the end of a phase where performance and documents (e.g., the business case) are assessed and a decision is made: **continue, continue with modifications, repeat the phase, or end the project/program**.
Product life cycle	The series of phases that represents the evolution of a product from concept through delivery, growth, maturity and retirement. A product life cycle may contain many projects.
Development approach	The method used to create and evolve the product, service or result (predictive, iterative, incremental, adaptive, hybrid).
Delivery cadence	The timing and frequency of deliveries: single delivery, multiple deliveries, periodic deliveries, or continuous delivery.

5.2 The spectrum of development approaches

Think of development approaches as a **continuum** from predictive to adaptive, not as two boxes.

	Predictive (plan-driven, "waterfall")	Iterative	Incremental	Adaptive / agile
Requirements	Defined up front and stable	Refined through repeated cycles (prototypes, feedback)	Defined for each increment	Elaborated frequently during delivery
Activities	Performed once for the whole project, in sequence	Repeated until correct	Performed once per increment	Repeated in short cycles (iterations/sprints) or continuous flow

	Predictive (plan-driven, "waterfall")	Iterative	Incremental	Adaptive / agile
Delivery	Single final delivery	Usually single final delivery (after iterations improve the solution)	Frequent smaller deliverables	Frequent, small, customer-valued increments
Change	Constrained as much as possible; managed by change control	Incorporated at periodic intervals	Incorporated at increment boundaries	Welcomed; incorporated in real time via the backlog
Stakeholder involvement	At specific milestones	Regular feedback on iterations	Regular, at each increment	Continuous
Risk and cost control	Detailed planning of mostly known considerations	Progressive elaboration with new information	—	Controlled as requirements and constraints emerge
Best when	Requirements are clear, technology is known, regulations demand documentation, changes are expensive (construction, manufacturing)	Complexity is high, frequent changes are expected, feedback improves the solution	Customers benefit from early partial value	High uncertainty, rapid change, innovative products, fast feedback is possible

- **Iterative** = improve the product through repeated cycles (e.g., prototype → feedback → refine). Optimises for *learning*.
- **Incremental** = deliver the product in usable pieces that add up (e.g., release module 1, then module 2). Optimises for *speed of delivery*.
- **Agile** approaches are both iterative *and* incremental.

> **Trap.** "Waterfall" does not mean "no change". Predictive projects handle change through **integrated change control**. Adaptive projects do not mean "no planning" — planning happens continuously and at several levels (vision, roadmap, release, iteration, daily).

Hybrid approaches

A **hybrid** approach combines predictive and adaptive elements. Common patterns:

1. **Agile development followed by predictive rollout** — software built in sprints, then a planned, phased deployment across 40 branches.
2. **Predictive project with adaptive components** — a construction project uses agile sprints for the building-management software.
3. **Mostly agile with a predictive component** — the product is built iteratively, but a regulatory submission follows a fixed, documented sequence.
4. **Phase-gated governance with agile delivery** — funding and governance use stage gates; work inside each stage is delivered in sprints.
5. **Iterative design, predictive build** — prototypes and user research early, then a plan-driven build.

> **Exam tip.** In hybrid scenarios, look for the answer that applies the *right* mechanism to the *right* part: change control for the baselined predictive component, backlog refinement for the adaptive component, and integrated reporting across both.

5.3 Delivery cadence

Cadence	Example
Single delivery	A bridge opened at the end of the project
Multiple deliveries	A regulatory programme delivering separate submissions at different times
Periodic deliveries	A fixed release every month or every quarter
Continuous delivery	Features released to users as soon as they are ready (DevOps pipelines)

5.4 Choosing a development approach

The Guide groups the considerations into three families:

1. Product, service or result

- *Degree of innovation* — novel products favour adaptive; well-understood products favour predictive.
- *Requirements certainty* — clear and stable → predictive; uncertain → adaptive.
- *Scope stability* — stable scope → predictive.
- *Ease of change* — if changes are cheap (software), adaptive works well; if changes are costly (concrete), predictive.
- *Delivery options* — if the product can be delivered in usable pieces, incremental/adaptive is possible.
- *Risk* — high-risk products benefit from incremental learning *or* from rigorous up-front analysis, depending on the nature of the risk.
- *Safety requirements* and *regulations* — strict documentation and verification often push toward predictive (or hybrid).

2. Project

- *Stakeholders* — availability of customer/product owner for frequent feedback.
- *Schedule constraints* — the need to deliver something early favours incremental.
- *Funding availability* — uncertain funding favours incremental releases that deliver value early (and can stop).

3. Organisation

- *Structure* — hierarchical, bureaucratic organisations often favour predictive; flatter ones support adaptive.
- *Culture* — trust, empowerment and tolerance of experimentation support adaptive.
- *Capability* — experience with agile, availability of coaches.
- *Team size and location* — adaptive approaches work best with small (often 3–9 people), dedicated, co-located or well-connected teams.

Helpful models

Stacey matrix (agreement vs certainty):

	Requirements agreed and clear	Requirements far from agreement
Technology certain	*Simple* → predictive	*Complicated* (political)
Technology uncertain	*Complicated* (technical)	*Complex* → adaptive; far corner = *chaotic*

Cynefin framework: *clear* (best practice; sense–categorise–respond), *complicated* (good practice via experts; sense–analyse–respond), *complex* (emergent practice; probe–sense–respond — use experiments and agile), *chaotic* (novel practice; act–sense–respond — stabilise first), and *confusion* (not yet known which domain applies).

Agile suitability filter (from the *Agile Practice Guide*): assess **culture** (buy-in, trust in the team, decision-making powers), **team** (size, experience, access to the customer/business) and **project** (likelihood of change, criticality of the product, possibility of incremental delivery). Results fall in the agile, hybrid or predictive zones.

5.5 Life cycles in practice — examples

Situation	Suitable approach	Why
Building a hospital wing	Predictive (with hybrid elements for IT systems)	Scope is defined by design drawings; changes after construction starts are very expensive; regulations require inspections
Developing a new mobile app for an uncertain market	Adaptive	High uncertainty; frequent feedback; low cost of change
Implementing an ERP package in 30 countries	Hybrid	Configuration can be iterative; data migration, training and go-lives are planned and phased
Running a one-off international conference	Predictive with iterative elements	Fixed date; marketing campaigns can be iterated
Developing a new drug	Predictive, phase-gated	Clinical trial phases and regulatory approvals require strict sequence and documentation

5.6 Planning in each approach

Level	Predictive	Adaptive
Strategic	Business case, charter	Product vision, roadmap
Overall plan	Integrated project management plan with baselines	Release plan (features targeted per release)
Detailed plan	Detailed schedule, rolling-wave refinement of later work	Iteration (sprint) plan, created by the team
Daily	Status updates against schedule	Daily stand-up (daily scrum)

Rolling wave planning — plan the near term in detail and the far term at a higher level, refining as information emerges — is a form of progressive elaboration used in predictive and hybrid projects.

Chapter 5 Review Questions

Answer all questions first, then check the answer key for Chapter 5 in Appendix B.

5.1 A team builds a prototype, gathers user feedback, refines the prototype, and repeats until the design is approved. The final product is delivered once. Which approach is this?

- **A.** Iterative
- **B.** Continuous delivery
- **C.** Incremental
- **D.** Predictive

5.2 A software product is delivered as module 1 in March, module 2 in June and module 3 in September; each module is usable on its own. Which term best describes this?

- **A.** Iterative development with a single delivery
- **B.** Predictive single delivery
- **C.** Phase gate review
- **D.** Incremental delivery

5.3 At the end of the design phase, the steering committee reviews performance and the business case and decides whether the project should continue. What is this review called?

- **A.** A risk audit
- **B.** A phase gate
- **C.** A retrospective
- **D.** A sprint review

5.4 Which project characteristic most strongly favours a predictive approach?

- **A.** Well-understood requirements, and changes after construction starts are very expensive
- **B.** The product can be released in small usable increments
- **C.** The market is very uncertain, and frequent customer feedback is essential to shape the product
- **D.** The team is small, co-located and experienced in Scrum

5.5 A bank is developing a new loan application. The user interface will be built in two-week sprints, while the integration with the core banking system must follow a fixed, documented and audited sequence. Which approach fits best?

- **A.** Hybrid
- **B.** Kanban only
- **C.** Pure predictive
- **D.** Pure adaptive

5.6 In the Cynefin framework, which response pattern is recommended for complex situations?

- **A.** Act, sense, respond — stabilise first
- **B.** Probe, sense, respond — run safe-to-fail experiments and adapt
- **C.** Sense, categorise, respond — apply the established best practice for the situation

- **D.** Sense, analyse, respond — ask experts

5.7 Which two factors should be assessed when using the agile suitability filter? (Choose two.)

- **A.** The likelihood of change in the project
- **B.** The colour coding of the status report
- **C.** The team's access to the customer or business representative
- **D.** The project manager's years since certification
- **E.** The number of pages in the contract

5.8 Which statement about predictive approaches is correct?

- **A.** Planning happens only during the first week
- **B.** Stakeholders are involved continuously in every iteration, so changes never need approval
- **C.** Changes are not permitted once the plan is approved
- **D.** Changes are possible but go through formal change control against the baselines

5.9 A project manager plans the next three months in detail and later phases only at milestone level, adding detail as the project progresses. What technique is this?

- **A.** Resource levelling
- **B.** Crashing
- **C.** Fast tracking
- **D.** Rolling wave planning

5.10 A company releases new features to users several times a day using an automated pipeline. Which delivery cadence is this?

- **A.** Multiple deliveries at fixed contractual dates
- **B.** Single delivery
- **C.** Continuous delivery
- **D.** Periodic delivery

5.11 On the Stacey matrix, where both requirements agreement and technology certainty are high, which approach is usually appropriate?

- **A.** Chaotic response
- **B.** Experimental probing
- **C.** Predictive
- **D.** Adaptive

5.12 Which planning artifact is the adaptive equivalent of a high-level schedule that shows which features are targeted for upcoming releases?

- **A.** Work breakdown structure dictionary
- **B.** Release plan or product roadmap
- **C.** Risk register
- **D.** Sprint retrospective notes

Chapter 6 — Focus Areas and the 40 Processes

After this chapter you should be able to:

- describe the purpose of the five focus areas and how they repeat during a project;
- place each of the 40 processes in its performance domain and focus area;
- translate Sixth-Edition process names into Eighth-Edition names;
- explain inputs, tools and techniques, and outputs (ITTOs) and the most common ones;
- list the components of the project management plan, the main project documents and the baselines;
- distinguish work performance data, information and reports.

6.1 The five focus areas

The Eighth Edition calls the former *process groups* **focus areas**. A focus area is a logical grouping of processes by the *nature of the work* — not a phase of the project. All five can occur within each phase, release or iteration, and they overlap.

Focus area	Purpose	Typical key outputs
Initiating	Define a new project or phase, obtain authorisation, align stakeholders on purpose	Project charter, assumption log, stakeholder register
Planning	Establish the scope, refine objectives and define how the work will be done	Project management plan, baselines, subsidiary plans, project documents
Executing	Perform the work defined in the plan, lead people, manage engagement and communications	Deliverables, work performance data, issue log, change requests
Monitoring and Controlling	Track, review and regulate progress and performance; identify needed changes; validate and control deliverables	Work performance information and reports, approved/rejected changes, accepted deliverables, forecasts
Closing	Formally complete the project, phase or contract	Final product transition, final report, archives, lessons learned

> **Exam tip. Monitoring and Controlling runs in parallel with all other focus areas** — from the first day to the last. And in adaptive approaches all five focus areas occur inside every iteration: planning (sprint planning), executing (development), monitoring and controlling (daily scrum, sprint review), closing (iteration closure) — and initiating when a new release or phase starts.

How many processes per focus area? **Initiating 2 · Planning 19 · Executing 8 · Monitoring and Controlling 10 · Closing 1 = 40.**

6.2 The 40 processes — domain × focus area

Domain	Initiating	Planning	Executing	Monitoring & Controlling	Closing
Governance (9)	Initiate Project or Phase	Integrate and Align Project Plans · Plan Sourcing Strategy	Manage Project Execution · Manage Quality Assurance · Manage Project Knowledge	Monitor and Control Project Performance · Assess and Implement Changes	Close Project or Phase
Scope (6)		Plan Scope Management · Elicit and Analyze Requirements · Define Scope · Develop Scope Structure		Validate Scope · Monitor and Control Scope	
Schedule (3)		Plan Schedule Management · Develop Schedule		Monitor and Control Schedule	
Finance (4)		Plan Financial Management · Estimate Costs · Develop Budget		Monitor and Control Finances	
Stakeholders (7)	Identify Stakeholders	Plan Stakeholder Engagement · Plan Communications Management	Manage Stakeholder Engagement · Manage Communications	Monitor Stakeholder Engagement · Monitor Communications	
Resources (5)		Plan Resource Management · Estimate Resources	Acquire Resources · Lead the Team	Monitor and Control Resourcing	
Risk (6)		Plan Risk Management · Identify Risks · Perform Risk Analysis · Plan Risk Responses	Implement Risk Responses	Monitor Risks	

Patterns that help you memorise the table:

- Every domain except Governance and Stakeholders has a **"Plan … Management"** process that creates its management plan (Plan Scope Management, Plan Schedule Management, Plan Financial Management, Plan Resource Management, Plan Risk Management); Stakeholders has *Plan Stakeholder Engagement* and *Plan Communications Management*, and Governance has *Integrate and Align Project Plans* (the integrated plan) and *Plan Sourcing Strategy*.
- Every domain has at least one **monitoring and controlling** process.
- Governance is the only domain with processes in **all five** focus areas.

- Only **Initiate Project or Phase** and **Identify Stakeholders** are initiating processes; only **Close Project or Phase** is a closing process.

6.3 From the Sixth Edition's 49 processes to the Eighth Edition's 40

Sixth Edition process(es)	Eighth Edition process
Develop Project Charter	**Initiate Project or Phase**
Develop Project Management Plan	**Integrate and Align Project Plans**
Plan Procurement Management	**Plan Sourcing Strategy**
Direct and Manage Project Work	**Manage Project Execution**
Manage Quality	**Manage Quality Assurance**
Manage Project Knowledge	Manage Project Knowledge
Monitor and Control Project Work	**Monitor and Control Project Performance**
Perform Integrated Change Control	**Assess and Implement Changes**
Close Project or Phase	Close Project or Phase
Plan Scope Management · Collect Requirements	Plan Scope Management · **Elicit and Analyze Requirements**
Define Scope + Plan Quality Management	**Define Scope** (now includes quality planning)
Create WBS	**Develop Scope Structure**
Validate Scope	Validate Scope
Control Scope + Control Quality	**Monitor and Control Scope** (now includes quality control)
Plan Schedule Management	Plan Schedule Management
Define Activities · Sequence Activities · Estimate Activity Durations · Develop Schedule	**Develop Schedule** (one process)
Control Schedule	**Monitor and Control Schedule**
Plan Cost Management · Estimate Costs · Determine Budget · Control Costs	**Plan Financial Management** · Estimate Costs · **Develop Budget** · **Monitor and Control Finances**
Identify Stakeholders · Plan Stakeholder Engagement · Manage Stakeholder Engagement · Monitor Stakeholder Engagement	Same names
Plan Communications Management · Manage Communications · Monitor Communications	Same names (now in the Stakeholders domain)
Plan Resource Management · Estimate Activity Resources	Plan Resource Management · **Estimate Resources**
Acquire Resources (+ Conduct Procurements)	**Acquire Resources** (obtaining internal and external resources; procurement detail in Appendix X4)

Sixth Edition process(es)	Eighth Edition process
Develop Team · Manage Team	**Lead the Team**
Control Resources (+ Control Procurements)	**Monitor and Control Resourcing**
Plan Risk Management · Identify Risks	Same names
Perform Qualitative Risk Analysis · Perform Quantitative Risk Analysis	**Perform Risk Analysis**
Plan Risk Responses · Implement Risk Responses · Monitor Risks	Same names

Arithmetic check: 49 − 3 (schedule merges) − 1 (risk analysis merge) − 1 (team merge) − 1 (quality planning) − 1 (quality control) − 2 (conduct and control procurements absorbed) = **40**.

> **What changed.** The Eighth Edition states that its processes are **non-prescriptive**: they describe work that is commonly needed, but the team decides — through tailoring — which processes to use, how formally, and how often. Don't expect the exam to ask you to recite inputs and outputs; expect it to test whether you know what a process is *for* and what it *produces*.

6.4 Anatomy of a process: ITTOs

Each process is described by:

- **Inputs** — what you need to perform it (documents, plans, data, environmental factors);
- **Tools and techniques** — how you transform the inputs (methods, analyses, meetings);
- **Outputs** — what the process produces (artifacts, deliverables, decisions, updates).

Outputs of one process are often inputs to another — this is how processes **interact**.

The most common inputs

Input	Notes
Project management plan (or a component of it)	The integrated plan and its subsidiary plans and baselines
Project documents	Logs, registers, estimates, schedules — documents that are *not* part of the plan
Enterprise environmental factors	Culture, market, regulations, infrastructure
Organisational process assets	Policies, templates, historical information, lessons learned
Business documents	Business case, benefits management plan (mainly in initiating)
Agreements	Contracts with customers or suppliers
Work performance data/information	Status data from execution and control

The most common tools and techniques (by family)

Family	Examples
Expert judgment	Advice from individuals or groups with specialised knowledge
Data gathering	Brainstorming, interviews, focus groups, questionnaires and surveys, benchmarking, checklists
Data analysis	Alternatives analysis, root-cause analysis, variance analysis, trend analysis, earned value analysis, what-if scenario analysis, reserve analysis, SWOT, assumption and constraint analysis
Data representation	Charts and diagrams, matrices, mind maps, affinity diagrams, stakeholder grids, histograms
Decision-making	Voting (unanimity, majority, plurality), autocratic decision, multicriteria decision analysis
Communication skills	Feedback, presentations, active listening
Interpersonal and team skills	Facilitation, conflict management, negotiation, influencing, leadership, emotional intelligence, meeting management
Estimating	Analogous, parametric, bottom-up, three-point
Meetings	Kickoff, planning, status, review, retrospective, change control board meetings
Project management information system (PMIS)	Scheduling software, dashboards, configuration management tools, AI-assisted analytics

6.5 The project management plan and project documents

The **project management plan** describes how the project will be executed, monitored and controlled, and closed. It integrates:

Subsidiary management plans — scope management plan, requirements management plan, schedule management plan, financial (cost) management plan, quality management plan (quality planning is part of defining scope in the Eighth Edition, but a quality plan is still a component), resource management plan, communications management plan, risk management plan, sourcing (procurement) strategy plan, stakeholder engagement plan.

Baselines — scope baseline, schedule baseline, cost baseline (together the **performance measurement baseline, PMB**).

Additional components — change management plan, configuration management plan, performance measurement baseline, project life cycle description, development approach, management reviews.

Project documents (not part of the plan, updated freely as the project progresses — no formal change control needed): activity attributes and list, assumption log, basis of estimates, change log, cost estimates, duration estimates, issue log, lessons learned register, milestone list, physical resource assignments, project calendars, project communications, project schedule, network diagram, project scope statement*, project team assignments, quality control measurements, quality metrics, quality report, requirements documentation, requirements traceability matrix, resource breakdown structure, resource calendars, resource requirements, risk register, risk report, schedule data, schedule forecasts, stakeholder register, team charter, test and evaluation documents.

* The project scope statement is also part of the scope baseline once approved.

> **Trap.** Updating the **issue log**, **risk register** or **lessons learned register** does **not** require a change request — they are project documents. Changing a **baseline** or the **project management plan** does.

The baselines

Baseline	Components	Changed only through
Scope baseline	Approved project scope statement + WBS (scope structure) + WBS dictionary	Formal change control
Schedule baseline	Approved version of the schedule model with baseline start and finish dates	Formal change control
Cost baseline	Approved time-phased budget, *excluding management reserves*, including contingency reserves	Formal change control
Performance measurement baseline (PMB)	Integrated scope-schedule-cost baseline used for earned value management	Formal change control

6.6 Data → information → reports

Term	Created in	Example
Work performance data	Executing (Manage Project Execution and other executing processes)	Raw observations: "Activity 4.2 is 60% complete; actual cost $12,000; 3 defects found"
Work performance information	Monitoring and controlling processes	Data compared with the plan and analysed: "CPI = 0.86; activity 4.2 is 5 days late; defect rate above threshold"
Work performance reports	Monitor and Control Project Performance	Compiled for decisions and communication: status reports, dashboards, heat maps, forecasts

> **Exam tip.** A question that says "the team reports that 70% of the work is complete" is describing **data**. Once compared with the plan ("we are 10% behind"), it becomes **information**. Assembled into a dashboard for the steering committee, it becomes a **report**.

6.7 Change requests: the four types

Executing and monitoring processes produce **change requests**, which go through *Assess and Implement Changes*:

Type	Purpose
Corrective action	Realign *current* performance with the plan (e.g., add a tester because testing is behind)

Type	Purpose
Preventive action	Ensure *future* performance stays aligned with the plan (e.g., train the team before a new technology is used)
Defect repair	Modify a non-conforming product or component
Update	Change formally controlled documents or plans to reflect modified or additional ideas or content

Chapter 6 Review Questions

Answer all questions first, then check the answer key for Chapter 6 in Appendix B.

6.1 How many processes does the Planning focus area contain in the Eighth Edition?

- **A.** 8
- **B.** 10
- **C.** 19
- **D.** 24

6.2 Which two processes are the only processes in the Initiating focus area?

- **A.** Initiate Project or Phase and Plan Scope Management
- **B.** Integrate and Align Project Plans and Identify Stakeholders
- **C.** Initiate Project or Phase and Identify Stakeholders
- **D.** Identify Stakeholders and Identify Risks

6.3 Which Eighth Edition process replaces the Sixth Edition's "Perform Integrated Change Control"?

- **A.** Monitor and Control Project Performance
- **B.** Assess and Implement Changes
- **C.** Monitor and Control Scope
- **D.** Manage Project Execution

6.4 In the Eighth Edition, which process includes the activities formerly performed in Define Activities, Sequence Activities and Estimate Activity Durations?

- **A.** Estimate Resources
- **B.** Develop Scope Structure
- **C.** Plan Schedule Management
- **D.** Develop Schedule

6.5 The team updates the risk register with a newly identified risk. What is required?

- **A.** A new version of the project charter
- **B.** None; the risk register is a project document updated as needed
- **C.** Approval from the sponsor before any update
- **D.** A change request approved by the change control board, since all documents are controlled

6.6 Which of the following is part of the scope baseline?

- **A.** The WBS dictionary

- **B.** The requirements traceability matrix
- **C.** The risk register
- **D.** The stakeholder register

6.7 "Testing is 40% complete and 12 defects have been logged." This statement is an example of:

- **A.** A work performance report
- **B.** Work performance data
- **C.** A change request
- **D.** Work performance information

6.8 A project manager notices that a key activity is slipping and asks a second developer to help so the activity finishes on its baseline date. What type of change request is this?

- **A.** Preventive action
- **B.** Update
- **C.** Defect repair
- **D.** Corrective action

6.9 Which two statements about the focus areas are correct? (Choose two.)

- **A.** Focus areas are mandatory and cannot be tailored
- **B.** In adaptive approaches the focus areas occur within each iteration
- **C.** Each focus area corresponds to one project phase that occurs only once
- **D.** Closing occurs only at the end of the whole project, never at the end of a phase
- **E.** Monitoring and Controlling activities occur throughout the project

6.10 Which performance domain is the only one with processes in all five focus areas?

- **A.** Scope
- **B.** Stakeholders
- **C.** Risk
- **D.** Governance

6.11 Which statement about the cost baseline is correct?

- **A.** It includes contingency reserves but excludes management reserves
- **B.** It includes both contingency and management reserves
- **C.** It excludes all reserves
- **D.** It includes management reserves but excludes contingency reserves

6.12 Match each Sixth Edition process with its Eighth Edition equivalent.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Create WBS	**A.** Lead the Team
2. Develop Team and Manage Team	**B.** Elicit and Analyze Requirements
3. Plan Cost Management	**C.** Develop Scope Structure
4. Collect Requirements	**D.** Plan Financial Management

6.13 Which process produces the work performance reports used for status reporting to the steering committee?

- **A.** Monitor and Control Project Performance
- **B.** Manage Project Execution
- **C.** Close Project or Phase
- **D.** Manage Communications

Chapter 7 — Governance Performance Domain

After this chapter you should be able to:

- explain what project governance is and what good governance looks like;
- describe the purpose, key inputs and outputs of the nine Governance processes;
- choose an appropriate contract type and sourcing approach;
- apply integrated change control in predictive, adaptive and hybrid contexts;
- close a project or phase correctly, including early termination.

7.1 What the Governance domain covers

The Governance domain addresses the **framework, functions and processes that guide project activities** so the project delivers its intended outcomes in line with organisational strategy, policies and obligations. It absorbs most of the former *Integration Management* knowledge area, plus quality assurance and the sourcing (procurement) strategy.

Good governance provides:

- **Clear decision rights** — who can decide what, and the thresholds (for example, the PM can approve schedule changes up to five days; the CCB beyond that; the steering committee for anything affecting the business case).
- **Accountability and oversight** — sponsor, steering committee, PMO and audits.
- **Escalation paths** — how issues, risks and decisions move up when they exceed authority.
- **Alignment** — the project stays linked to strategy and the business case through phase gates and reviews.
- **Integration** — plans, work and changes fit together as a coherent whole.
- **Compliance** — legal, regulatory, contractual and policy obligations are met (see Chapter 17).

Signs of effective governance (desired outcomes): decisions are timely and made at the right level; the project remains aligned with objectives; stakeholders trust the information they receive; changes are controlled without paralysing delivery; knowledge is captured and reused; the project or phase closes cleanly with value transitioned.

> **Exam tip.** Governance is **not** bureaucracy for its own sake. The best answer tailors governance to the context: lightweight and empowering for small adaptive teams, more formal for regulated, high-risk or contract-based work.

7.2 Initiate Project or Phase *(Initiating)*

Purpose: formally authorise the project (or the next phase), define its purpose and success criteria, and give the PM authority.

Key inputs	Key tools & techniques	Key outputs
Business case, benefits management plan, agreements, EEFs, OPAs	Expert judgment, data gathering (brainstorming, focus groups, interviews), interpersonal skills (facilitation, conflict management, meeting management), meetings	**Project charter**, **assumption log**

Charter contents: project purpose; measurable objectives and success criteria; high-level requirements; high-level description, boundaries and key deliverables; overall project risk; summary milestone schedule; pre-approved financial resources; key stakeholder list; project approval requirements (what constitutes success, who signs off); exit criteria; assigned PM, responsibility and authority level; name and authority of the sponsor.

Assumption log: records high-level strategic and operational assumptions and constraints identified at initiation; later refined with detailed assumptions.

In adaptive contexts the equivalent artifacts are a **product vision** (often an "elevator pitch"), a **project/product charter** or **team charter**, a **product roadmap** and an initial **product backlog**. Phase or release initiation re-confirms the vision and business value.

> **Trap.** The PM should be **assigned as early as possible** — ideally during charter development — but the PM never authorises their own project. The **sponsor** issues the charter.

7.3 Integrate and Align Project Plans *(Planning)*

Purpose: define, prepare and coordinate all plan components and consolidate them into an **integrated project management plan** aligned with governance, the development approach and stakeholder expectations.

- **Outputs:** the **project management plan** (subsidiary plans, baselines, change and configuration management plans, life cycle and development approach).
- **Techniques:** expert judgment, data gathering (checklists, brainstorming), interpersonal skills (conflict management, facilitation), meetings — especially the **kickoff meeting**.
- **Kickoff meeting:** marks the end of planning and the start of execution in predictive projects (small projects may hold it right after initiation). It communicates objectives, gains commitment, explains roles and responsibilities. In adaptive projects a similar launch (often called *iteration zero*, *chartering workshop* or *inception*) aligns the team on vision, ways of working and the initial backlog.
- **Progressive elaboration:** the plan is refined as the project progresses; once baselined, changes to it go through change control.
- **Plan delivery:** choose how the product will be delivered — releases, increments, milestones — and integrate adaptive release plans with predictive milestones in hybrid projects.

7.4 Plan Sourcing Strategy *(Planning)*

Purpose: decide **what to obtain from outside** the organisation, **how** (contract type, selection method) and **from whom**, and document it in the **sourcing strategy plan** (the Sixth Edition's procurement management plan). Appendix X4 (Procurement) expands on strategic implications, make-or-buy and the project manager's role.

Make-or-buy analysis

Compare the cost, risk, capability, schedule and strategic implications of producing internally versus buying. Consider total cost (purchase price + indirect costs + life-cycle costs), core competence, intellectual property, confidentiality, capacity, and the effect on the team.

Example: Renting a machine costs $500/day; buying costs $18,000 plus $100/day to operate. Break-even: 18,000 + 100d = 500d → d = 45 days. If you need it for more than 45 days, buying is cheaper.

Contract types

Family	Type	How the seller is paid	Who carries the cost risk
Fixed price (FP)	**Firm fixed price (FFP)**	Fixed total price	**Seller** (most)
	Fixed price incentive fee (FPIF)	Fixed price + incentive for meeting targets; ceiling price; cost sharing ratio	Seller (shared up to the ceiling)
	Fixed price with economic price adjustment (FP-EPA)	Fixed price adjusted for defined changes (inflation, commodity prices) — for multi-year contracts	Seller (with protection against market changes)
Cost-reimbursable (CR)	**Cost plus fixed fee (CPFF)**	Allowable costs + fixed fee (usually % of *estimated* cost)	**Buyer**
	Cost plus incentive fee (CPIF)	Costs + fee that depends on performance against targets; cost overrun/underrun shared by ratio (e.g., 80/20)	Buyer (shared)
	Cost plus award fee (CPAF)	Costs + award fee based on the buyer's subjective evaluation of performance	Buyer
Time and materials (T&M)	—	Rate per hour/day + materials at cost; often a not-to-exceed limit	Buyer (mid-way between FP and CR)

How to choose:

- **Scope well defined and stable** → fixed price.
- **Scope uncertain or evolving** → cost-reimbursable or T&M (with controls such as not-to-exceed caps).
- **Small amounts of staff augmentation, or work starting before a statement of work is complete** → T&M.
- **Buyer wants to motivate performance** → incentive or award fee.

Agile contracting options (from the *Agile Practice Guide*): fixed-price increments (per user story or sprint), not-to-exceed T&M, graduated fixed price (rate varies with early/late delivery), early cancellation option, dynamic scope option (swap features of equal size), team augmentation, multi-tiered structures (master agreement + statements of work).

Source selection methods

Least cost; qualifications only; quality-based/highest technical score; **quality and cost-based** (weighted scoring); sole source (single supplier, with justification); fixed budget (scope adapted to a stated budget).

Procurement documents

Document	Use
Request for information (RFI)	Gather information on what sellers can offer — early market research
Request for quotation (RFQ)	Obtain prices for well-defined, standard items
Request for proposal (RFP)	Obtain proposals for complex work where sellers propose a solution
Procurement statement of work (SOW)	Describes the procured item in enough detail for sellers to determine if they can provide it (performance, functional or design SOW)
Terms of reference (TOR)	SOW equivalent for services
Source selection criteria	How proposals will be evaluated (technical approach, price, capability, past performance, financial stability, sustainability, IP rights)
Independent cost estimate	Buyer's own estimate to benchmark proposals

> **Exam tip.** Procurement decisions are governed by **organisational policies and legal requirements**. The PM usually works *with* the procurement or contracts department, which holds the authority to sign contracts. A PM who promises a supplier the contract informally is creating legal and ethical problems.

7.5 Manage Project Execution *(Executing)*

Purpose: lead and perform the work defined in the project management plan, and implement **approved** changes, to achieve the project's objectives.

- **Outputs: deliverables**, **work performance data**, **issue log**, **change requests**, updates to plans and documents.
- **Approved change requests** (corrective actions, preventive actions, defect repairs) are *implemented* here.
- The **issue log** records current problems, owners, due dates and status.
- The PMIS and meetings support execution.

> **Trap.** A team member who implements a change the customer asked for — without approval — is creating **scope creep**. In a predictive context, the request must go through *Assess and Implement Changes*; in an adaptive context, it goes to the **product owner** for backlog prioritisation.

7.6 Manage Quality Assurance *(Executing)*

Purpose: translate the quality requirements and policies into executable quality activities and **confirm that the processes being used will produce deliverables that meet requirements**. Focus is on the *process* (assurance), whereas inspecting deliverables is quality *control* (done within *Monitor and Control Scope*).

Key techniques: **quality audits**, **process analysis**, checklists, **root-cause analysis**, design for X, problem solving, cause-and-effect diagrams, flowcharts, test and evaluation documents. Key outputs: **quality reports**, test and evaluation documents, change requests (process improvements). Details in Chapter 9.

7.7 Manage Project Knowledge *(Executing)*

Purpose: use existing knowledge and create new knowledge to achieve objectives and contribute to organisational learning.

Explicit knowledge	Tacit knowledge
Easily codified in words, pictures, numbers (manuals, lessons learned, procedures)	Personal, hard to express (experience, insights, "know-how")
Shared through documents and repositories (information management)	Shared through **conversation and interaction**: communities of practice, mentoring, job shadowing, storytelling, workshops, pairing, knowledge fairs

- **Key output: lessons learned register** — created early and updated *throughout* the project (not only at the end). At closure it is transferred to the organisational **lessons learned repository** (an OPA).
- **Knowledge transfer** to operations and to future projects is an explicit PMP task (People domain, task 7).
- In agile teams, retrospectives, pairing, mob programming and shared code ownership spread knowledge continuously.

7.8 Monitor and Control Project Performance *(Monitoring and Controlling)*

Purpose: track, review and report overall progress toward the objectives defined in the plan, and recommend corrective or preventive actions.

- Combines work performance information from every control process (scope, schedule, finance, resources, risk, stakeholders, communications).
- Techniques: **variance analysis**, **trend analysis**, **earned value analysis**, root-cause analysis, alternatives analysis, cost-benefit analysis, decision-making, meetings.
- Outputs: **work performance reports** (status reports, dashboards, forecasts), change requests.
- Agile/hybrid equivalents: burnup/burndown charts, cumulative flow diagrams, velocity trends, information radiators, sprint reviews.

7.9 Assess and Implement Changes *(Monitoring and Controlling)*

Purpose: review all change requests; approve, defer or reject them; manage changes to deliverables, project documents and the project management plan; and communicate decisions. This is **integrated change control**.

The change control flow (predictive)

1. **Identify** the need for a change (anyone can raise it).
2. **Document** it in a change request (written; verbal requests are captured in writing).
3. **Evaluate the impact** on scope, schedule, cost, quality, risk, resources, stakeholders and the business case — *before* deciding (holistic view).
4. **Explore options** (alternatives, trade-offs).
5. **Decide** — by the person or body with the authority (PM within thresholds, CCB, sponsor, steering committee, customer for contract changes).
6. **Update** the plan, baselines and documents; record in the **change log**.
7. **Communicate** the decision to affected stakeholders.
8. **Implement** (in Manage Project Execution) and **verify** the change was implemented correctly.

Configuration management supports change control by managing the *specifications of the product and documents*: **configuration identification** (what items are controlled), **configuration status accounting** (recording and reporting their state), and **configuration verification and audit** (confirming the composition is correct and changes were implemented as approved).

Change in adaptive contexts

- Change is expected and welcomed. New ideas and changed requirements become **backlog items** that the **product owner** prioritises; the *current* sprint goal is protected (the team and PO negotiate if something truly urgent arises).
- Changes to the **overall** constraints (release date, budget, contract, fundamental vision) still follow governance, and may need sponsor or steering approval.

> **Exam tip — the classic sequence for a change request in a predictive project:** *evaluate impact → explore options → submit to CCB → update plan and log → communicate → implement.* The wrong answers usually skip impact analysis, go straight to the sponsor, implement before approval, or refuse to consider the change at all.

7.10 Close Project or Phase *(Closing)*

Purpose: finalise all activities for the project, phase or contract; archive information; release resources; and ensure value is transitioned.

Closing activities:

- Confirm all work is completed and deliverables are **formally accepted** (accepted deliverables come from *Validate Scope*).
- **Transfer** the final product, service or result to operations or the customer (with training, documentation, support arrangements).

- **Close procurements**: verify all contract work, settle claims, make final payments, issue formal written notice of contract completion, archive procurement records.
- Measure stakeholder satisfaction; complete the **final report** (summary of performance against objectives: scope, quality, cost, schedule, benefits, risks/issues).
- Finalise and transfer **lessons learned**; update OPAs; **archive** project documents.
- **Release resources** (team and physical) and celebrate/recognise contributions.
- Hand over **benefit tracking** to the benefit owner.

Early termination: if the project is cancelled, the PM still performs closure: document the reason, the status of deliverables (completed and incomplete), transfer partial work, close contracts per their termination clauses, capture lessons learned, release resources.

> **Trap.** "The customer has verbally accepted the product; release the team" — wrong. Obtain **formal written acceptance**, complete administrative closure and lessons learned **before** releasing resources.

7.11 Governance in adaptive and hybrid environments

- **Lightweight, value-based governance:** funding by increment or by product (rather than one big up-front approval), with continuation decisions based on demonstrated value.
- **Guardrails instead of gates:** clear boundaries (budget, compliance rules, architecture standards) within which teams decide.
- **Frequent transparency:** sprint reviews and information radiators replace some formal status reports.
- **Hybrid governance:** phase gates for investment decisions; sprints inside phases; a single integrated view of progress and risk for the steering committee.
- **The PMO** may evolve into a **value delivery office (VDO)** or agile centre of excellence that coaches teams rather than policing them (Appendix X2; see Chapter 17).

Chapter 7 Review Questions

Answer all questions first, then check the answer key for Chapter 7 in Appendix B.

7.1 A project's scope is unclear, and the buyer wants the seller to start immediately while requirements are being developed. Which contract type is most appropriate?

- **A.** Firm fixed price
- **B.** Time and materials with a not-to-exceed limit
- **C.** Fixed price with economic price adjustment
- **D.** Fixed price incentive fee with a ceiling price

7.2 Under which contract type does the seller bear the greatest cost risk?

- **A.** Firm fixed price
- **B.** Cost plus award fee
- **C.** Cost plus fixed fee
- **D.** Time and materials

7.3 A customer emails a developer directly asking for an extra report. The developer adds it without telling anyone. In a predictive project, what does this represent and what should have happened?

- **A.** A defect repair that needs no approval
- **B.** Scope creep; it should go through integrated change control
- **C.** Gold plating; the developer should be disciplined immediately for agreeing to it
- **D.** Good customer service that should be encouraged

7.4 A change request has been submitted to alter a key deliverable in a predictive project. What should the project manager do first?

- **A.** Submit it directly to the change control board without any analysis, to save time
- **B.** Reject it because the scope baseline is approved
- **C.** Implement it to keep the customer satisfied
- **D.** Evaluate its impact on scope, schedule, cost, quality, risk and value

7.5 What is the main difference between explicit and tacit knowledge?

- **A.** Explicit knowledge cannot be shared outside the team
- **B.** There is no difference; both terms describe documentation
- **C.** Explicit knowledge can be codified; tacit knowledge is personal and shared through interaction
- **D.** Tacit knowledge is always written down in the lessons learned register, while explicit knowledge is not

7.6 When should the lessons learned register be updated?

- **A.** Only after a major failure
- **B.** Only during the final closure meeting, when the whole team can reflect on the project
- **C.** Only when the sponsor requests it
- **D.** Throughout the project, then transferred to the organisation's repository at closure

7.7 The customer has informally said they are happy with the final product. The functional managers want their people back immediately. What should the project manager do first?

- **A.** Start the next project with the same team
- **B.** Release all team members immediately
- **C.** Obtain formal acceptance before completing closure
- **D.** Archive the documents and close the project, since the customer is happy

7.8 Which process confirms that the project's processes will produce deliverables that meet quality requirements, for example through quality audits?

- **A.** Manage Project Knowledge
- **B.** Validate Scope
- **C.** Manage Quality Assurance
- **D.** Monitor and Control Scope

7.9 A buyer has a well-defined scope but wants to motivate the seller to finish early and under budget, sharing savings according to an agreed ratio up to a ceiling price. Which contract type fits?

- **A.** Fixed price incentive fee
- **B.** Time and materials

- **C.** Firm fixed price
- **D.** Cost plus fixed fee

7.10 The procurement department is preparing to buy standard laptops in large quantities, where price is the main differentiator. Which procurement document is most appropriate?

- **A.** Request for proposal
- **B.** Request for quotation
- **C.** Terms of reference for consulting services
- **D.** Request for information

7.11 Which two activities are part of closing a procurement? (Choose two.)

- **A.** Holding a bidder conference
- **B.** Creating the procurement statement of work
- **C.** Issuing a new request for proposal
- **D.** Settling open claims and making final payments
- **E.** Providing formal written notice that the contract is complete

7.12 A sponsor cancels a project halfway through because of a strategic change. What should the project manager do?

- **A.** Ask the team to finish the most important deliverables without funding
- **B.** Perform closure: document the status, transfer completed work, close contracts and capture lessons
- **C.** Continue working until the original end date to recover the investment
- **D.** Stop all work immediately and leave the project documentation exactly as it is, since the project has been cancelled

7.13 In an adaptive project, a stakeholder requests a new feature in the middle of a sprint. What is the most appropriate handling?

- **A.** Ask the developers to add it to the current sprint immediately, since the stakeholder is important
- **B.** Submit a formal change request to the change control board
- **C.** Refuse the request because scope is fixed
- **D.** Add it to the product backlog for the product owner to prioritise

7.14 Which element of configuration management confirms that the product's composition is correct and that approved changes were implemented?

- **A.** Configuration verification and audit
- **B.** Configuration status accounting
- **C.** Configuration identification
- **D.** Analysis of the change log by the PMO

7.15 Renting equipment costs $400 per day. Buying costs $12,000 plus $100 per day for operation. Beyond how many days of use does buying become cheaper?

- **A.** 24 days
- **B.** 30 days
- **C.** 40 days
- **D.** 120 days

Chapter 8 — Scope Performance Domain

After this chapter you should be able to:

- explain the six Scope processes and their key outputs;
- elicit, analyse, prioritise and trace requirements;
- build a WBS (scope structure) and a product backlog, and know when each is appropriate;
- distinguish validating scope from controlling quality, and scope creep from gold plating;
- manage scope in predictive, adaptive and hybrid projects.

8.1 What the Scope domain covers

The Scope domain addresses **defining, structuring, validating and controlling the work and the deliverables** required to achieve the project's objectives — and in the Eighth Edition it also carries **quality planning** (in *Define Scope*) and **quality control** (in *Monitor and Control Scope*).

Two kinds of scope:

Product scope	Project scope
The features and functions that characterise the product, service or result	The work performed to deliver the product with the specified features and functions
Measured against the **requirements**	Measured against the **project management plan** (scope baseline)

Desired outcomes: stakeholders share an understanding of what will be delivered; requirements are clear, prioritised and traceable; deliverables meet acceptance criteria; scope changes are controlled; delivered scope produces the expected value.

8.2 Plan Scope Management *(Planning)*

Creates the **scope management plan** (how scope will be defined, structured, validated and controlled; how the WBS/backlog will be created and maintained; how formal acceptance will be obtained) and the **requirements management plan** (how requirements will be elicited, analysed, documented, prioritised, traced, and how changes to them are handled; configuration management of requirements).

8.3 Elicit and Analyze Requirements *(Planning)*

Purpose: determine, document and manage stakeholder needs and requirements. (Formerly *Collect Requirements*; the new name emphasises that requirements are *drawn out* and *analysed*, not just collected.)

Types of requirements

Type	Description	Example
Business	Higher-level needs of the organisation	"Reduce customer churn by 10%"

Type	Description	Example
Stakeholder	Needs of a stakeholder or group	"Branch managers need a weekly churn report"
Solution — functional	Behaviours of the product	"The system sends a renewal reminder 30 days before expiry"
Solution — non-functional	Environmental conditions or qualities (performance, security, usability, reliability, sustainability)	"Pages load in under 2 seconds for 10,000 users"
Transition and readiness	Temporary capabilities to move from current to future state	Data conversion, training
Project	Actions, processes or conditions the project must meet	Milestone dates, contractual obligations
Quality	Conditions or criteria for successful completion of a deliverable	Test coverage, acceptance thresholds

Elicitation techniques

Technique	Best for
Interviews	Confidential or detailed information from individuals and experts
Focus groups	Prequalified stakeholders and SMEs discussing expectations of a product
Facilitated workshops (e.g., **JAD** in software, **QFD** / *voice of the customer* in manufacturing)	Cross-functional sessions that define requirements quickly and reconcile differences
Questionnaires and surveys	Large, dispersed audiences; quick statistics
Brainstorming	Generating many ideas
Benchmarking	Comparing with best-in-class practices
Observation / job shadowing	When people struggle to articulate their needs
Prototypes, wireframes, storyboards	Getting early feedback on a tangible model
Document analysis	Existing contracts, regulations, process descriptions
Context diagrams	Showing the system's scope, inputs, outputs and actors
Affinity diagrams, mind mapping	Grouping ideas into themes
Nominal group technique	Brainstorming followed by ranking/voting
Story mapping (adaptive)	Arranging user stories along the user journey and by release

Prioritisation techniques

- **MoSCoW** — Must have, Should have, Could have, Won't have (this time).
- **Kano model** — basic (must-be), performance (one-dimensional), excitement (delighters), indifferent and reverse features. Over time delighters become expected.

- **Cost of delay / WSJF** (weighted shortest job first = cost of delay ÷ job size).
- **100-point method**, **paired comparison**, **multicriteria weighting**.

Requirements traceability matrix (RTM)

A grid linking each requirement to its **origin** (business need, objective) and forward to the **deliverables, design, tests and acceptance**. It ensures each requirement adds business value and is delivered and tested, and helps assess the impact of changes.

> **Exam tip.** If stakeholders disagree about requirements, the best first move is usually to **facilitate a workshop** to reach agreement — not to choose the requirement of the most senior stakeholder, and not to add everything.

8.4 Define Scope *(Planning)*

Purpose: develop a detailed description of the project and product — what is included and what is **excluded** — and (in the Eighth Edition) plan how quality will be achieved and verified.

Project scope statement contents:

- product scope description (progressively elaborated);
- **deliverables**;
- **acceptance criteria** (conditions that must be met before deliverables are accepted);
- **project exclusions** (what is out of scope — prevents misunderstandings);
- **constraints** and **assumptions** (also in the assumption log).

Techniques: expert judgment, **product analysis** (product breakdown, systems analysis, value engineering, requirements analysis), **alternatives analysis**, multicriteria decision analysis, facilitation.

Quality planning within Define Scope: identify quality requirements and standards, define **quality metrics** (what is measured and the tolerance), prepare the **quality management plan** elements, plan **cost of quality** trade-offs, design test and evaluation approaches. See Chapter 9.

8.5 Develop Scope Structure *(Planning)*

Purpose: subdivide deliverables and work into smaller, more manageable components. In predictive projects the classic structure is the **work breakdown structure (WBS)**; in adaptive projects it is a **product backlog** organised into themes, epics, features and user stories (often visualised as a story map).

WBS essentials

- **Deliverable-oriented** hierarchical decomposition of the **total scope** of work.
- **100% rule**: the WBS includes all the work — and only the work — of the project, including project management work. The sum of the children equals the parent.
- **Work package**: the lowest level of the WBS, for which cost and duration can be estimated and managed. (A common heuristic is the 8/80 rule: between 8 and 80 hours of effort.)
- **Control account**: a management control point where scope, budget and schedule are integrated and compared with earned value; contains one or more work packages.

- **Planning package**: below a control account, with known work content but not yet detailed activities (supports rolling wave planning).
- **Code of accounts**: unique identifiers for each WBS element (e.g., 1.2.3).
- Decomposition can be organised by **phases**, **major deliverables**, subcomponents, or organisational units; different branches may be decomposed to different levels.
- Too much decomposition wastes effort; too little loses control.

WBS dictionary: detailed description of each WBS element — code of account, description of work, assumptions and constraints, responsible organisation, schedule milestones, associated activities, resources required, cost estimates, quality requirements, acceptance criteria, technical references, agreement information.

Scope baseline = approved scope statement + WBS + WBS dictionary.

Product backlog essentials (adaptive)

- An ordered list of everything that might be needed in the product; the single source of work for the team.
- Owned and ordered by the **product owner**, by value, risk, dependencies and learning.
- Items near the top are small and detailed ("ready"); items further down are larger and less detailed (**epics** to be split later) — the backlog is **progressively elaborated** through **backlog refinement**.
- **User story** format: *"As a , I want , so that ."* Good stories are **INVEST**: Independent, Negotiable, Valuable, Estimable, Small, Testable. A story has the **3 Cs**: Card, Conversation, Confirmation (acceptance criteria).
- **Definition of ready (DoR)**: conditions a backlog item must meet before the team pulls it into a sprint.
- **Definition of done (DoD)**: shared quality criteria that every increment must meet (coded, tested, reviewed, documented, deployable...). **Acceptance criteria** are specific to each story; the **DoD** applies to all.
- **MVP** (minimum viable product): the smallest product that lets you learn whether the solution delivers value. **MMF** (minimum marketable feature): the smallest piece of functionality that delivers value to users.

8.6 Validate Scope *(Monitoring and Controlling)*

Purpose: formalise acceptance of completed deliverables by the customer or sponsor.

- Input: **verified deliverables** (deliverables that passed quality control).
- Technique: **inspection** with the customer (reviews, walkthroughs, demonstrations), decision-making (voting).
- Output: **accepted deliverables** (signed off), change requests for those not accepted, work performance information.
- Performed periodically throughout the project — at the end of each phase or major deliverable, not just once.
- In adaptive projects, the **sprint review** (demonstration to stakeholders) and the product owner's acceptance of stories against acceptance criteria serve this purpose.

Quality control (part of Monitor and Control Scope)	Validate Scope
Checks **correctness** — does the deliverable meet the quality requirements?	Checks **acceptance** — does the customer formally accept it?
Usually performed by the team/quality staff, **first**	Performed with the customer/sponsor, **after** quality control
Output: **verified** deliverables	Output: **accepted** deliverables

8.7 Monitor and Control Scope *(Monitoring and Controlling)*

Purpose: monitor the status of project and product scope, manage changes to the scope baseline, and — in the Eighth Edition — **control quality** of deliverables.

- **Variance analysis**: compare actual scope delivered with the baseline; determine cause and degree of difference.
- **Trend analysis**: is performance improving or deteriorating?
- **Quality control**: inspection, testing, statistical sampling, control charts, checklists — producing **quality control measurements** and **verified deliverables**.
- Outputs: work performance information, change requests, updates.

Scope creep vs gold plating

Scope creep	Gold plating
Uncontrolled expansion of scope **without adjustments** to time, cost and resources — often driven by customer or stakeholder requests	The **team** adds extra features or quality beyond what was required, on its own initiative, often to impress the customer
Prevent with clear scope statements, exclusions, change control, backlog discipline	Prevent with clear acceptance criteria, DoD, culture of delivering exactly what is valued

Both are undesirable. Gold plating increases cost and risk, and may introduce defects, without agreed value.

8.8 Scope in hybrid projects

- Use a **WBS for the overall structure** (phases, major deliverables, predictive components) and **backlogs** for the adaptive components — for example, the WBS element "Customer portal" is delivered through a backlog of user stories.
- Keep **one source of truth for each component** and a traceability link between requirements, backlog items and contractual deliverables.
- The **scope baseline** may define *outcomes and capacity* (e.g., "a portal meeting these objectives within six sprints") while detailed features remain flexible.

Chapter 8 Review Questions

Answer all questions first, then check the answer key for Chapter 8 in Appendix B.

8.1 What does the 100% rule state for a WBS?

- **A.** The WBS includes all of the project's work, and only that work
- **B.** Each work package must take exactly 100 hours
- **C.** Every deliverable must have 100% test coverage
- **D.** The WBS must be 100% complete before the project charter can be approved by the sponsor

8.2 What is the lowest level of the WBS at which cost and duration can be reliably estimated and managed?

- **A.** Planning package
- **B.** Control account
- **C.** Work package
- **D.** Activity

8.3 A deliverable has passed internal testing. The project manager now meets the customer to review it and obtain sign-off. Which process is being performed?

- **A.** Monitor and Control Scope
- **B.** Define Scope
- **C.** Manage Quality Assurance
- **D.** Validate Scope

8.4 A developer adds an animation the customer never asked for because "they will love it". This is an example of:

- **A.** A preventive action
- **B.** Scope creep
- **C.** Gold plating
- **D.** Progressive elaboration

8.5 Which document links each requirement to its business objective and to the deliverables and tests that satisfy it?

- **A.** Requirements traceability matrix
- **B.** Stakeholder register
- **C.** WBS dictionary
- **D.** Issue log

8.6 Two key stakeholders insist on conflicting requirements for the same report. What should the project manager do first?

- **A.** Facilitate a workshop with both stakeholders to reach agreement
- **B.** Implement both versions of the report
- **C.** Remove the report from the scope
- **D.** Implement the requirement of the more senior stakeholder, since seniority decides

8.7 Which statement correctly distinguishes acceptance criteria from the definition of done?

- **A.** They are identical terms
- **B.** Acceptance criteria are story-specific; the definition of done applies to every item

- **C.** The definition of done is written by the customer for each story; acceptance criteria are team-wide
- **D.** Acceptance criteria apply only to predictive projects

8.8 Which section of the project scope statement helps most to prevent stakeholders from assuming that certain work is included?

- **A.** Project exclusions
- **B.** Acceptance criteria
- **C.** Product scope description
- **D.** Constraints

8.9 Which two characteristics are part of the INVEST criteria for good user stories? (Choose two.)

- **A.** Valuable to the project manager only
- **B.** Negotiable
- **C.** Verified by the sponsor
- **D.** Inexpensive
- **E.** Testable

8.10 "The system shall support 5,000 concurrent users with a response time under two seconds." This is an example of which requirement type?

- **A.** Transition requirement
- **B.** Non-functional requirement
- **C.** Functional requirement
- **D.** Business requirement

8.11 In the Kano model, what tends to happen to "excitement" features (delighters) over time?

- **A.** They always remain delighters
- **B.** They become reverse features that customers actively dislike over time
- **C.** They become expected basic features as customers get used to them
- **D.** They are removed from the product backlog

8.12 Who orders the product backlog in Scrum?

- **A.** The developers, by technical preference
- **B.** The product owner
- **C.** The project sponsor
- **D.** The scrum master

8.13 Where is quality control performed in the Eighth Edition process structure?

- **A.** Within Manage Quality Assurance
- **B.** Within Plan Scope Management
- **C.** Within Validate Scope
- **D.** Within Monitor and Control Scope

8.14 A product owner wants to learn quickly whether customers will use a new subscription service before investing in the full solution. What should the team deliver first?

- **A.** A requirements traceability matrix

- **B.** A detailed WBS dictionary
- **C.** The complete product with all features
- **D.** A minimum viable product

Chapter 9 — Quality: Embedded Across Governance and Scope

After this chapter you should be able to:

- explain where quality planning, assurance and control live in the Eighth Edition;
- distinguish quality from grade, and precision from accuracy;
- calculate and classify the cost of quality;
- use the seven basic quality tools and interpret a control chart;
- apply quality practices in adaptive teams (definition of done, test automation, continuous integration).

9.1 Quality in the Eighth Edition

Quality is the degree to which a set of inherent characteristics fulfils requirements. Two classic definitions summarise it: **conformance to requirements** (Crosby) and **fitness for use** (Juran). Quality must satisfy both the *stated* requirements and the *real needs* of stakeholders.

There is no Quality performance domain. Following the principle **Embed quality into processes and deliverables**:

Quality activity	Eighth Edition home	Focus
Quality planning — identify standards, requirements, metrics; plan cost of quality; design tests	**Define Scope** (Scope domain)	Decide what "good" means and how it will be achieved and verified
Quality assurance — confirm processes are adequate and followed; improve them	**Manage Quality Assurance** (Governance domain)	The **process**
Quality control — inspect and test deliverables; measure results	**Monitor and Control Scope** (Scope domain)	The **product**

The PMP exam task *Plan and optimise quality of products/deliverables* (Process domain) covers all three.

9.2 Key concepts

Quality vs grade

- **Quality** = how well the product meets requirements. Low quality is **always a problem**.
- **Grade** = a category assigned to products with the same functional use but different technical characteristics (e.g., economy vs business class). Low grade **may be acceptable** if it meets the requirements.
- *Example:* a low-grade software product with few features and no defects is acceptable; a high-grade product full of defects is not.

Precision vs accuracy

- **Precision** = repeated measurements are clustered closely (consistency).
- **Accuracy** = the measured value is close to the true value (correctness).
- Precise measurements are not necessarily accurate, and vice versa.

Prevention over inspection — quality is planned, designed and built in, not inspected in. The cost of preventing mistakes is generally much lower than the cost of correcting them when found by inspection or by customers.

Continuous improvement — the **Plan-Do-Check-Act (PDCA)** cycle (Shewhart, popularised by Deming) underlies modern quality management. **Kaizen** = continuous small improvements by everyone.

Management responsibility — success requires participation of all team members, but management provides the resources and environment for quality (Deming estimated most quality problems originate in the system, not in individual workers).

Mutually beneficial partnership with suppliers — quality of procured items affects quality of the whole.

Quality theorists you may meet in questions

Name	Key idea
W. Edwards Deming	PDCA, 14 points for management, most problems are system problems; reduce variation
Joseph Juran	Fitness for use; quality trilogy (planning, control, improvement); Pareto principle (80/20) applied to quality
Philip Crosby	Conformance to requirements; zero defects; "quality is free" (cost of non-conformance)
Walter Shewhart	Statistical process control; control charts; Plan-Do-Study/Check-Act
Kaoru Ishikawa	Cause-and-effect (fishbone) diagram; quality circles
Genichi Taguchi	Quality loss function; design of experiments

Quality methodologies

- **Total Quality Management (TQM)** — organisation-wide, customer-focused, continuous improvement.
- **Six Sigma** — reduce variation and defects to 3.4 defects per million opportunities; **DMAIC** (Define, Measure, Analyse, Improve, Control) to improve existing processes; DMADV for new designs.
- **Lean** — eliminate waste and maximise flow (see Chapter 15).
- **ISO 9000 family** — quality management system standards.

9.3 Cost of quality (COQ)

Cost of conformance (money spent to *avoid* failures)	Cost of non-conformance (money spent *because of* failures)
Prevention costs — build a quality product: training, documenting processes, equipment, time to do it right, design reviews	**Internal failure costs** — failures found *by the project*: rework, scrap

Cost of conformance (money spent to *avoid* failures)	Cost of non-conformance (money spent *because of* failures)
Appraisal costs — assess the quality: testing, destructive testing loss, inspections	**External failure costs** — failures found *by the customer*: liabilities, warranty work, lost business, reputation

> **Exam tip.** Spending on prevention and appraisal is usually justified because external failures are the most expensive. A question that asks "which is the most expensive cost of quality?" is pointing at **external failure**. Also note: **warranty claims** = external failure; **inspection** = appraisal; **training** = prevention; **rework before delivery** = internal failure.

9.4 Quality metrics and the quality management plan

A **quality metric** describes a product or project attribute and how it is measured, with an allowable variation — the **tolerance**. Examples: defect density, failure rate, on-time performance, availability (99.9%), customer satisfaction score, test coverage, mean time between failures.

The **quality management plan** (planned within Define Scope) describes: quality standards adopted; quality objectives; roles and responsibilities; deliverables and processes subject to review; quality control and assurance activities; tools; procedures for handling non-conformance, corrective actions and continuous improvement.

9.5 The seven basic quality tools

Tool	What it shows	Typical use
Cause-and-effect (fishbone, Ishikawa) diagram	Possible causes grouped in categories (e.g., people, methods, machines, materials, measurement, environment) leading to an effect	Root-cause analysis; often combined with the **5 Whys**
Flowchart (process map)	Steps, decisions and flow of a process (SIPOC: suppliers, inputs, process, outputs, customers)	Find inefficiencies and failure points
Check sheet (tally sheet)	Counts of occurrences by category	Collect data on defects efficiently
Pareto diagram	Vertical bars ordered by frequency, with a cumulative line	Focus on the **vital few** causes that produce most problems (80/20 rule)
Histogram	Distribution of data (bar chart of frequencies)	Show shape and spread of measurements
Control chart	Process data over time against control limits	Determine whether a process is **stable/predictable**
Scatter diagram	Relationship between two variables	Identify **correlation** (not proof of causation)

Other useful tools: **affinity diagrams**, **matrix diagrams**, **mind maps**, **logical data models**, **design of experiments (DOE)**, **statistical sampling**, **benchmarking**, **audits**, **checklists**.

Control charts in detail

- **Mean** — the centre line of the process.
- **Upper and lower control limits (UCL/LCL)** — typically set at **±3 sigma** from the mean; they reflect the *voice of the process* and are calculated from process data.
- **Specification limits** — set by the *customer* or requirements (voice of the customer). A process can be in control but still not meet specifications, if the specification limits are tighter than the control limits.
- **Out of control** when:
  - a single point falls **outside** a control limit; or
  - the **rule of seven**: seven consecutive points fall on one side of the mean (a non-random trend), even if all are within limits.
- **Common-cause (random) variation** is inherent in the process; **special-cause (assignable) variation** has a specific, identifiable cause that should be investigated.

Statistical sampling

- **Attribute sampling** — the result conforms or does not (pass/fail).
- **Variable sampling** — the result is rated on a continuous scale measuring the degree of conformity.
- Sampling is used when inspecting everything is too costly or destructive.

Sigma levels (for reference)

± sigma	% of values within limits
1σ	68.27%
2σ	95.45%
3σ	99.73%
6σ	99.9999998% (Six Sigma programmes target 3.4 defects per million, allowing a 1.5σ shift)

9.6 Quality assurance in practice (Manage Quality Assurance)

- **Quality audits** — structured, independent reviews to determine whether project activities comply with policies, processes and procedures; identify good practices, gaps and non-conformities; share good practices; support process improvement.
- **Process analysis** — identify improvements, non-value-adding activities, constraints.
- **Design for X (DfX)** — design to optimise a particular characteristic (cost, reliability, manufacturability, serviceability, sustainability).
- **Problem solving** — define the problem, identify root cause, generate solutions, choose, implement, verify.
- Output: **quality reports** (issues, recommendations for corrective actions), **test and evaluation documents**, change requests.

9.7 Quality in adaptive teams

- **Definition of done** — a shared, explicit quality standard for every increment.

- **Acceptance criteria** for each story, often written as examples (behaviour-driven development: *Given – When – Then*).
- **Test-driven development (TDD)**, **automated testing**, **continuous integration and delivery (CI/CD)** — build quality in and get fast feedback.
- **Pair programming**, **mob programming**, **code reviews** — real-time inspection and knowledge sharing.
- **Refactoring** — continuously improve the design to keep technical debt low.
- **Retrospectives** — inspect and adapt the process every iteration (continuous improvement).
- **Small batch sizes** — reveal problems early; reduce the cost of fixing them.
- **Escaped defects** — a key quality metric: defects found after release.

> **Trap.** "The team is behind; skip testing this sprint and catch up later" is never the right answer. Lowering the definition of done creates technical debt and violates the principle of embedding quality. The right answer usually involves the product owner adjusting **scope**, not quality.

Chapter 9 Review Questions

Answer all questions first, then check the answer key for Chapter 9 in Appendix B.

9.1 A software product has few features but no defects and meets all requirements. Another product has many features but frequent crashes. Which statement is correct?

- **A.** The first is low grade but high quality; the second is high grade but low quality
- **B.** Both products are low quality
- **C.** Grade and quality mean the same thing
- **D.** The first product is low quality because it has few features and only basic functions

9.2 Which cost of quality category does the cost of handling warranty claims belong to?

- **A.** External failure costs
- **B.** Internal failure costs
- **C.** Appraisal costs
- **D.** Prevention costs

9.3 Seven consecutive measurements on a control chart fall above the mean but within the control limits. What should the project manager conclude?

- **A.** The control limits were calculated incorrectly
- **B.** The specification limits must be widened
- **C.** The process may be out of control and the cause should be investigated
- **D.** The process is in control, because all of the points are still within the control limits

9.4 A team wants to focus its improvement effort on the few causes responsible for most defects. Which tool is most appropriate?

- **A.** Flowchart
- **B.** Control chart
- **C.** Pareto diagram
- **D.** Scatter diagram

9.5 Which activity is an example of quality assurance rather than quality control?

- **A.** Measuring the dimensions of a delivered component against the drawing
- **B.** Running test cases on a completed module
- **C.** Inspecting a batch of manufactured parts
- **D.** Auditing whether the team follows the agreed testing process

9.6 Measurements of a part are tightly clustered but consistently 2 mm away from the true value. How would you describe them?

- **A.** Accurate but not precise
- **B.** Precise but not accurate
- **C.** Neither precise nor accurate
- **D.** Both precise and accurate

9.7 Who sets the specification limits used alongside a control chart?

- **A.** The scrum master
- **B.** The customer or the requirements
- **C.** The process data, at three standard deviations from the mean
- **D.** The project management office, at random

9.8 A team member says, "We are behind schedule; let's skip unit testing for this sprint." What is the best response from the project manager or scrum master?

- **A.** Ask the sponsor to approve skipping tests
- **B.** Agree, as long as all of the testing is done before the final release to customers
- **C.** Let the testers decide alone
- **D.** Keep the definition of done and adjust the sprint scope with the product owner

9.9 Which two items are prevention costs? (Choose two.)

- **A.** Product recalls
- **B.** Rework of defective units before shipment
- **C.** Training the team in a new coding standard
- **D.** Design reviews before construction
- **E.** Final inspection of finished goods

9.10 Which quality tool would best show whether there is a relationship between the number of training hours and the number of defects per developer?

- **A.** Cause-and-effect diagram
- **B.** Histogram
- **C.** Scatter diagram
- **D.** Check sheet

9.11 In the Eighth Edition, where are quality metrics and the quality management plan primarily developed?

- **A.** As part of Acquire Resources
- **B.** As part of Define Scope
- **C.** As part of Manage Quality Assurance

- **D.** As part of Close Project or Phase

9.12 Which quality theorist is most associated with the idea that "quality is free" and the goal of zero defects?

- **A.** Walter Shewhart
- **B.** Joseph Juran
- **C.** Kaoru Ishikawa
- **D.** Philip Crosby

Chapter 10 — Schedule Performance Domain

After this chapter you should be able to:

- describe the three Schedule processes and the schedule management plan;
- define activities, sequence them with dependencies, leads and lags, and estimate durations (including PERT);
- calculate the critical path, total float and free float with forward and backward passes;
- choose between crashing, fast tracking, resource levelling and resource smoothing;
- plan and track schedules in adaptive work (release planning, velocity, burn charts, flow metrics).

10.1 What the Schedule domain covers

The Schedule domain addresses **planning, developing and controlling the timing of the work** so the project delivers its outcomes when they are needed. It has three processes:

Process	Focus area	Purpose
Plan Schedule Management	Planning	Establish policies, procedures and documentation for planning, developing, managing, executing and controlling the schedule
Develop Schedule	Planning	Define activities, sequence them, estimate durations and resources, and build the schedule model and baseline (or release/iteration plans)
Monitor and Control Schedule	Monitoring and Controlling	Monitor status, manage changes to the schedule baseline, forecast completion

10.2 Plan Schedule Management

The **schedule management plan** specifies: the scheduling methodology and tool; schedule model development and maintenance; level of accuracy (rounding of durations, contingency); **units of measure** (hours, days, story points); **organisational procedure links** (WBS framework); **control thresholds** (e.g., ±10% variance before action); **rules of performance measurement** (EVM rules, % complete methods); reporting formats; release and iteration lengths for adaptive work.

10.3 Develop Schedule — step by step

Step 1: Define activities

Decompose **work packages** into **activities** (the work needed to produce the deliverables). Outputs: **activity list**, **activity attributes** (ID, description, predecessors, successors, lead/lag, resources, constraints, location), **milestone list** (milestones have **zero duration** — they mark significant points or events). Use **rolling wave planning** for later work.

Step 2: Sequence activities — the precedence diagramming method (PDM)

PDM uses **activity-on-node (AON)** diagrams with four relationship types:

Relationship	Meaning	Example	Frequency
Finish-to-start (FS)	B cannot start until A finishes	Pour foundation, then build walls	**Most common**
Start-to-start (SS)	B cannot start until A starts	Start levelling concrete after pouring starts	Common
Finish-to-finish (FF)	B cannot finish until A finishes	Finish editing after writing finishes	Common
Start-to-finish (SF)	B cannot finish until A starts	Old security shift cannot end until the new shift starts	**Rare**

Dependency types:

Type	Description
Mandatory (hard logic)	Required by contract or inherent in the nature of the work (you can't test code that hasn't been written)
Discretionary (preferred / soft logic)	Based on best practice or preference; can be changed — **the first candidates for fast tracking**
External	Involve a relationship with non-project activities (permit from a regulator, vendor delivery) — usually outside the team's control
Internal	Between project activities, usually within the team's control

A dependency can be both mandatory and external, etc.

Leads and lags:

- **Lead** = the successor can start *before* the predecessor finishes (FS −2 days) — an acceleration.
- **Lag** = a required *delay* between activities (FS +3 days: wait three days for concrete to cure).

Step 3: Estimate resources and durations

Technique	How it works	When to use
Expert judgment	Experience of people who did similar work	Always useful
Analogous (top-down)	Use the actual duration/cost of a similar past project, adjusted	Early, little detail; quick but less accurate
Parametric	Statistical relationship × quantity (e.g., 2 hours per m^2 × 500 m^2)	When reliable unit rates exist; can be very accurate
Three-point	Optimistic (O), most likely (M), pessimistic (P)	When uncertainty is significant
Bottom-up	Estimate each lower-level component and aggregate	When detail is available; most accurate, most time-consuming
Data analysis: alternatives, reserve analysis	Compare options; add contingency reserves for identified risks	Throughout

Technique	How it works	When to use
Decision-making (e.g., fist of five), Delphi	Build consensus, reduce bias	Team estimates

Three-point formulas

- **Triangular distribution:** E = (O + M + P) ÷ 3
- **Beta / PERT distribution:** E = (O + 4M + P) ÷ 6
- **Standard deviation (PERT):** σ = (P − O) ÷ 6
- Variance = σ^2; for a path, add variances (not standard deviations), then take the square root.

Example: O = 4, M = 6, P = 14 days. PERT E = (4 + 24 + 14) ÷ 6 = **7 days**; σ = (14 − 4) ÷ 6 = **1.67 days**. Triangular E = 24 ÷ 3 = 8 days. Range at ±1σ: 5.33 to 8.67 days (≈68% confidence); ±2σ ≈ 95%; ±3σ ≈ 99.7%.

Remember: *effort* (person-hours of work) is not *duration* (calendar time). Adding people does not reduce duration linearly (law of diminishing returns, communication overhead — Brooks's law: adding people to a late software project makes it later). **Parkinson's law**: work expands to fill the time available. **Student syndrome**: people start at the last possible moment.

Step 4: Build the schedule model — critical path method (CPM)

- **Forward pass** → early start (ES) and early finish (EF). Convention used here: ES of the first activity = 0; EF = ES + duration; ES of a successor = the **largest** EF of its predecessors.
- **Backward pass** → late finish (LF) and late start (LS). LF of the last activity = its EF (project end); LS = LF − duration; LF of a predecessor = the **smallest** LS of its successors.
- **Total float (slack)** = LS − ES = LF − EF — how long an activity can be delayed without delaying the **project end date**.
- **Free float** = ES of the earliest successor − EF of the activity — how long it can be delayed without delaying **any successor's early start**.
- **Critical path** = the **longest path** through the network; it determines the shortest possible project duration; activities on it usually have **zero total float** (negative float means the path is already late against a constraint date).
- **Near-critical path** = a path with little float; watch it closely. A project can have more than one critical path — which increases risk.

Worked example

Activity	Duration (days)	Predecessor
A	3	—
B	5	A
C	2	A
D	4	C
E	3	B, D

Paths: **A-B-E** = 3 + 5 + 3 = **11**; A-C-D-E = 3 + 2 + 4 + 3 = 12. Critical path = **A-C-D-E (12 days)**.

Forward pass: A (0–3); B (3–8); C (3–5); D (5–9); E starts at max(8, 9) = 9, finishes 12. Backward pass: E (LS 9, LF 12); B (LF 9, LS 4); D (LF 9, LS 5); C (LF 5, LS 3); A (LF = min(4, 3) = 3, LS 0). Float: A 0, C 0, D 0, E 0, **B: LS − ES = 4 − 3 = 1 day** (free float of B = ES of E − EF of B = 9 − 8 = 1).

Step 5: Optimise — resource optimisation and compression

Technique	What it does	Effect
Resource levelling	Adjusts start and finish dates to balance resource demand with supply (e.g., one specialist can't do two activities at once)	**Can change the critical path and lengthen the project**
Resource smoothing	Adjusts activities **within their float** so resource requirements don't exceed limits	Critical path and end date **do not change**; may not optimise all resources
Crashing	Adds resources (overtime, extra people, faster equipment) to critical-path activities to shorten duration at the **lowest additional cost**	**Increases cost**; works only on critical activities where extra resources help
Fast tracking	Performs activities in **parallel** that were planned in sequence (usually by changing discretionary dependencies or using leads)	**Increases risk and potential rework**; little or no added cost

Crashing logic: crash the critical activity with the **lowest cost per day saved** first, one day at a time, re-checking the critical path because another path may become critical.

> **Exam tip.** "The sponsor wants the project to finish two weeks early, without increasing the budget" → **fast tracking**. "...without increasing risk, money is available" → **crashing**. Neither helps if applied to non-critical activities.

Other schedule-model tools: **what-if scenario analysis**, **Monte Carlo simulation** (probability distribution of completion dates, e.g., "80% chance of finishing by 30 June"), **agile release planning**, the **critical chain method** (resource-constrained critical path with **buffers**: project buffer at the end of the chain, feeding buffers where non-critical chains join, resource buffers; durations stripped of individual padding).

Step 6: Present and baseline the schedule

- **Schedule baseline** — the approved version of the schedule model; changed only through change control.
- Presentations: **bar (Gantt) charts** (easy to read, show durations and dates), **milestone charts** (only key dates — good for executives), **project schedule network diagrams** (logic and critical path), **calendars**, **roadmaps**.
- **Schedule data** — milestones, activities, attributes, assumptions, constraints, resource requirements, alternative schedules, contingency reserves.

10.4 Scheduling in adaptive projects

Concept	Description
Timeboxing	Fixed periods (sprints/iterations of 1–4 weeks, meetings with maximum durations)
Release planning	Determine features for each release based on the roadmap, priorities and velocity
Iteration (sprint) planning	The team selects backlog items it can complete in the iteration and plans the work
Story points	Relative estimates of size/complexity/uncertainty (often Fibonacci: 1, 2, 3, 5, 8, 13, 21)
Planning poker	Team estimation by simultaneously revealing cards; discuss outliers; re-vote
Affinity estimating / T-shirt sizing	Quickly group many items by relative size (S, M, L, XL)
Velocity	Story points completed per iteration; used by **the team** for forecasting — never to compare teams
Capacity	Available team time in an iteration (holidays, part-time members)
On-demand scheduling (Kanban/pull)	Work is pulled from the backlog when capacity is available, limited by WIP limits

Forecasting with velocity: remaining backlog ÷ average velocity = number of iterations remaining. *Example:* 240 points remaining ÷ 30 points per sprint = 8 sprints; with 2-week sprints = 16 weeks.

Tracking charts

- **Burndown chart** — remaining work vs time. A line above the ideal line means behind.
- **Burnup chart** — completed work vs time, plus a total-scope line; shows **scope changes** clearly (the scope line moves up).
- **Cumulative flow diagram (CFD)** — bands of work in each state over time; widening bands show **bottlenecks**; band thickness = WIP; horizontal distance ≈ cycle time.
- **Lead time** = time from request to delivery (customer view). **Cycle time** = time from work start to completion (team view). **Throughput** = items completed per period.
- **Little's Law**: average cycle time = WIP ÷ throughput. *Example:* 12 items in progress, 4 completed per week → cycle time = 3 weeks. To reduce cycle time, **reduce WIP**.

10.5 Monitor and Control Schedule

- Determine current status, compare with the baseline (variance analysis), assess trends, forecast completion.
- **Earned value schedule measures**: SV = EV − PV; SPI = EV ÷ PV (see Chapter 11).
- Techniques: performance reviews, trend analysis, **critical path analysis** (is the critical path changing?), what-if analysis, resource optimisation, leads and lags, schedule compression, **iteration burndown** and **velocity trend** in adaptive work.

- Outputs: work performance information, **schedule forecasts**, change requests (to the baseline), updates.
- Agile focus: are we delivering value at a sustainable pace? Retrospectives address impediments; the product owner adjusts scope to meet dates.

> **Trap.** A non-critical activity that is late is not necessarily a problem — check its **float** before taking action. Conversely, a critical activity that slips by one day delays the project by one day unless you recover it.

Chapter 10 Review Questions

Answer all questions first, then check the answer key for Chapter 10 in Appendix B.

10.1 An activity has an optimistic estimate of 6 days, a most likely estimate of 9 days and a pessimistic estimate of 18 days. Using PERT (beta distribution), what is the expected duration in days?

(Fill in the blank.)

Answer: _____

10.2 Using the same estimates (O = 6, M = 9, P = 18), what is the PERT standard deviation?

- **A.** 1.5 days
- **B.** 2 days
- **C.** 3 days
- **D.** 4 days

10.3 The sponsor asks to shorten the project without adding cost, and the team is willing to accept more risk. Which technique should the project manager consider first?

- **A.** Resource smoothing
- **B.** Adding a project buffer
- **C.** Crashing the activities on the critical path
- **D.** Fast tracking critical-path activities

10.4 Which technique adjusts activities within their available float so that resource limits are respected without changing the critical path?

- **A.** Crashing
- **B.** Resource smoothing
- **C.** Resource levelling
- **D.** Fast tracking

10.5 Activities: A (4 days, no predecessor), B (6 days, after A), C (3 days, after A), D (5 days, after C), E (2 days, after B and D). What is the length of the critical path?

- **A.** 12 days
- **B.** 13 days
- **C.** 14 days
- **D.** 20 days

10.6 In the same network (A 4, B 6 after A, C 3 after A, D 5 after C, E 2 after B and D), what is the total float of activity B?

- **A.** 2 days
- **B.** 1 day
- **C.** 4 days
- **D.** 0 days

10.7 Testing cannot start until the code is written. What type of dependency is this?

- **A.** Start-to-start
- **B.** Discretionary
- **C.** External
- **D.** Mandatory

10.8 A team has 12 work items in progress and completes on average 3 items per week. Using Little's Law, what is the average cycle time?

- **A.** 0.25 weeks
- **B.** 3 weeks
- **C.** 4 weeks
- **D.** 36 weeks

10.9 A product backlog contains 300 story points of remaining work. The team's average velocity is 25 points per two-week sprint. Approximately how long will it take to complete the backlog?

- **A.** 12 weeks
- **B.** 15 weeks
- **C.** 24 weeks
- **D.** 30 weeks

10.10 Which chart makes scope increases most visible to stakeholders in an agile project?

- **A.** A histogram of defects found per sprint
- **B.** Burndown chart without a scope line
- **C.** Milestone chart
- **D.** Burnup chart with a total-scope line

10.11 A manager wants to compare the productivity of two Scrum teams using their velocities. What should the project manager advise?

- **A.** Velocity is team-specific and should not be used to compare teams
- **B.** Normalise both to hours and rank the developers
- **C.** Ask the teams to double their velocity
- **D.** Compare them directly; the higher velocity indicates the better team

10.12 Which two statements about the critical path are correct? (Choose two.)

- **A.** It is the shortest path through the network
- **B.** It is the path with the most activities
- **C.** It never changes once the baseline is approved
- **D.** It is the longest path through the schedule network

- **E.** Activities on it usually have zero total float

10.13 A cumulative flow diagram shows the "Testing" band growing wider each week while "Done" grows slowly. What does this indicate?

- **A.** A bottleneck in testing
- **B.** The team is increasing its velocity
- **C.** Testing has no work in progress
- **D.** Scope is decreasing

10.14 Which relationship type is the least commonly used in precedence diagramming?

- **A.** Start-to-start
- **B.** Start-to-finish
- **C.** Finish-to-finish
- **D.** Finish-to-start

Chapter 11 — Finance Performance Domain

After this chapter you should be able to:

- describe the four Finance processes and the financial management plan;
- choose estimating techniques and state typical accuracy ranges;
- build a budget with contingency and management reserves and explain the cost baseline;
- perform earned value calculations and forecasts, and interpret the results;
- evaluate projects with financial measures (NPV, IRR, ROI, payback, BCR) and explain funding concepts.

11.1 What the Finance domain covers

The Finance domain addresses **the planning, estimating, budgeting, funding and control of the project's financial resources**, and the link between money spent and value created. It replaces the Sixth Edition's Cost Management and broadens it to **funding, financial value and financial governance**.

Process	Focus area	Key output
Plan Financial Management	Planning	Financial management plan
Estimate Costs	Planning	Cost estimates, basis of estimates
Develop Budget	Planning	Cost baseline, project funding requirements
Monitor and Control Finances	Monitoring and Controlling	Work performance information, cost forecasts, change requests

Desired outcomes: costs are estimated realistically; funding is available when needed; spending is tracked against value delivered; variances are detected early and acted on; financial decisions support the business case.

11.2 Plan Financial Management

The **financial management plan** (formerly cost management plan) defines: **units of measure** (currency, staff-hours); **level of precision** (rounding, e.g., to the nearest $100); **level of accuracy** (acceptable range, e.g., ±10%, including contingency); **organisational procedure links** (control accounts in the WBS); **control thresholds** (variance that triggers action); **rules of performance measurement** (EVM techniques, how % complete is measured); funding sources and timing; currency and exchange-rate handling; reporting formats; approval authorities for spending.

11.3 Estimate Costs

Cost concepts

Concept	Meaning
Direct costs	Attributable directly to the project (team salaries, materials)
Indirect costs	Shared across projects (overheads such as office rent, utilities)
Fixed costs	Do not change with volume (equipment set-up)
Variable costs	Change with volume (materials per unit, hourly labour)
Sunk costs	Already spent and cannot be recovered — **must not influence** future decisions
Opportunity cost	The value of the best alternative given up by choosing one option
Life-cycle costing	Total cost of ownership over the product's life (acquisition, operation, maintenance, disposal)
Depreciation	Allocation of an asset's cost over its useful life (straight-line or accelerated)
Cost of quality	Conformance + non-conformance costs (Chapter 9)

Estimating techniques

Same family as duration estimating: **analogous**, **parametric**, **bottom-up**, **three-point** (PERT), expert judgment, **reserve analysis**, **cost of quality**, **alternatives analysis**, decision-making, PMIS, and (in adaptive work) estimating by **team cost per iteration** (burn rate) or **cost per story point**.

Accuracy ranges

Estimate	Typical range	When
Rough order of magnitude (ROM)	**−25% to +75%**	Initiation
Budget estimate	−10% to +25%	Early planning
Definitive estimate	**−5% to +10%**	Detailed planning

The output includes **cost estimates** and the **basis of estimates** (documentation of how the estimate was derived, assumptions, constraints, range, confidence level).

11.4 Develop Budget

Cost aggregation: activity estimates → work package estimates → control accounts → project.

Layer	Contents
Activity and work package cost estimates	From Estimate Costs
+ Contingency reserves	For **identified** risks (known-unknowns); controlled by the PM within the plan's rules
= Cost baseline	Approved, time-phased budget (the **S-curve**); basis for EVM; = **BAC**

Layer	Contents
+ Management reserves	For **unidentified** risks (unknown-unknowns); controlled by management/sponsor; using them requires a change to the cost baseline
= Project budget	Total funds authorised

> **Exam tip.** The **cost baseline includes contingency reserve but not management reserve**. Earned value is measured against the cost baseline (the PMB), so management reserve is **not** part of BAC.

Funding limit reconciliation: compare planned spending with funding limits (e.g., the organisation can release only $200,000 per quarter); reschedule work if necessary. **Project funding requirements** are derived from the cost baseline plus management reserve and are often provided in **incremental, step-like amounts**, while the expenditure curve is smooth.

Funding sources may include internal capital budgets, external financing, grants, customer payments, public–private partnerships. **CAPEX** (capital expenditures, creating assets) vs **OPEX** (operational expenditures, running costs) treatment matters to finance departments and can affect how agile work is funded.

Adaptive funding: many organisations fund **persistent teams or products** for a period (e.g., a quarter) and decide continuation based on value delivered (incremental or "venture capital" style funding).

11.5 Monitor and Control Finances — earned value management (EVM)

EVM integrates scope, schedule and cost to measure performance and forecast results.

The three basic values

Term	Definition	Question it answers
PV — planned value	Authorised budget assigned to the work **scheduled** to be done by now	How much work *should* be done?
EV — earned value	Budgeted value of the work **actually completed** by now	How much work *is* done (in budget terms)?
AC — actual cost	Actual cost incurred for the work completed by now	How much did the completed work *cost*?
BAC — budget at completion	Total planned value of the project (the cost baseline)	What is the total budget?

Variances and indices

Measure	Formula	Interpretation
Cost variance	**CV = EV − AC**	Negative = over budget; positive = under budget
Schedule variance	**SV = EV − PV**	Negative = behind schedule; positive = ahead

Measure	Formula	Interpretation
Cost performance index	**CPI = EV ÷ AC**	< 1 = over budget (getting less than $1 of work per $1); > 1 = under budget
Schedule performance index	**SPI = EV ÷ PV**	< 1 = behind; > 1 = ahead
Variance at completion	**VAC = BAC − EAC**	Negative = expected overrun

Memory aid: **EV comes first** in every variance and every index. Negative/less than 1 is bad.

Forecasting: estimate at completion (EAC)

Situation	Formula
Current cost performance will **continue** (most common default)	**EAC = BAC ÷ CPI**
Past variance was **atypical**; future work at the planned rate	**EAC = AC + (BAC − EV)**
Both cost and schedule performance influence the remaining work	**EAC = AC + [(BAC − EV) ÷ (CPI × SPI)]**
The original estimate is **fundamentally flawed**; re-estimate	**EAC = AC + bottom-up ETC**

Estimate to complete: ETC = EAC − AC (how much more will we spend).

To-complete performance index (TCPI) — the cost performance needed on the remaining work to meet a target:

- To meet the **BAC**: **TCPI = (BAC − EV) ÷ (BAC − AC)**
- To meet a new approved **EAC**: TCPI = (BAC − EV) ÷ (EAC − AC)
- TCPI > 1 → you must be **more efficient** than planned for the rest of the project.

Worked example

A project has BAC = $200,000, planned to finish in 10 months at an even spend rate. At the end of month 4, the project is 30% complete and has spent $70,000.

- PV = 40% × 200,000 = **$80,000** (4 of 10 months)
- EV = 30% × 200,000 = **$60,000**
- AC = **$70,000**
- CV = 60,000 − 70,000 = **−$10,000** (over budget)
- SV = 60,000 − 80,000 = **−$20,000** (behind schedule)
- CPI = 60,000 ÷ 70,000 = **0.857**
- SPI = 60,000 ÷ 80,000 = **0.75**
- EAC (typical) = 200,000 ÷ 0.857 ≈ **$233,333**
- ETC = 233,333 − 70,000 ≈ **$163,333**
- VAC = 200,000 − 233,333 ≈ **−$33,333**
- TCPI (to meet BAC) = (200,000 − 60,000) ÷ (200,000 − 70,000) = 140,000 ÷ 130,000 ≈ **1.077**

Interpretation: the project is over budget and behind schedule; at the current efficiency it will overrun by about $33,000; to finish within the original budget the team would need to work about 7.7% more efficiently than planned — unrealistic unless something changes. The PM should analyse root causes

and present options (scope trade-offs, corrective actions, re-baseline through change control).

Measuring percent complete (rules of performance measurement)

- **0/100** — credit only when complete (conservative; for short tasks).
- **50/50** or **20/80** — partial credit at start, the rest at completion.
- **Weighted milestones**, **physical % complete**, **units completed**.
- Avoid subjective "90% done" estimates — the 90% syndrome.

Other control techniques

Variance analysis, **trend analysis**, **reserve analysis** (are remaining reserves adequate? release unused contingency when risks pass), **forecasting**, **burn rate** tracking in agile (team cost per sprint × sprints remaining), **agile EVM** (EV based on story points completed vs planned per release).

11.6 Financial measures for selecting and justifying projects

Measure	Formula / meaning	Decision rule
Present value (PV)	$PV = FV \div (1 + r)^n$	Converts future money into today's money
Net present value (NPV)	Sum of discounted inflows − discounted outflows (initial investment)	**Higher is better; NPV > 0 is acceptable.** NPV already accounts for the project's duration — do not adjust for years again
Internal rate of return (IRR)	Discount rate at which NPV = 0	**Higher is better**; compare with the cost of capital (hurdle rate)
Return on investment (ROI)	(Net benefit ÷ cost) × 100%	Higher is better
Payback period	Time to recover the initial investment from net cash inflows	**Shorter is better**; ignores time value of money (unless discounted) and benefits after payback
Benefit–cost ratio (BCR)	Benefits ÷ costs (present values)	**> 1** means benefits exceed costs; higher is better
Opportunity cost	Value of the option not chosen	When choosing project A (NPV 50k) over B (NPV 40k), the opportunity cost is **40k**
Economic value added (EVA)	Net operating profit after tax − (capital × cost of capital)	Positive means value is created

PV example: 121,000*receivedin2yearsat*10100,000**.

Payback example: invest $300,000; net inflows $75,000 per year → payback = **4 years**.

> **Trap. Sunk costs** are irrelevant: "We have already spent $2 million, so we must continue" is not a valid reason. Decide based on future costs and benefits.

Chapter 11 Review Questions

Answer all questions first, then check the answer key for Chapter 11 in Appendix B.

11.1 A project has EV = $45,000 and AC = $50,000. What is the CPI? (Give two decimals.)

(Fill in the blank.)

Answer: _____

11.2 BAC = $500,000, EV = $200,000, AC = $250,000, PV = $220,000. Assuming current cost performance continues, what is the EAC?

- **A.** $450,000
- **B.** $520,000
- **C.** $550,000
- **D.** $625,000

11.3 Using the same data (BAC $500,000, EV $200,000, AC $250,000, PV $220,000), what is the schedule variance?

- **A.** −$20,000
- **B.** $20,000
- **C.** −$30,000
- **D.** −$50,000

11.4 Using the same data (BAC $500,000, EV $200,000, AC $250,000), what is the TCPI needed to finish within the BAC?

- **A.** 0.83
- **B.** 1.00
- **C.** 1.20
- **D.** 1.25

11.5 Which reserve is included in the cost baseline?

- **A.** Management reserve
- **B.** Contingency reserve
- **C.** Neither reserve
- **D.** Both contingency and management reserves

11.6 A project has an SPI of 1.10 and a CPI of 0.85. How is the project performing?

- **A.** Ahead of schedule and over budget
- **B.** Behind schedule and over budget
- **C.** Behind schedule and under budget
- **D.** Ahead of schedule and under budget

11.7 During initiation, the sponsor asks for an estimate. Little information is available. What accuracy range is typical?

- **A.** −10% to +10%
- **B.** −5% to +10%
- **C.** Exact, because the budget is fixed
- **D.** −25% to +75%

11.8 Project A has an NPV of $120,000 over 5 years, and Project B has an NPV of $95,000 over 2 years. Only one can be selected. Which should be chosen based on NPV, and what is the opportunity cost?

- **A.** Project A; opportunity cost $95,000
- **B.** Project A; opportunity cost $25,000
- **C.** Project B; opportunity cost $25,000
- **D.** Project B, because it is shorter; opportunity cost $120,000

11.9 Variances from a one-off supplier error are considered atypical and the remaining work is expected to proceed at the budgeted rate. AC = $80,000, EV = $70,000 and BAC = $300,000. What is the EAC?

- **A.** $230,000
- **B.** $300,000
- **C.** $310,000
- **D.** $342,857

11.10 A steering committee member argues that a failing project must continue because $3 million has already been spent. Which concept should the project manager explain?

- **A.** Depreciation makes the project cheaper over time
- **B.** Opportunity cost requires the project to continue until the end
- **C.** Sunk costs should not drive decisions about future investment
- **D.** Management reserve covers all past spending

11.11 Which two items are true about management reserve? (Choose two.)

- **A.** It is used to calculate CPI
- **B.** The project manager may spend it freely for identified risks
- **C.** It addresses unknown-unknowns
- **D.** It is part of the cost baseline and BAC
- **E.** Using it normally requires approval from management and a change to the cost baseline

11.12 An investment of $400,000 generates net cash inflows of $100,000 per year. What is the simple payback period?

- **A.** 2.5 years
- **B.** 4 years
- **C.** 5 years
- **D.** 40 years

11.13 What is the present value of $242,000 to be received in two years if the discount rate is 10%?

- **A.** $193,600
- **B.** $200,000
- **C.** $217,800
- **D.** $220,000

11.14 The spending curve of a project rises smoothly, but the organisation releases funds quarterly. What is the most appropriate action during Develop Budget?

- **A.** Ask the team to work unpaid when funds run out
- **B.** Remove contingency reserves

- **C.** Ignore the funding schedule, because the cost baseline has already been approved by the sponsor
- **D.** Perform funding limit reconciliation and adjust the schedule of work if needed

Chapter 12 — Stakeholders Performance Domain (including Communications)

After this chapter you should be able to:

- identify, analyse and prioritise stakeholders using common models;
- plan, manage and monitor stakeholder engagement, including expectation alignment;
- plan, manage and monitor communications, and calculate communication channels;
- choose communication methods (interactive, push, pull) and technologies appropriate to the audience;
- engage stakeholders effectively in predictive, adaptive and hybrid projects.

12.1 What the Stakeholders domain covers

The Stakeholders domain addresses **identifying, understanding, engaging and communicating with the people and organisations who affect or are affected by the project**. In the Eighth Edition it contains both stakeholder engagement and communications (seven processes):

Focus area	Processes
Initiating	**Identify Stakeholders**
Planning	**Plan Stakeholder Engagement**, **Plan Communications Management**
Executing	**Manage Stakeholder Engagement**, **Manage Communications**
Monitoring and Controlling	**Monitor Stakeholder Engagement**, **Monitor Communications**

Desired outcomes: a productive working relationship with stakeholders; agreement on objectives; stakeholders who are supporters are satisfied; opposition does not undermine the project; information reaches the right people, at the right time, in the right format; expectations are aligned with what will be delivered.

> **Exam tip.** Stakeholder engagement is **continuous**. Identify stakeholders early and **re-identify** them throughout (at each phase, when the organisation changes, when someone new appears). A stakeholder who "suddenly appears" in the middle of a project is a sign that identification was incomplete — add them to the register and analyse them.

12.2 Identify Stakeholders *(Initiating)*

Purpose: identify stakeholders regularly and analyse their interests, involvement, interdependencies, influence and potential impact on success.

- **Inputs:** charter, business documents, agreements, EEFs, OPAs.
- **Techniques:** brainstorming, interviews, questionnaires and surveys, **stakeholder analysis**, document analysis, **data representation** (stakeholder mapping).

- **Output:** the **stakeholder register** — identification information (name, position, contact, role), **assessment information** (major requirements, expectations, potential for influence, phase of most interest), **stakeholder classification** (internal/external, impact, influence, power, interest, direction of influence).

Stakeholder analysis models

Model	Dimensions	Use
Power/interest grid	Power (authority) × interest (concern)	High power–high interest: **manage closely**; high power–low interest: **keep satisfied**; low power–high interest: **keep informed**; low power–low interest: **monitor**
Power/influence grid	Authority × active involvement	Similar quadrants
Impact/influence grid	Ability to cause changes × active involvement	Similar quadrants
Salience model	**Power** (ability to impose will), **urgency** (need for immediate attention — time-sensitive or high-stakes), **legitimacy** (appropriateness of involvement)	Stakeholders with all three ("definitive") get most attention; useful for large, complex communities
Stakeholder cube	Combines grid dimensions in 3-D	Refines classification
Directions of influence	**Upward** (senior management, sponsor), **downward** (team, specialists), **outward** (suppliers, regulators, public, customers), **sideward** (peers, other PMs)	Tailor communications and engagement
Prioritisation	Rank by impact, interest, power	When there are many stakeholders

12.3 Plan Stakeholder Engagement *(Planning)*

Purpose: develop approaches to involve stakeholders based on their needs, expectations, interests and potential impact.

Stakeholder engagement assessment matrix

Compares **current (C)** and **desired (D)** engagement levels:

Stakeholder	Unaware	Resistant	Neutral	Supportive	Leading
Finance director		C		D	
Union representative	C		D		
Head of sales				C	D

- **Unaware** — not aware of the project and its potential impacts.
- **Resistant** — aware but resistant to the project or its changes.
- **Neutral** — aware but neither supportive nor resistant.
- **Supportive** — aware and supportive of the work and its outcomes.

- **Leading** — aware and actively engaged in ensuring success.

The gap between C and D drives engagement actions. Not every stakeholder needs to be "leading" — the desired level depends on their role.

Output: the **stakeholder engagement plan** — strategies and actions for productive involvement; may be formal or informal, detailed or broad. It can contain sensitive information (e.g., notes about resistant stakeholders), so distribution must be handled carefully.

12.4 Manage Stakeholder Engagement *(Executing)*

Purpose: communicate and work with stakeholders to meet their needs and expectations, address issues and foster appropriate involvement.

- Engage stakeholders at the right stages; manage expectations through negotiation and communication; address risks and concerns; clarify and resolve issues.
- **Techniques:** communication skills (feedback), **interpersonal skills** (conflict management, cultural awareness, negotiation, observation/conversation, political awareness), ground rules, meetings.
- **Outputs:** change requests, issue log updates, updates to the stakeholder register and engagement plan.

Aligning and managing expectations (PMP People tasks 5 and 6)

- **Surface expectations early** — interviews, workshops, product vision sessions; record them (stakeholder register, requirements, assumptions).
- **Find common ground** — facilitate workshops; use objective criteria (business case, value, strategy).
- **Make trade-offs explicit** — show the impact of competing demands on scope, schedule, cost and risk.
- **Confirm agreement** — documented decisions, acceptance criteria, working agreements.
- **Monitor satisfaction** — feedback, surveys, demos, informal conversations.
- **Reset expectations transparently** when reality changes — early, honestly, with options.

12.5 Monitor Stakeholder Engagement *(Monitoring and Controlling)*

Purpose: monitor relationships and tailor engagement strategies. Techniques: alternatives analysis, root-cause analysis, stakeholder analysis, **engagement assessment matrix** (has C moved toward D?), feedback, active listening, cultural awareness, leadership, networking, political awareness, meetings. Outputs: work performance information, change requests, updates.

12.6 Plan Communications Management *(Planning)*

Purpose: develop an appropriate approach and plan for communication activities based on the information needs of each stakeholder or group, available assets and the needs of the project.

Communication requirements analysis

Determine the information needs of stakeholders: who needs what information, when, how often, in what format, who provides it, and why. Consider organisational charts, stakeholder responsibilities, disciplines involved, logistics (locations, time zones), internal and external information needs, legal requirements.

Number of communication channels

Channels = n (n − 1) ÷ 2, where n = number of people (including the PM).

Example: a team of 10 has 10 × 9 ÷ 2 = **45** channels. If it grows to 15, there are 15 × 14 ÷ 2 = 105 channels — **60 more**. Communication complexity grows much faster than team size.

Communication models

Basic sender–receiver model: the **sender** encodes the message (thoughts → words, symbols) and transmits it through a **medium/channel**; **noise** (distance, technology, culture, language, distractions) may interfere; the **receiver** decodes it and **acknowledges** receipt; then provides **feedback/response**. The sender is responsible for clear encoding and for confirming the message was understood; the receiver for decoding completely and acknowledging.

Interactive model adds the importance of feedback loops; **cross-cultural communication** recognises that the sender's and receiver's cultures, generations, experiences and emotional states affect understanding.

Gulf of execution and evaluation (Norman): the gap between what a user intends and what the system allows, and between what the system shows and what the user understands — relevant for designing communications and products.

Communication methods

Method	Description	Examples	Best for
Interactive	Real-time, multidirectional exchange	Meetings, phone/video calls, instant messaging, workshops	Immediate understanding, complex or sensitive topics, decisions
Push	Sent to specific recipients; ensures distribution but not understanding	Letters, memos, emails, reports, voicemails, press releases	Targeted information that must reach people
Pull	Recipients access the content at their own discretion	Intranet, wikis, knowledge repositories, e-learning, dashboards	Large volumes of information or large audiences

Other communication dimensions

- **Formal** (reports, briefings, contracts) vs **informal** (emails, ad-hoc conversations).
- **Written** vs **oral**; **verbal** vs **non-verbal** (most of the meaning in face-to-face communication is carried non-verbally).
- **Internal** vs **external**; **official** vs **unofficial**; **hierarchical** (upward, downward) vs **horizontal** (peers).
- **Communication technology** choices depend on urgency, availability, ease of use, project environment (co-located vs virtual), sensitivity and confidentiality.

Communication skills

- **Active listening** — paying full attention, paraphrasing, asking clarifying questions.
- **Feedback** — timely, specific, constructive.
- **Presentations**, **meeting management** (agenda, timebox, facilitation, minutes, action items).
- The **5 Cs of written communication**: correct grammar and spelling, concise expression, clear purpose and expression for the reader, coherent logical flow, controlling flow of words and ideas.

Output: the **communications management plan** — stakeholder communication requirements; information to be communicated (language, format, content, detail); reason for distribution; timeframe and frequency; person responsible; person authorised to release confidential information; recipients; methods and technologies; resources (time and budget); escalation process; glossary; flow charts; constraints (legislation, regulation, technology).

12.7 Manage Communications *(Executing)*

Purpose: ensure timely and appropriate collection, creation, distribution, storage, retrieval, management, monitoring and disposition of project information.

- Uses: communication technology, methods, skills, PMIS, **project reporting** (status, progress, forecasts), interpersonal skills, meetings.
- Output: **project communications** (performance reports, deliverables status, schedule progress, costs incurred, presentations).
- In adaptive teams: **information radiators** (big visible charts, task boards, burn charts), daily stand-ups, sprint reviews, transparent backlogs, team chat channels, **osmotic communication** in co-located teams.

12.8 Monitor Communications *(Monitoring and Controlling)*

Purpose: ensure the information needs of the project and its stakeholders are met. Check whether communications had the desired effect (Did people read and understand? Are stakeholders acting on it?). Techniques: stakeholder engagement assessment matrix, observation, surveys, feedback, meetings. Outputs: work performance information, change requests (e.g., to the communications plan).

> **Trap.** "Send more emails" is rarely the answer to a communication problem. Check **what** the stakeholder needs, **how** they prefer to receive it, and whether the communications plan should change.

12.9 Stakeholders in adaptive and hybrid projects

- The **product owner** represents stakeholders' interests and the business; stakeholders still participate directly in **sprint reviews**, backlog refinement and user research.
- **Frequent demonstrations** align expectations with reality better than documents.
- **Transparency** — anyone can see the backlog, the board and the burn charts.
- In hybrid projects, combine formal reporting for governance bodies with agile transparency for users and the team, and make sure **one consistent story** is told.

Chapter 12 Review Questions

Answer all questions first, then check the answer key for Chapter 12 in Appendix B.

12.1 A project team grows from 8 to 12 members (the project manager is included in these counts). How many additional communication channels are created?

- **A.** 4
- **B.** 28
- **C.** 38
- **D.** 66

12.2 A stakeholder has high power but low interest in the project. According to the power/interest grid, what is the appropriate strategy?

- **A.** Monitor with minimum effort
- **B.** Keep informed
- **C.** Keep satisfied
- **D.** Manage closely

12.3 Which three attributes are used in the salience model to classify stakeholders?

- **A.** Power, urgency and legitimacy
- **B.** Urgency, budget and authority
- **C.** Impact, influence and interest
- **D.** Power, interest and influence

12.4 The stakeholder engagement assessment matrix shows the IT security manager as "Resistant" now and "Supportive" as desired. What should the project manager do next?

- **A.** Escalate to the sponsor to override the manager
- **B.** Plan engagement actions to understand and address the manager's concerns
- **C.** Change the desired level to "Resistant" to close the gap
- **D.** Remove the IT security manager from the stakeholder register, since the manager is resistant

12.5 A project manager needs to share a 300-page technical specification with 400 stakeholders worldwide who will consult it as needed. Which communication method is most appropriate?

- **A.** Interactive communication in a single meeting
- **B.** Push communication by sending the file to everyone by email
- **C.** Informal hallway conversations
- **D.** Pull communication via a shared repository

12.6 An important decision about a scope trade-off must be made by several stakeholders who disagree. Which communication method is best?

- **A.** Interactive communication, such as a facilitated meeting
- **B.** Pull communication, such as a wiki page
- **C.** Push communication, such as a detailed memo sent to everyone involved
- **D.** A monthly newsletter

12.7 Midway through a project, a regional manager the team had never heard of complains that the new process will affect her staff. What should the project manager do first?

- **A.** Submit a change request to remove her staff from scope
- **B.** Ignore the complaint because stakeholder identification ended in initiating
- **C.** Add her to the stakeholder register and analyse her interests
- **D.** Ask the sponsor to tell her the decision is final

12.8 In the basic communication model, who is responsible for confirming that the message was understood correctly?

- **A.** The medium
- **B.** The project sponsor, who is accountable for all communication
- **C.** Only the receiver
- **D.** The sender, by checking for feedback

12.9 Which two items would you expect in a communications management plan? (Choose two.)

- **A.** The contract's price adjustment formula
- **B.** The frequency and format of each communication
- **C.** The quality control measurements
- **D.** The person responsible for communicating each type of information
- **E.** The critical path of the schedule

12.10 A stakeholder complains that she never reads the weekly status email because it is too long. What should the project manager do?

- **A.** Ask the sponsor to require her to read it
- **B.** Discuss her information needs and adjust the communication approach
- **C.** Send the weekly email twice a week instead, so that she is more likely to notice and read it
- **D.** Remove her from the distribution list

12.11 In an agile team, which practice best supports transparent, low-effort communication of progress to anyone who walks by?

- **A.** A password-protected file on the PM's laptop
- **B.** Information radiators such as a visible task board and burndown chart
- **C.** Detailed minutes sent after each meeting
- **D.** A confidential monthly report sent only to the sponsor and the steering committee

12.12 Which direction of influence describes communications with suppliers, regulators and the public?

- **A.** Upward
- **B.** Downward
- **C.** Sideward
- **D.** Outward

12.13 Why should the stakeholder engagement plan sometimes have restricted distribution?

- **A.** It may contain sensitive assessments of stakeholders' attitudes
- **B.** It is the same document as the project charter and must be signed first by the sponsor
- **C.** Agile projects do not allow stakeholders to see plans

- **D.** It contains the project's bank account details

Chapter 13 — Resources Performance Domain (including Procurement)

After this chapter you should be able to:

- describe the five Resources processes and their key outputs;
- plan team and physical resources with RAMs/RACI charts, team charters and resource breakdown structures;
- acquire internal and external resources, including the main steps of conducting a procurement;
- lead the team (development, performance, recognition) — the detailed people skills are in Chapter 16;
- monitor and control physical resources and contracts, including claims and incentive-fee calculations.

13.1 What the Resources domain covers

The Resources domain addresses **identifying, acquiring, leading and managing the people, physical and virtual resources** needed to complete the project. It covers the team *and* equipment, materials, facilities, software and infrastructure, and — together with *Plan Sourcing Strategy* in Governance and Appendix X4 — the acquisition and control of externally sourced resources.

Process	Focus area	Key outputs
Plan Resource Management	Planning	Resource management plan, team charter
Estimate Resources	Planning	Resource requirements, basis of estimates, resource breakdown structure
Acquire Resources	Executing	Physical resource assignments, project team assignments, resource calendars, (for external sourcing) selected sellers and agreements
Lead the Team	Executing	Team performance assessments, change requests, updates (formerly *Develop Team* and *Manage Team*)
Monitor and Control Resourcing	Monitoring and Controlling	Work performance information, change requests (physical/virtual resources, and contracts)

Desired outcomes: the right resources are available at the right time and place; the team is high-performing, engaged and growing; resources are used efficiently and sustainably; suppliers deliver as agreed.

13.2 Plan Resource Management *(Planning)*

Resource management plan contents: identification of resources; how to acquire them; **roles and responsibilities** (role, authority, responsibility, competence); project organisation charts; project team resource management (how to define, staff, manage and release); training; team development methods;

resource control; recognition plan.

Data representation tools

Tool	Description
Hierarchical charts — organisational breakdown structure (OBS), resource breakdown structure (RBS), WBS	Show reporting relationships, resource categories and types
Responsibility assignment matrix (RAM)	Grid linking work packages/activities to people or groups
RACI chart	A RAM with **R**esponsible (does the work), **A**ccountable (owns the outcome — **only one A per task**), **C**onsulted (two-way input), **I**nformed (one-way updates)
Text-oriented formats	Position descriptions, role-responsibility-authority forms

Team charter

A document created **by the team** that establishes values, agreements and operating guidelines: team values, communication guidelines, **decision-making criteria and process**, **conflict resolution process**, meeting guidelines, team agreements (ground rules), working hours across time zones. The team charter is most effective when the team writes it together and revisits it.

13.3 Estimate Resources *(Planning)*

Estimate team resources and the type and quantities of materials, equipment and supplies. Techniques: bottom-up, analogous and parametric estimating, alternatives analysis (in-house vs outsource, manual vs automated, different skill levels), PMIS. Outputs: **resource requirements**, **basis of estimates**, **resource breakdown structure** (hierarchy of resources by category — labour, material, equipment, supplies — and type).

13.4 Acquire Resources *(Executing)*

Purpose: obtain the team members, facilities, equipment, materials, supplies and other resources needed — from inside the organisation or from external sources.

Internal acquisition

- **Pre-assignment** — team members selected in advance (e.g., named in the proposal or charter).
- **Negotiation** — with functional managers, other project managers, external organisations. The PM's influence is critical in matrix organisations.
- **Virtual teams** — expand the talent pool, reduce travel, enable people with mobility limitations or in different time zones; require strong communication planning and technology.
- **Multicriteria decision analysis** — weigh availability, cost, ability, experience, knowledge, skills, attitude, international factors.
- **Halo effect** — assuming someone is good at everything because they are good at one thing (e.g., promoting the best developer to PM). Avoid it in selection.

- If preferred resources are not available, use **alternative resources** (with lower competence or higher cost), document the impact and consider risk.

External acquisition — conducting a procurement

When resources or work are sourced externally, the steps are typically:

1. **Advertise** and distribute **bid documents** (RFI/RFQ/RFP, procurement SOW) to prospective sellers.
2. Hold a **bidder conference** (pre-bid or vendor conference) so all prospective sellers receive the **same information** at the same time — fairness and transparency. Questions and answers are shared with all bidders.
3. **Receive proposals** and evaluate them against the **source selection criteria** using a **weighting system**, **screening** (must-meet criteria), independent estimates, and expert review — often by a proposal evaluation committee.
4. **Negotiate** — clarify structure, requirements and terms; the **procurement/contracts department** usually leads, and the PM supports.
5. **Award** the **agreement** (contract) — a mutually binding legal agreement that obligates the seller to provide the specified products/services and the buyer to pay.

Typical contract components: statement of work or deliverables; schedule and milestones; performance reporting; period of performance; roles and responsibilities; seller's place of performance; pricing and payment terms; inspection, quality and acceptance criteria; **warranty**; product support; **limitation of liability**; fees and **retainage** (amount withheld until completion); **penalties** and **liquidated damages** (pre-agreed compensation for specific breaches such as late delivery); **incentives**; insurance and performance bonds; subcontractor approvals; **change request handling**; **termination clause** and **alternative dispute resolution (ADR)** mechanisms; **force majeure** (unforeseeable circumstances that excuse non-performance); **indemnification**; confidentiality and **intellectual property** provisions.

> **Exam tip.** A contract is a **legal** document: changes to it must follow the **contract change control system** and usually require the authority of the procurement/contracts administrator. The PM cannot change a contract by informal agreement with the seller's project manager. Also, the **buyer's PM should not direct the seller's employees** as if they were their own staff — work through the contract and the seller's management.

13.5 Lead the Team *(Executing)*

This process merges the former *Develop Team* (improving competencies, interaction and environment) and *Manage Team* (tracking performance, providing feedback, resolving issues, managing changes). Key ideas — expanded in Chapter 16:

- **Team development** — training, team-building, recognition and rewards, co-location (war rooms, *tight matrix*) or effective virtual collaboration, communication technology, individual and team assessments (personality and style assessments, 360° feedback).
- **Stages of team development (Tuckman):** forming, storming, norming, performing, adjourning.
- **Motivation** — intrinsic and extrinsic, autonomy-mastery-purpose; recognition should be **timely**, **specific** and appropriate to the person and culture; win-lose (zero-sum) rewards that only a few can earn can hurt team cohesion.

- **Team performance assessments** — evaluate improvements in skills, competencies, team behaviour, turnover rate, cohesiveness.
- **Conflict management**, **decision-making**, **emotional intelligence**, **influencing**, **leadership**.
- **Self-organising teams** in adaptive contexts; the leader is a **servant leader** who removes impediments, shields the team, coaches and facilitates.
- **Performance problems** — address them **privately**, early, with specific feedback, focusing on behaviour and support; involve the functional manager when appropriate (it may affect their appraisal).

13.6 Monitor and Control Resourcing *(Monitoring and Controlling)*

Purpose: ensure that the **physical and virtual resources** (equipment, materials, facilities, software, infrastructure) assigned and allocated are available as planned, monitor planned vs actual use, and take corrective action. It also covers monitoring **externally sourced** resources and contract performance. (Team performance itself is handled through *Lead the Team*.)

- Techniques: data analysis (alternatives, cost-benefit, performance reviews, trend analysis), problem solving, negotiation, influencing, PMIS.
- Physical resource issues: late deliveries, wrong quantities, defective materials, equipment failures, scarcity, price increases, environmental waste.
- **Just-in-time (JIT)** inventory reduces holding costs but requires reliable suppliers.

Controlling contracts (procurement control)

- **Contract administration** — manage the relationship, monitor performance against the agreement, approve payments, manage changes.
- **Performance reviews** and **inspections/audits** — verify the seller's work and processes.
- **Earned value analysis** of seller performance; **trend analysis**.
- **Claims administration** — a **claim** (dispute, appeal) arises when buyer and seller cannot agree on compensation for a change or on whether a change occurred. Preferred resolution order: **negotiation** → **alternative dispute resolution (ADR)** such as **mediation** (a neutral third party helps the parties agree) or **arbitration** (a neutral third party decides; often binding) → **litigation** (courts) only as a last resort.
- **Records management** — maintain complete records of contract and procurement activities (they may be needed in disputes or audits).
- **Payment systems** — pay according to contract terms, upon verification of work.
- **Closing a contract** — see Chapter 7: verify deliverables accepted, settle claims, final payment, formal written notice of completion, archive records, lessons learned. A contract may also be terminated early **for cause** (breach) or **for convenience** (as allowed by the termination clause).

> **Trap.** If a seller is not performing, the first step is usually to **review the contract and the seller's performance data, then discuss with the seller** (through the formal channel) — not to terminate the contract or withhold all payments immediately.

13.7 Incentive contracts — calculations you may see

Cost plus incentive fee (CPIF) example: target cost $100,000; target fee 10,000; *shareratio* * * 80/20 * * (*buyer*/*seller*); *actualcost* * *90,000**.

- Underrun = $10,000; seller's share = 20% × 10,000 = $2,000.
- Fee = 10,000 + 2,000 = **12,000 * *;*totalpricepaid* = 90,000 + 12,000 = * *102,000**.
- If actual cost were $110,000: overrun 10,000; *seller'sshare*−2,000; fee $8,000; total $118,000.

Fixed price incentive fee (FPIF) example: target cost $200,000; target profit $20,000 (target price $220,000); share ratio 70/30; **ceiling price $240,000**; actual cost $210,000.

- Overrun $10,000; seller absorbs 30% = $3,000; profit = 20,000 − 3,000 = 17,000; *price* = 210,000 + 17,000 = * *227,000** (below the ceiling, so payable).
- **Point of total assumption (PTA)** = ((ceiling price − target price) ÷ buyer's share ratio) + target cost = ((240,000 − 220,000) ÷ 0.70) + 200,000 ≈ **$228,571**. Above this cost, the seller bears **all** additional costs.

13.8 Resources in adaptive and hybrid projects

- Small (often 3–9 people), stable, **dedicated**, **cross-functional** teams; **T-shaped** people (deep in one skill, broad across others) and generalising specialists reduce bottlenecks.
- Teams are **self-organising**; work is pulled, not assigned.
- Co-location or strong virtual collaboration practices; **sustainable pace**.
- Vendors may join as part of blended teams under agile-friendly contracts (Chapter 7).
- In hybrid projects, physical resources for predictive components (heavy equipment, materials) are planned and controlled in the traditional way, while team capacity for adaptive components is managed through velocity and capacity planning.

Chapter 13 Review Questions

Answer all questions first, then check the answer key for Chapter 13 in Appendix B.

13.1 In a RACI chart, how many people should be Accountable for a single activity?

- **A.** Everyone who is Responsible
- **B.** At least two, for redundancy
- **C.** None; accountability belongs to the team
- **D.** Exactly one

13.2 Who should create the team charter?

- **A.** The functional managers
- **B.** The team members together, facilitated by the project manager
- **C.** The sponsor alone
- **D.** The PMO, as a standard template that the team is not allowed to change

13.3 During procurement, one prospective seller calls the project manager privately asking for clarification of a requirement. What should the project manager do?

- **A.** Disqualify the seller immediately
- **B.** Answer through the formal process so that all sellers receive the same information
- **C.** Ignore the question
- **D.** Answer privately by phone, to help the seller prepare a better and more accurate proposal

13.4 A seller and buyer cannot agree on compensation for a change the seller says was requested. Direct negotiation has failed. What is generally the preferred next step?

- **A.** Termination of the contract for cause, followed by a new procurement
- **B.** Stopping all payments to the seller
- **C.** Alternative dispute resolution, such as mediation or arbitration
- **D.** Immediate litigation

13.5 CPIF contract: target cost $200,000, target fee $15,000, share ratio 80/20 (buyer/seller), actual cost $180,000. What is the total amount the buyer pays, in dollars?

(Fill in the blank.)

Answer: _____

13.6 A team member performs poorly on several tasks. What should the project manager do first?

- **A.** Replace the team member
- **B.** Report the team member to the functional manager immediately, with evidence of the poor work
- **C.** Discuss it privately with the team member to understand the causes
- **D.** Raise the issue in the next team meeting

13.7 Which process is primarily concerned with ensuring that equipment, materials and facilities are available as planned and used as intended?

- **A.** Monitor and Control Resourcing
- **B.** Lead the Team
- **C.** Manage Project Knowledge
- **D.** Estimate Resources

13.8 The project manager learns that the seller's lead engineer is working inefficiently. The buyer's PM wants to instruct the engineer directly on how to work. What is the best approach?

- **A.** Raise the concern through the contract channels with the seller's management
- **B.** Withhold payment until the engineer is replaced
- **C.** Give the engineer direct daily instructions, as if the engineer were part of the internal team
- **D.** Terminate the contract for convenience

13.9 What is the purpose of a bidder conference?

- **A.** To give all prospective sellers the same, clear understanding of the procurement
- **B.** To close the procurement
- **C.** To negotiate the final price and the detailed contract terms with the preferred seller
- **D.** To select the winning seller

13.10 Which two items are typically included in a team charter? (Choose two.)

- **A.** Conflict resolution process
- **B.** Risk probability and impact scales
- **C.** Procurement statement of work
- **D.** Decision-making process
- **E.** Earned value thresholds

13.11 The best software developer is promoted to project manager because "if she is great at coding, she will be great at managing". This reasoning reflects:

- **A.** Resource smoothing
- **B.** Expectancy theory
- **C.** Pre-assignment
- **D.** The halo effect

13.12 FPIF contract: target cost $100,000, target profit $10,000, target price $110,000, ceiling price $125,000, buyer/seller share ratio 75/25. What is the point of total assumption?

- **A.** $110,000
- **B.** $115,000
- **C.** $120,000
- **D.** $125,000

13.13 In a contract, what is retainage?

- **A.** A bonus paid for early delivery
- **B.** An amount withheld from each payment until the work is completed satisfactorily
- **C.** The seller's profit margin
- **D.** The penalty for late delivery, agreed in advance and deducted from each of the payments

Chapter 14 — Risk Performance Domain

After this chapter you should be able to:

- distinguish individual risks, overall project risk, issues and uncertainty;
- describe the six Risk processes and their outputs (risk management plan, risk register, risk report);
- analyse risks qualitatively and quantitatively (probability-impact, EMV, decision trees, Monte Carlo, sensitivity);
- select response strategies for threats and opportunities, including contingency and fallback plans;
- monitor risks and manage reserves in predictive and adaptive contexts.

14.1 Key concepts

Term	Meaning
Risk	An uncertain event or condition that, if it occurs, has a **positive or negative** effect on one or more objectives
Threat	A risk with a negative effect
Opportunity	A risk with a positive effect
Individual project risk	A specific uncertain event or condition
Overall project risk	The effect of uncertainty on the project **as a whole** — from all sources, including individual risks — representing the exposure of stakeholders to variations in outcome
Issue	A current condition or situation that is **already happening** and must be resolved (logged in the **issue log**)
Risk appetite	The degree of uncertainty an organisation or individual is willing to accept in anticipation of a reward
Risk threshold	The measure of acceptable variation around an objective (e.g., cost overrun > 5% is unacceptable) — reflects appetite
Risk tolerance	Older term close to threshold: the degree of risk that can be withstood
Known-unknowns	Identified risks → contingency reserve
Unknown-unknowns	Unidentified risks → management reserve
Residual risk	Risk remaining after responses are implemented
Secondary risk	A new risk that arises **as a direct result** of implementing a risk response
Trigger condition	An event or situation indicating a risk is about to occur
Workaround	An **unplanned** response to a risk that has occurred (or to an issue) — not planned in advance
Contingency plan	A planned response executed if a trigger occurs

Term	Meaning
Fallback plan	A plan B used if the contingency plan (primary response) is not effective
Risk owner	The person responsible for monitoring a risk and implementing its response

Other forms of uncertainty the Guide asks you to recognise: **ambiguity** (unclear situation — use expert judgment, experiments, progressive elaboration), **complexity** (many interacting parts — use systems thinking, simulation, decoupling) and **volatility** (rapid, unpredictable change — use alternatives analysis, reserves). The acronym **VUCA** (volatility, uncertainty, complexity, ambiguity) summarises them. **Resilience** — the ability to absorb and recover from impacts — is built through reserves, flexibility, redundancy and learning.

14.2 Plan Risk Management *(Planning)*

Risk management plan contents: risk strategy; methodology; roles and responsibilities; funding (reserves); timing (when and how often risk activities happen); **risk categories** (often a **risk breakdown structure, RBS** — e.g., technical, management, commercial, external); **stakeholder risk appetite**; **definitions of risk probability and impact** (scales); **probability and impact matrix** (with rules for high/medium/low); reporting formats; tracking.

14.3 Identify Risks *(Planning)*

Purpose: identify individual risks and sources of overall project risk, and document their characteristics. Identification is **iterative** — everyone on the team should be encouraged to identify risks at any time.

Techniques: expert judgment; **brainstorming**; **checklists** (from historical information — useful but never exhaustive); **interviews**; **root-cause analysis**; **assumption and constraint analysis** (inaccurate, unstable or incomplete assumptions are risks); **SWOT analysis** (strengths, weaknesses, opportunities, threats); **document analysis**; **prompt lists** — e.g., **PESTLE** (political, economic, social, technological, legal, environmental), **TECOP** (technical, environmental, commercial, operational, political), **VUCA**; facilitation; meetings (risk workshops); **Delphi technique** (anonymous expert rounds to reach consensus, reducing bias).

Risk statement (metalanguage): *"Because of , may occur, which would lead to ."* Example: "Because the vendor is a new supplier, delivery of the servers may be late, which would delay testing by three weeks."

Outputs: the **risk register** (list of identified risks, potential risk owners, potential responses, later: causes, categories, probability, impact, priority, triggers, responses, residual and secondary risks) and the **risk report** (sources of overall project risk and summary information on individual risks — updated throughout).

14.4 Perform Risk Analysis *(Planning)*

In the Eighth Edition this single process covers both **qualitative** and **quantitative** analysis.

Qualitative analysis (always performed; fast, subjective)

- **Probability and impact assessment** on defined scales (e.g., 1–5 or very low to very high).

- **Probability and impact matrix** — combine both to rank risks (high, medium, low).
- **Assessment of other risk parameters: urgency** (how soon a response is needed), **proximity** (how soon the risk could affect objectives), **dormancy** (time between occurrence and discovery of its impact), **manageability**, **controllability**, **detectability**, **connectivity** (relationships with other risks), **strategic impact**, **propinquity** (how much the risk matters to stakeholders).
- **Risk data quality assessment** — how accurate and reliable is the data about the risk?
- **Risk categorisation** — by source (RBS), affected WBS element, phase — reveals common root causes.
- **Data representation** — heat maps, hierarchical charts (bubble charts showing three parameters).
- Output: updated risk register (priorities, owners), assumption log, issue log, risk report.

Quantitative analysis (only when justified — high-value, complex, high-risk projects)

Technique	Use
Monte Carlo simulation	Run thousands of iterations of the schedule or cost model using probability distributions → probability of meeting targets (S-curve; e.g., "P80 completion date is 15 March")
Sensitivity analysis (tornado diagram)	Which individual risks or uncertain variables have the **most potential impact** on outcomes (longest bar at the top)
Decision tree analysis with **expected monetary value (EMV)**	Choose the best alternative when outcomes are uncertain
Influence diagrams	Graphical representation of decision situations, uncertainties and outcomes
Criticality analysis	How often an activity appears on the critical path across simulation iterations

Expected monetary value (EMV) = probability × impact. Opportunities have **positive** values, threats **negative**. The EMV of a set of risks is the sum.

Example: Risk A: 30% chance of a $50,000 *delay penalty* → *EMV* = −15,000. Opportunity B: 20% chance of a $40,000 *early – completion bonus* → *EMV* = +8,000. Net EMV = **−$7,000** → a contingency reserve of about $7,000 could be justified for these two items.

Decision tree example: Option 1 — build in-house: cost $200,000; 40% chance of an extra $60,000 *problem*. *Expected cost* = $200,000 + (0.4×60,000) = **$224,000**. Option 2 — buy: cost $230,000; 10% chance of an extra $20,000. *Expected cost* = $230,000 + $2,000 = **$232,000**. → Choose **build** (lower expected cost).

14.5 Plan Risk Responses *(Planning)*

Strategies for threats

Strategy	Description	Example
Escalate	The threat is **outside the project's scope** or the PM's authority; pass it to the appropriate owner (program, portfolio, other part of the organisation). Once escalated, it is no longer monitored by the project (though it may be noted)	A risk to the company's market position spotted by the project

Strategy	Description	Example
Avoid	Eliminate the threat or protect the project from its impact (change the plan, extend the schedule, reduce scope, use a proven approach)	Drop an unproven technology
Transfer	Shift ownership and impact to a **third party** (insurance, warranties, performance bonds, fixed-price contracts, outsourcing). Usually involves paying a premium. The risk is not eliminated	Buy insurance
Mitigate	**Reduce probability and/or impact** to an acceptable level	Extra testing, prototypes, redundant suppliers
Accept	Acknowledge the threat without proactive action. **Active acceptance** = establish a contingency reserve (time, money, resources); **passive acceptance** = no action other than periodic review	Accept small weather delays with a schedule reserve

Strategies for opportunities

Strategy	Description	Example
Escalate	Opportunity outside the project's scope or authority → pass to the right owner	Savings for the whole company
Exploit	Make sure the opportunity **definitely happens** (raise probability to 100%)	Assign the most talented resources to finish early and earn a bonus
Share	Allocate ownership to a third party best able to capture the benefit (joint ventures, partnerships, teaming agreements)	Partner with a firm that has a complementary technology
Enhance	**Increase probability and/or impact**	Add resources to an activity to finish early
Accept	Take advantage if it occurs, without pursuing it actively	—

Overall project risk responses

Avoid (e.g., cancel or re-scope), exploit, transfer/share, mitigate/enhance, accept (active: overall contingency reserve).

Other response elements

- **Contingent response strategies** — designed for use only if certain events occur (with **triggers**).
- **Fallback plans** — if the primary response fails.
- **Risk owner** for each risk, and **risk action owners** for specific actions.
- **Contingency reserves** (time and cost) calculated with EMV, simulation or percentages.
- **Secondary risks** and **residual risks** documented.
- **Cost-benefit analysis** of responses — a response should not cost more than the risk it addresses (unless required by compliance or values).

14.6 Implement Risk Responses *(Executing)*

Purpose: implement the agreed risk response plans. A common failure is that responses are planned but **never executed** — risk owners must act. Outputs: change requests, updates to the risk register and risk report, issue log, lessons learned.

14.7 Monitor Risks *(Monitoring and Controlling)*

Purpose: monitor implementation of responses, track identified risks, identify new risks, evaluate risk process effectiveness.

- **Risk reassessment** — regularly review existing risks, close outdated ones, identify new ones.
- **Risk audits** — examine the effectiveness of the risk management **process** and of responses.
- **Reserve analysis** — compare remaining reserves with remaining risk; release reserves when risks pass.
- **Technical performance analysis** — compare technical achievements with the plan (e.g., weight, transaction speed).
- **Trend and variance analysis**, **meetings** (risk reviews).
- When a **risk occurs**, implement the planned response; it becomes an **issue** to manage. If no response was planned, apply a **workaround** and then update the risk register and lessons learned.

> **Exam tip.** A trigger for an identified risk occurs → **implement the contingency plan** defined in the risk register (involving the risk owner). An unidentified problem appears → it is an **issue**: manage it (workaround, issue log, possibly a change request) and then **identify any new risks**.

14.8 Risk in adaptive and hybrid projects

- **Short iterations, frequent delivery and feedback** reduce risk by validating assumptions early.
- **Risk-adjusted backlog** — risk responses become backlog items that the product owner prioritises alongside features; high-risk items are often tackled early.
- **Spikes** — timeboxed research or experiments to reduce technical or design uncertainty.
- **Daily stand-ups and retrospectives** surface risks and impediments continuously; information radiators make them visible.
- **Cross-functional teams** and **sustainable pace** reduce people-related risks.
- **Incremental funding** limits financial exposure.
- In hybrid projects, maintain an integrated risk register across predictive and adaptive components, and watch the **interfaces** between them.

Chapter 14 Review Questions

Answer all questions first, then check the answer key for Chapter 14 in Appendix B.

14.1 A risk has a 25% probability of causing a $80,000 loss. An opportunity has a 40% probability of producing a $30,000 saving. What is the combined expected monetary value?

- **A.** +$12,000
- **B.** −$32,000
- **C.** −$20,000
- **D.** −$8,000

14.2 The project team buys insurance against fire damage to the construction site. Which risk response strategy is this?

- **A.** Transfer

- **B.** Mitigate
- **C.** Accept
- **D.** Avoid

14.3 A team assigns its most experienced engineers to a module to make sure it is completed early, guaranteeing a client bonus. Which opportunity strategy is this?

- **A.** Enhance
- **B.** Exploit
- **C.** Accept
- **D.** Share

14.4 A risk trigger for a previously identified risk has occurred. What should the project manager do?

- **A.** Close the risk in the register
- **B.** Begin identifying risks for the first time
- **C.** Perform a new Monte Carlo simulation of the whole project before doing anything else
- **D.** Ensure the risk owner implements the planned contingency response

14.5 An unexpected problem occurs that was never identified as a risk. The team applies an unplanned fix to keep working. What is this fix called?

- **A.** Risk mitigation
- **B.** A fallback plan
- **C.** A workaround
- **D.** A contingency plan

14.6 Which technique would best show which uncertain variables have the greatest effect on the project's total cost?

- **A.** Sensitivity analysis displayed as a tornado diagram
- **B.** A probability and impact matrix of the individual risks
- **C.** Risk breakdown structure
- **D.** Checklist analysis

14.7 The project team identifies a risk that a new government policy could reduce the organisation's overall market share — far beyond the project's objectives. What should the project manager do?

- **A.** Accept the risk passively and ignore it
- **B.** Transfer it by buying insurance
- **C.** Escalate the risk to the appropriate owner in the organisation
- **D.** Mitigate the risk using the project's own contingency reserve and resources

14.8 After implementing a response to reduce the chance of a server failure, a small chance of failure remains. What is this remaining risk called?

- **A.** Secondary risk
- **B.** Residual risk
- **C.** Trigger
- **D.** Issue

14.9 Option X has an initial cost of $100,000 and a 30% chance of an additional $50,000 cost. Option Y has an initial cost of $110,000 and a 20% chance of an additional $20,000 cost. Based on expected cost, which option should be chosen?

- **A.** Option X, expected cost $115,000
- **B.** Option X, expected cost $100,000
- **C.** Option Y, expected cost $130,000
- **D.** Option Y, expected cost $114,000

14.10 Which two practices help manage risk in adaptive projects? (Choose two.)

- **A.** Postponing all risk discussions to the final retrospective
- **B.** Including risk response activities in the backlog and prioritising them with features
- **C.** Eliminating demonstrations to save time
- **D.** Using timeboxed spikes to reduce technical uncertainty
- **E.** Freezing the backlog for the whole release

14.11 What is the main purpose of a risk audit?

- **A.** To calculate the project's critical path
- **B.** To evaluate how effective the risk process and responses are
- **C.** To transfer risks to the auditor
- **D.** To identify every possible risk on the project for the first time, with the auditors

14.12 Which statement best describes risk appetite and risk threshold?

- **A.** Appetite is the uncertainty accepted for reward; thresholds are measurable limits reflecting it
- **B.** They are identical terms
- **C.** Thresholds are set by the team alone without any stakeholder input, and the organisation's appetite is irrelevant
- **D.** Appetite applies only to opportunities and thresholds only to threats

14.13 A Monte Carlo simulation shows a 50% probability of finishing by 1 May and an 85% probability of finishing by 20 May. The sponsor wants high confidence. What should the project manager communicate?

- **A.** The schedule must be crashed or fast-tracked until the 1 May date has a confidence level of 100%
- **B.** The simulation is irrelevant because it is probabilistic
- **C.** The project will certainly finish on 1 May
- **D.** A target near 20 May gives 85% confidence; 1 May leaves a 50% chance of being late

14.14 In the Eighth Edition, which process covers both qualitative and quantitative risk analysis?

- **A.** Monitor Risks
- **B.** Plan Risk Management
- **C.** Perform Risk Analysis
- **D.** Identify Risks

Chapter 15 — Agile and Hybrid Delivery in Depth

After this chapter you should be able to:

- state the four values and twelve principles of the Agile Manifesto and apply them to scenarios;
- explain Scrum roles, events, artifacts and commitments, and the role of a servant leader;
- apply Kanban, Lean and XP practices;
- recognise scaling frameworks and their key ideas;
- estimate, plan, measure and contract in agile and hybrid environments;
- choose the "agile answer" in exam scenarios.

About 60% of PMP questions are agile or hybrid. This chapter collects the concepts most frequently tested.

15.1 The Agile Manifesto (2001)

Four values — *we value the items on the left more than the items on the right* (the right still has value):

1. **Individuals and interactions** over processes and tools
2. **Working software** (working product) over comprehensive documentation
3. **Customer collaboration** over contract negotiation
4. **Responding to change** over following a plan

Twelve principles (paraphrased):

1. Highest priority: satisfy the customer through **early and continuous delivery** of valuable software.
2. **Welcome changing requirements**, even late; harness change for the customer's competitive advantage.
3. **Deliver working software frequently**, from a couple of weeks to a couple of months, preferring the shorter timescale.
4. Business people and developers **work together daily**.
5. Build projects around **motivated individuals**; give them the environment and support they need and **trust** them.
6. **Face-to-face conversation** is the most efficient and effective method of conveying information.
7. **Working software is the primary measure of progress.**
8. Agile processes promote **sustainable development** — a constant pace indefinitely.
9. Continuous attention to **technical excellence and good design** enhances agility.
10. **Simplicity** — the art of maximising the amount of work *not* done — is essential.
11. The best architectures, requirements and designs emerge from **self-organising teams**.
12. At regular intervals the team **reflects** on how to become more effective and adjusts its behaviour.

Agile mindset: being agile (values and principles) matters more than doing agile (practices). Agile is an umbrella term for many approaches; **Lean** thinking is a superset that includes agile and Kanban.

15.2 Scrum

Scrum is a lightweight framework for developing complex products, based on **empiricism** (transparency, inspection, adaptation) and **lean thinking**. Scrum values: **commitment, focus, openness, respect, courage**.

Scrum team (accountabilities)

Role	Accountabilities
Product owner (PO)	Maximise the value of the product; develop and communicate the **product goal**; create, **order** and clearly communicate **product backlog** items; **accept** or reject work; the single voice of the stakeholders for the team. One person, not a committee
Scrum master (SM)	**Servant leader** and coach: helps everyone understand Scrum theory and practice; facilitates events as needed; **removes impediments**; shields the team from interruptions; helps the PO with backlog techniques; helps the organisation adopt Scrum
Developers	Everyone who creates the increment (any discipline); **self-managing**: decide **how** to do the work; create the **sprint backlog**; hold each other accountable; adhere to the **definition of done**

A Scrum team is typically **10 or fewer** people, cross-functional, with no sub-teams or hierarchies.

Events (all timeboxed)

Event	Timebox (for a one-month sprint; shorter for shorter sprints)	Purpose
Sprint	1 month or less (often 2 weeks); a container for all events	Create a valuable, useful "done" increment. No changes that endanger the **sprint goal**; quality does not decrease; scope may be clarified and renegotiated with the PO
Sprint planning	Max 8 hours	*Why* (sprint goal), *what* (selected items), *how* (plan) — produces the sprint backlog
Daily scrum	15 minutes, same time and place	Developers inspect progress toward the sprint goal and adapt the plan. **Not** a status meeting for the PM. Detailed problem-solving happens after
Sprint review	Max 4 hours	Inspect the increment with **stakeholders**, gather feedback, adapt the product backlog — a working session, not just a demo
Sprint retrospective	Max 3 hours	Inspect how the last sprint went (people, interactions, processes, tools, DoD) and plan **improvements**, which may be added to the next sprint backlog

Sprint cancellation: only the **product owner** can cancel a sprint, typically when the sprint goal becomes obsolete.

Backlog refinement is an ongoing activity (often ~10% of capacity) to break down, clarify, estimate and order items.

Artifacts and their commitments

Artifact	Commitment	Description
Product backlog	**Product goal**	Emergent, ordered list of what is needed to improve the product
Sprint backlog	**Sprint goal**	Selected items + the plan for delivering them; owned by developers
Increment	**Definition of done**	A concrete stepping stone toward the product goal; must be usable; multiple increments may be created in a sprint

15.3 Kanban

Kanban (Japanese: "visual signal") is a method for managing flow of work. Core practices:

1. **Visualise the workflow** (Kanban board with columns such as To do / In progress / Review / Done).
2. **Limit work in progress (WIP)** — WIP limits per column create a **pull system** and expose bottlenecks.
3. **Manage flow** — measure and improve **lead time**, **cycle time** and **throughput** (cumulative flow diagrams).
4. **Make policies explicit** (e.g., definition of done per column).
5. **Implement feedback loops** (replenishment meetings, reviews).
6. **Improve collaboratively, evolve experimentally.**

Kanban has **no prescribed roles or iterations**; it starts with what you do now and evolves. It suits support/operations work with continuous incoming requests. **Scrumban** combines Scrum events with Kanban flow and WIP limits.

15.4 Lean

Lean thinking (from the Toyota Production System) focuses on **maximising customer value while minimising waste**. Principles: identify value, map the **value stream**, create **flow**, establish **pull**, pursue **perfection** (continuous improvement).

Seven wastes in software development (Poppendieck): partially done work, extra features (gold plating), relearning, handoffs, task switching, delays, defects. (Manufacturing equivalents: inventory, overproduction, extra processing, transportation, motion, waiting, defects — plus unused talent.)

Value stream mapping — visualise every step from request to delivery, distinguishing value-adding time from waiting time, to eliminate delays and waste.

15.5 Extreme Programming (XP)

XP focuses on technical excellence and frequent releases. Practices: **pair programming**, **test-driven development (TDD)** — write the test first (red–green–refactor), **continuous integration**, **refactoring**, **simple design**, **collective code ownership**, **coding standards**, **small releases**, **sustainable pace** (40-hour week), **whole team** (customer on-site), **planning game**, **system metaphor**. XP roles include coach, customer, programmer, tracker, tester. XP values: communication, simplicity, feedback, courage, respect.

15.6 Other approaches and scaling frameworks

Approach	Key ideas
Feature-driven development (FDD)	Develop an overall model, build a features list, plan/design/build **by feature**; chief programmer, class owners
DSDM (Dynamic Systems Development Method)	Fixes **time, cost and quality**; scope varies (**MoSCoW**); eight principles; strong governance — popular for agile with corporate rigor
Crystal	A family of methodologies sized by team size and criticality (Clear, Yellow, Orange, Red...); emphasises people and communication
Disciplined Agile (DA)	PMI's toolkit for choosing your "way of working" (WoW); context-sensitive; principles such as *delight customers*, *be awesome*, *context counts*, *choice is good*, *optimise flow*, *enterprise awareness*
SAFe® (Scaled Agile Framework)	Organises many teams into **Agile Release Trains (ARTs)** of 50–125 people; **program increment (PI) planning** (typically every 8–12 weeks); roles such as Release Train Engineer; portfolio, large solution and essential configurations
Scrum of Scrums	Representatives of each team meet to coordinate dependencies and impediments across teams
LeSS (Large-Scale Scrum)	Many teams, **one product owner and one product backlog**; minimal extra roles
Nexus	Scrum.org's framework for 3–9 Scrum teams with a Nexus integration team
Enterprise Scrum / Scrum@Scale	Scaling Scrum principles to the enterprise

15.7 Estimating and planning in agile

- **Relative estimating** with **story points** (size, complexity, uncertainty) — faster and more consistent than absolute hours.
- **Planning poker** (wideband Delphi): each estimator reveals a card simultaneously; discuss the highest and lowest; re-vote until convergence — avoids anchoring.
- **Affinity estimating** and **T-shirt sizing** for many items or early, coarse estimates.
- **Ideal days**, **#NoEstimates** (count items, use throughput).
- **Velocity** = points completed per iteration (only "done" items count); used to forecast; **not** a performance target.
- **Capacity planning** — adjust for holidays and availability.
- **Levels of planning (the planning onion)**: strategy → portfolio → product (vision, roadmap) → release → iteration → day.
- **Iteration zero / sprint zero** — set-up work (environment, architecture runway, team formation). Use with care: it should still deliver some value.
- **Spike** — a timeboxed research activity to answer a question or reduce risk; an **architectural spike** explores technical approach.

15.8 Agile metrics and visual management

Metric / chart	Use	Caution
Velocity	Forecasting for the team	Not for comparing teams or evaluating individuals
Burndown	Remaining work in the sprint/release	Hides scope changes
Burnup	Completed work vs total scope	Shows scope growth
Cumulative flow diagram	Flow, WIP and bottlenecks	—
Lead time / cycle time / throughput	Predictability and speed	Reduce WIP to improve
Escaped defects, **technical debt**	Quality	—
Business value delivered, **customer satisfaction (NPS)**, **feature usage**	Outcomes	The most important measures
Team happiness / health checks	Sustainability	—

Information radiators (big visible charts) make these transparent. **Earned value can be adapted** to agile by measuring planned vs completed story points against release budgets.

15.9 Servant leadership and the agile project manager

Servant leaders put the needs of the team first. Their behaviours:

- **Remove impediments** (or empower the team to remove them);
- **Shield the team** from interruptions and diversions;
- **Facilitate** collaboration, events and decisions — rather than direct;
- **Coach and mentor**; grow the team's capability; encourage self-organisation;
- **Promote transparency** and **psychological safety**;
- **Communicate the vision** and keep the team focused on value.

The project manager's role in agile environments may be to support the team, coordinate across teams and with the wider organisation, manage budgets and contracts, handle governance, and act as a servant leader — but **not** to assign tasks or make the product owner's decisions.

15.10 Agile contracts

Recap from Chapter 7: **fixed-price increments**, **not-to-exceed T&M**, **graduated fixed price**, **early cancellation option** (customer can stop when enough value is delivered, paying a fee), **dynamic scope option** (swap features of equal size), **team augmentation**, **multi-tiered structure** (master service agreement + statements of work for each release), **capacity-based** contracts. The agile value *customer collaboration over contract negotiation* means the contract should enable collaboration, not replace it.

15.11 Hybrid delivery — making it work

- Agree on **which parts** of the project use which approach, and why (tailoring).
- Integrate plans: **milestones** from the predictive plan linked to **releases** from the adaptive plan.
- Synchronise at **integration points** (e.g., hardware ready for software integration in sprint 6).
- Use **consistent governance and reporting** (e.g., a combined dashboard: milestone status + burnup + risks).
- Manage **dependencies** between components explicitly (dependency boards, Scrum of Scrums).
- Educate stakeholders about the differences (e.g., why the adaptive scope is flexible while the date is fixed).

15.12 The "agile answer" — common exam patterns

Scenario	Usually the best answer
The team is in conflict over a technical approach	Let the team discuss and decide; facilitate if needed
A stakeholder asks a developer directly for a new feature	Redirect the request to the **product owner** for the backlog
The PO is not available to answer questions	Coach the PO on the importance of availability; agree on time slots; escalate to management only if it persists
A team member keeps talking about solutions in the daily scrum	Ask to take the discussion offline after the stand-up
Velocity dropped	Explore causes with the team in the **retrospective**; don't push for more points
Management wants the team to commit to more work	Explain capacity and velocity; protect sustainable pace; negotiate scope with the PO
Defects escaping to production	Strengthen the **definition of done**, automation and TDD; discuss in the retrospective
New team just formed	Facilitate a **team charter**/working agreements; coach Scrum practices
The team wants to change its process	Support experiments agreed in the retrospective
The organisation demands detailed upfront plans for an agile project	Educate on adaptive planning; provide roadmap/release plans and transparency; tailor reporting

Chapter 15 Review Questions

Answer all questions first, then check the answer key for Chapter 15 in Appendix B.

15.1 Which statement reflects the Agile Manifesto value "Working software over comprehensive documentation"?

- **A.** Documentation is forbidden in agile projects
- **B.** Working software is optional, as long as the documentation is complete and approved

- **C.** Only the product owner writes documentation
- **D.** Documentation still has value, but delivering a working product is valued more

15.2 Who is accountable for ordering the product backlog and maximising the value of the product in Scrum?

- **A.** The scrum master
- **B.** The developers
- **C.** The project manager
- **D.** The product owner

15.3 During the daily scrum, two developers start a detailed technical debate. What should the scrum master do?

- **A.** Extend the daily scrum until the technical debate has been fully resolved by the two developers
- **B.** Suggest they continue after the daily scrum so the event stays focused
- **C.** Decide the technical approach for them
- **D.** Cancel the daily scrum for the rest of the sprint

15.4 Who can cancel a sprint in Scrum?

- **A.** The product owner
- **B.** The scrum master
- **C.** Any developer
- **D.** The sponsor, at any time without the product owner

15.5 In Kanban, what is the main purpose of a work-in-progress (WIP) limit?

- **A.** To ensure that every team member is always busy with at least one work item
- **B.** To fix the length of each iteration
- **C.** To create a pull system that improves flow and exposes bottlenecks
- **D.** To estimate story points

15.6 Which XP practice involves writing an automated test before writing the code that makes it pass?

- **A.** Test-driven development
- **B.** Refactoring
- **C.** Pair programming
- **D.** Continuous integration

15.7 A large organisation coordinates eight agile teams working on one product with one product owner and one product backlog, adding as few extra roles as possible. Which framework does this describe?

- **A.** Kanban
- **B.** LeSS
- **C.** Crystal Clear
- **D.** Feature-driven development

15.8 Which two behaviours are typical of a servant leader? (Choose two.)

- **A.** Removing impediments that block the team
- **B.** Reporting individual velocity to managers
- **C.** Shielding the team from interruptions so they can focus

- **D.** Assigning tasks to each developer every morning
- **E.** Deciding the technical design for the team

15.9 Which Lean waste is created when developers frequently switch between several projects at once?

- **A.** Task switching
- **B.** Handoffs
- **C.** Relearning
- **D.** Extra features

15.10 In DSDM, which elements are typically fixed while scope varies?

- **A.** Scope and cost
- **B.** Scope and time
- **C.** Time, cost and quality
- **D.** Scope, quality and team

15.11 The sprint review is best described as:

- **A.** A status meeting for the project manager
- **B.** A working session where stakeholders inspect the increment and adapt the backlog
- **C.** The formal signing of the contract
- **D.** A meeting where the team reflects on its process and agrees improvements for the next sprint

15.12 During planning poker, estimates for a story range from 2 to 13 points. What should happen next?

- **A.** Take the average of all estimates
- **B.** Use the highest estimate to be safe
- **C.** The scrum master chooses the final estimate, since the scrum master facilitates the session
- **D.** The lowest and highest estimators explain their reasoning, then the team re-estimates

15.13 In a hybrid project, the hardware is delivered predictively and the software in sprints. Which practice best integrates the two?

- **A.** Forcing the software team to use a detailed Gantt chart for every task, so that both parts look the same
- **B.** Running the two parts with no coordination
- **C.** Aligning release plans with hardware milestones at defined integration points
- **D.** Delaying all software work until the hardware is complete

15.14 What is the purpose of a spike in agile development?

- **A.** A defect found in production
- **B.** A short meeting held to celebrate a successful release with the stakeholders
- **C.** A timeboxed investigation to reduce uncertainty before committing
- **D.** A sudden increase in velocity

Chapter 16 — Leading People: Vision, Teams, Conflict, Motivation, Influence and Ethics

After this chapter you should be able to:

- build a common vision and team agreements;
- choose a leadership style and source of power suited to the situation;
- apply motivation theories and team development models;
- select a conflict resolution approach based on the situation;
- use emotional intelligence, negotiation, decision-making and coaching techniques;
- apply the PMI Code of Ethics and Professional Conduct to dilemmas.

The People domain is 33% of the exam. Many of its questions have no "process" answer — they test judgment about human behaviour.

16.1 Developing a common vision (People task 1)

A shared vision aligns the team and stakeholders on **why** the project exists and **what success looks like**. Tools:

- **Project/product vision statement** — short, inspiring, memorable.
- **Elevator pitch** — *For* (target customer) *who* (need), *the* (product) *is a* (category) *that* (key benefit). *Unlike* (alternative), *our product* (differentiator).
- **Product box** — the team designs the "box" the product would be sold in, forcing focus on key benefits.
- **Project charter / team charter** — purpose, objectives, success criteria, values and working agreements.
- **Chartering workshops / inception** — collaborative sessions with stakeholders and team.
- **Roadmaps** — show how the vision will be reached over time.
- **Revisit the vision** regularly (sprint reviews, phase gates), especially when the team or context changes.

> **Exam tip.** If the team members disagree about priorities or "don't know why we are doing this", the best answer usually brings people back to the **vision, objectives and business value** — often by facilitating a session with the product owner or sponsor.

16.2 Leadership styles

Style	Description	When useful
Servant leadership	Serves the team first: removes impediments, coaches, facilitates, empowers	Default for agile teams and knowledge work

Style	Description	When useful
Transformational	Inspires through vision, idealised attributes, individual consideration, intellectual stimulation	Change initiatives, motivating innovation
Transactional	Focuses on goals, feedback and rewards for performance (management by exception)	Routine work, clear targets
Charismatic	Inspires through energy and conviction	Rallying people; risk of dependence on the leader
Interactional	Combination of transactional, transformational and charismatic	Many situations
Laissez-faire	Hands-off; team makes decisions	Highly skilled, self-directed teams
Directive / autocratic / command-and-control	Leader decides and directs	Crises, safety emergencies, inexperienced teams in urgent situations
Participative / democratic	Involves the team in decisions	Building buy-in
Situational leadership (Hersey & Blanchard)	Adapt style to follower **competence** and **commitment**	Always — matching style to readiness

Situational Leadership® II (Blanchard) maps four styles to four development levels:

Development level	Typical state	Style
D1 — low competence, high commitment ("enthusiastic beginner")	New, eager	**S1 Directing** — high direction, low support
D2 — some competence, low commitment ("disillusioned learner")	Struggling	**S2 Coaching** — high direction, high support
D3 — moderate-high competence, variable commitment ("capable but cautious")	Able but hesitant	**S3 Supporting** — low direction, high support
D4 — high competence, high commitment ("self-reliant achiever")	Expert	**S4 Delegating** — low direction, low support

Leadership vs management: management focuses on processes, systems, control and short-term results ("doing things right"); leadership focuses on people, vision, inspiration and change ("doing the right things"). Project managers need both.

16.3 Sources of power

Power	Based on
Formal / legitimate / positional	The PM's assigned position (charter)
Reward	Ability to give rewards
Penalty / coercive	Ability to impose negative consequences — **least effective** in the long term

Power	Based on
Expert	Knowledge, skills, experience — **among the most effective**
Referent	Respect, admiration, credibility — people want to follow you — **among the most effective**
Informational	Control over gathering or distribution of information
Situational	Gained because of unique circumstances (e.g., a crisis)
Relational	Networks, connections and alliances
Personal / charismatic	Charm, attraction
Ingratiating	Flattery or other common ground (manipulative — avoid)
Guilt-based / persuasive / avoiding	Other forms described in PMI materials

> **Exam tip.** The PM's most effective forms of power are generally **expert** and **referent**; the least effective is **coercive/penalty**. In matrix organisations the PM has limited formal power, so influence, negotiation and relationship-building matter more.

16.4 Motivation theories

Theory	Key idea	Practical meaning
Maslow's hierarchy of needs	Physiological → safety → social/belonging → esteem → self-actualisation. Lower needs must be met before higher ones motivate	A worried employee (job security) won't be motivated by an award ceremony
Herzberg's two-factor theory	**Hygiene factors** (salary, working conditions, policies, job security, supervision) prevent dissatisfaction but **do not motivate**; **motivators** (achievement, recognition, responsibility, growth, the work itself) create satisfaction	Fix hygiene problems, then use motivators
McGregor's Theory X and Y	X: people dislike work, must be controlled (micromanagement). Y: people are self-motivated and seek responsibility (empowerment)	Agile assumes Theory Y
Ouchi's Theory Z	Long-term employment, collective decision-making, concern for employee well-being, loyalty	Japanese management style
McClelland's achievement (acquired needs) theory	Needs for **achievement**, **affiliation** and **power** differ by person	Give achievers challenging goals, affiliators teamwork, power-seekers leadership opportunities
Vroom's expectancy theory	Motivation = **expectancy** (effort → performance) × **instrumentality** (performance → reward) × **valence** (value of reward)	Rewards must be achievable and valued

Theory	Key idea	Practical meaning
Daniel Pink — Drive	Intrinsic motivation from **autonomy, mastery and purpose**; extrinsic rewards work for simple tasks but can hurt creative work	Empower teams, support learning, connect work to purpose
Equity theory	People compare their input/outcome ratio with others'	Perceived unfairness demotivates

Recognition and rewards: timely, specific, sincere, appropriate to the person and culture (some people dislike public praise). Avoid **win-lose** rewards that only one person can win when you want teamwork.

16.5 Team development

Tuckman's ladder

1. **Forming** — team meets, learns about the project and roles; people are polite, independent, unsure. *Leader:* directive; clarify goals, roles, ground rules.
2. **Storming** — conflicts emerge over approaches and roles; if not handled constructively, the team can regress. *Leader:* coach, facilitate conflict resolution.
3. **Norming** — team adjusts behaviours, develops trust and working habits. *Leader:* support, facilitate.
4. **Performing** — high-functioning, interdependent, self-managing. *Leader:* delegate, remove impediments.
5. **Adjourning** — work completed, team released (also called *mourning*). *Leader:* recognise, celebrate, capture lessons, support transition.

Teams can move back to an earlier stage when membership or circumstances change.

Drexler/Sibbet team performance model

Seven stages: **orientation** (why am I here?), **trust building** (who are you?), **goal clarification** (what are we doing?), **commitment** (how will we do it?), **implementation** (who does what, when, where?), **high performance** (wow!), **renewal** (why continue?).

Building high-performing teams

- Clear vision and objectives, **shared ownership**, **trust**, **open communication**, **constructive disagreement**, **psychological safety**, **empowerment**, **recognition**, **collaboration**, accountability, **continuous learning** (training, cross-training, mentoring, pairing).
- **Team charter / working agreements**; **ground rules** (e.g., cameras on in decision meetings, response times).
- **Co-location** (war room) or effective **virtual team** practices: overlap hours, clear communication protocols, digital collaboration tools, occasional in-person meetings, inclusive meeting practices across time zones and cultures.
- **Diversity, equity and inclusion** — diverse teams make better decisions when everyone is heard; leaders ensure inclusive participation.

16.6 Conflict management

Common sources of project conflict (in descending frequency in classic research): **schedules**, **project priorities**, **resources**, technical opinions, administrative procedures, cost, personalities. Personality conflicts are usually the *least* frequent root cause.

Conflict is normal and can be healthy (constructive debate improves solutions). Handle it early, privately when possible, with focus on issues not personalities, and on the present and future, not the past.

Five resolution techniques (Thomas–Kilmann / PMI)

Technique	Description	Result	When appropriate
Collaborate / problem-solve	Incorporate multiple viewpoints; work openly to find a solution that satisfies all; requires trust and time	**Win-win**; consensus and commitment	Usually the **best** when time permits and the issue is important
Compromise / reconcile	Each party gives up something to reach a temporary or partial solution	**Lose-lose** (or partial win-win)	When collaboration fails, parties have equal power, or time is short
Smooth / accommodate	Emphasise areas of agreement, downplay differences; concede one's position	Lose-win; temporary peace	To preserve relationships, when the issue matters more to the other party
Force / direct	Push one viewpoint at the expense of others (using power)	Win-lose	Emergencies, safety, legal or ethical issues, when unpopular decisions are needed quickly
Withdraw / avoid	Retreat or postpone	Lose-lose (temporary)	When emotions are high and a cooling-off period helps, or the issue is trivial — never a long-term solution

Levels of conflict (Speed Leas): 1 — problem to solve (information sharing, collaboration); 2 — disagreement (personal protection appears; support and empowerment help); 3 — contest (winning matters; use accommodation, negotiation); 4 — crusade/fight (protecting one's own group; diplomacy needed); 5 — world war (destroy the other; do whatever is necessary to prevent harm). The approach depends on the level.

> **Exam tip.** When two team members argue, the PM usually **first lets them try to resolve it themselves**, then **facilitates a private conversation** focused on the issue (collaborate). Escalating to functional managers or HR is for persistent or serious issues (harassment, ethics violations). In agile teams, the team is encouraged to resolve conflicts itself, with the scrum master/servant leader facilitating.

16.7 Emotional intelligence (EI)

The ability to identify, assess and manage one's own emotions and those of others. Goleman's model:

Component	Focus
Self-awareness	Recognising one's own emotions, strengths, weaknesses
Self-regulation (self-management)	Controlling impulses, staying calm, adapting
Motivation	Inner drive toward goals
Empathy	Understanding others' feelings and perspectives
Social skills	Managing relationships, building networks, influencing

Self-awareness and self-regulation are about **yourself**; empathy and social skills about **others**. High EI leaders handle stress, conflict and change better.

16.8 Decision-making

Method	Description
Unanimity	Everyone agrees (Delphi can help)
Majority	More than 50% support
Plurality	The largest block decides even without a majority
Autocratic	One person decides (appropriate in emergencies or when it is clearly that person's decision)
Consensus	Everyone can live with and support the decision
Fist of five / fist-to-five	Each person shows 0–5 fingers; low counts trigger discussion
Roman voting	Thumbs up, sideways, down
Multicriteria decision analysis	Weighted criteria matrix to evaluate options
Delphi	Anonymous rounds with experts; reduces bias and groupthink
Nominal group technique	Ideas generated individually, then ranked

Divergent then convergent thinking: generate options broadly (brainstorm), then narrow down. **Decide at the last responsible moment** for uncertain decisions (agile).

16.9 Negotiation and influencing

- Aim for **win-win** outcomes (principled negotiation — Fisher & Ury: separate people from the problem; focus on interests, not positions; invent options for mutual gain; insist on objective criteria).
- Know your **BATNA** (best alternative to a negotiated agreement) — your walk-away point; and the **ZOPA** (zone of possible agreement).
- **Influencing** without authority: build relationships and trust, understand others' interests, use data and logic, reciprocity, coalitions, credibility (expert/referent power).
- **Political awareness** — understand power structures and informal networks.

16.10 Coaching, mentoring and knowledge transfer (People task 7)

- **Coaching** — helps a person develop their own solutions through questions and feedback (performance-oriented, often short-term). **OSCAR model**: Outcome, Situation, Choices/consequences, Actions, Review.
- **Mentoring** — a more experienced person shares knowledge and guidance (career-oriented, longer-term).
- **Training** — formal learning (classroom, e-learning, on-the-job); planned in the resource management plan; costs included in the budget.
- **Knowledge transfer techniques** — pairing, job shadowing, communities of practice, lessons learned sessions, documentation and wikis, demos, "lunch and learn", handover workshops with operations, rotating roles, storytelling.
- Knowledge transfer happens **throughout** the project (not only at the end) and into **operations** and **future projects**.

16.11 Cultural awareness

- Cultures differ on power distance, individualism/collectivism, uncertainty avoidance, time orientation, communication style (direct/indirect, high-context/low-context) — e.g., Hofstede's dimensions.
- Practices: learn about team members' cultures, avoid assumptions and stereotypes, use clear language, confirm understanding, respect holidays and time zones, adapt recognition and feedback styles.

16.12 The PMI Code of Ethics and Professional Conduct

The Code describes expectations for PMI members and certification holders in four values. Each has **aspirational standards** (what we strive for) and **mandatory standards** (firm requirements; violations can lead to disciplinary action).

Value	Aspirational examples	Mandatory examples
Responsibility	Make decisions in the best interests of society, safety and environment; accept only assignments consistent with our skills; fulfil commitments; own errors	**Follow policies, rules, laws**; **report unethical or illegal conduct** (backed by facts); bring violations to the attention of appropriate authorities
Respect	Learn about others' norms and customs; listen; approach conflicts directly; conduct ourselves professionally	**Negotiate in good faith**; do not exercise power to influence others for personal benefit; do not act abusively; respect others' property rights
Fairness	Be transparent in decision-making; examine impartiality; provide equal access to information; equal opportunities	**Disclose conflicts of interest** and refrain from decisions when conflicted; do not hire/fire, reward/punish based on personal considerations (favouritism, nepotism, bribery); do not discriminate

Value	Aspirational examples	Mandatory examples
Honesty	Seek to understand the truth; be truthful in communications; provide accurate, timely information; make commitments in good faith	**Do not engage in deceptive behaviour** (false statements, half-truths, information out of context); do not act dishonestly for personal gain

Typical ethical scenarios and responses:

- Gifts from vendors: follow company policy and the law; decline gifts that could influence decisions; disclose.
- Discovering a team member's falsified test results: address it, report through appropriate channels, correct the record.
- Being asked to misreport status: refuse; report accurately; discuss with the sponsor.
- Conflict of interest (e.g., a relative's company bids): **disclose** it and remove yourself from the decision.
- Local customs (e.g., facilitation payments) vs law: **follow the law** (anti-bribery laws apply even abroad) and company policy.
- Copyright/software licences: respect intellectual property.

Chapter 16 Review Questions

Answer all questions first, then check the answer key for Chapter 16 in Appendix B.

16.1 Two team members have been arguing about which testing tool to use, and the disagreement is delaying work. What should the project manager do first?

- **A.** Meet with both to understand their positions and help them find a solution
- **B.** Report both team members to their functional managers
- **C.** Choose the tool and instruct them to use it
- **D.** Tell them to stop arguing and wait until the issue resolves itself without any intervention

16.2 Which conflict resolution technique is most appropriate when an immediate decision is needed to address a safety hazard on site?

- **A.** Withdraw/avoid
- **B.** Collaborate/problem-solve
- **C.** Smooth/accommodate
- **D.** Force/direct

16.3 According to Herzberg, which of the following is a hygiene factor rather than a motivator?

- **A.** Responsibility for meaningful work
- **B.** Opportunities for growth
- **C.** Salary and working conditions
- **D.** Recognition for achievement

16.4 A newly formed team is polite but unsure of their roles and the project's goals. Which stage of team development is this, and what should the leader emphasise?

- **A.** Storming; let conflicts escalate naturally

- **B.** Forming; clarify goals, roles and ground rules
- **C.** Adjourning; celebrate success
- **D.** Performing; delegate everything to the team at once

16.5 A highly skilled and motivated team member is assigned a task she has done many times. According to situational leadership, which style is most appropriate?

- **A.** Supporting with high direction
- **B.** Coaching
- **C.** Directing
- **D.** Delegating

16.6 Which sources of power are generally considered the most effective for a project manager?

- **A.** Expert and referent power
- **B.** Ingratiating and penalty power
- **C.** Coercive and legitimate power
- **D.** Situational and coercive power

16.7 A project manager learns that a company owned by her brother has submitted a bid for the project's contract. What should she do?

- **A.** Help her brother improve the bid
- **B.** Reject the bid immediately without telling anyone
- **C.** Evaluate the bid fairly herself, because she is an ethical person and knows the work
- **D.** Disclose the conflict and stay out of the selection decision

16.8 Which component of emotional intelligence is primarily about understanding the feelings and perspectives of others?

- **A.** Self-regulation
- **B.** Motivation
- **C.** Empathy
- **D.** Self-awareness

16.9 According to Vroom's expectancy theory, why might a team ignore a generous bonus offered for finishing two months early?

- **A.** The team is in the norming stage, so it is not yet ready for targets
- **B.** The team believes the target is unachievable, so expectancy is low
- **C.** Monetary rewards always demotivate
- **D.** Bonuses are hygiene factors

16.10 Which two actions best support knowledge transfer from the project team to the operations team? (Choose two.)

- **A.** Releasing the team before the handover
- **B.** Pairing operations staff with developers during the final iterations
- **C.** Sending the full document archive by email on the last day
- **D.** Keeping key knowledge with a single expert to protect it
- **E.** Job shadowing and joint handover workshops before go-live

16.11 The team regularly debates the purpose of the product, and members prioritise different features. What should the project manager do?

- **A.** Decide the priorities personally
- **B.** Facilitate a session with the product owner and team to clarify the vision
- **C.** Ignore the debate because conflict is healthy
- **D.** Ask each developer to work on the features they personally prefer, to keep them motivated

16.12 In negotiation, what is a BATNA?

- **A.** The best alternative to a negotiated agreement, i.e. the walk-away option
- **B.** A type of contract
- **C.** The zone of possible agreement
- **D.** The budget available at the negotiation stage for the whole of the contract

16.13 A team member from a culture with high respect for hierarchy rarely speaks in meetings. What should the project manager do?

- **A.** Assume the person has nothing to contribute
- **B.** Offer ways to contribute that suit the person, such as one-to-one or written input
- **C.** Publicly insist that the person speaks up in every meeting, so that the team hears all views
- **D.** Remove the person from meetings

16.14 Which statement describes the difference between coaching and mentoring?

- **A.** Mentoring is only for poor performers
- **B.** Coaching is always done by human resources, while mentoring is always done by the project sponsor
- **C.** Coaching helps a person find their own solutions; mentoring shares the mentor's experience
- **D.** They are identical

Chapter 17 — The Business Environment: Compliance, Issues, Change, Improvement, Sustainability, AI and PMOs

After this chapter you should be able to:

- plan and manage compliance with laws, regulations, standards and policies;
- manage issues and remove impediments using the right escalation path;
- plan benefits realisation and transition to operations;
- support organisational change using recognised change models;
- evaluate and respond to changes in the external business environment;
- drive continuous improvement;
- integrate sustainability (ESG) into project decisions;
- use artificial intelligence responsibly in project work;
- describe the types and value of project management offices.

The Business Environment domain is now **26%** of the exam. Governance and change control were covered in Chapter 7, and risk in Chapter 14. This chapter covers the remaining tasks.

17.1 Compliance (Business Environment task 2)

Compliance = meeting the requirements imposed by laws, regulations, standards, contracts and organisational policies. Examples: data protection (e.g., GDPR), health and safety, environmental permits, accessibility, financial controls (e.g., SOX), industry standards (ISO, GMP in pharma, aviation regulations), labour law, anti-bribery and corruption laws, export controls, sustainability reporting rules.

Managing compliance:

1. **Identify** compliance requirements early (legal, regulatory, contractual, policy) with experts — legal, compliance, quality, security.
2. **Classify** them by category and risk (what happens if we fail?).
3. **Plan** how compliance will be achieved and demonstrated — build requirements into scope, acceptance criteria, the definition of done, the quality plan and the schedule (e.g., audit dates, regulatory submissions).
4. **Assign owners** and integrate into governance (compliance checkpoints at phase gates).
5. **Monitor and audit** — compliance audits, inspections, evidence collection.
6. **Analyse the consequences of non-compliance** (fines, shutdown, reputational damage, safety incidents) and treat them as risks.
7. **Address non-compliance immediately** — stop the non-compliant activity if needed, inform the right authorities internally, correct, and prevent recurrence.

> **Exam tip.** Compliance is **not negotiable** and not something to "trade off" for schedule. If a regulation changes during the project, assess the impact and process the needed changes — don't ignore it, and don't wait until the end. If you discover a violation, **don't hide it**: report it through the proper channels.

17.2 Issues and impediments (Business Environment task 4)

	Risk	Issue	Impediment (blocker)
Definition	Uncertain future event	Current problem that must be resolved	Anything that slows or blocks the team's progress
Where recorded	Risk register	**Issue log**	Impediment board / issue log; raised in daily stand-ups
Typical owner	Risk owner	Issue owner (assigned)	Scrum master / servant leader, or whoever can remove it

Issue log contents: issue type, who raised it and when, description, priority, **owner**, target resolution date, status, final solution.

Handling issues and impediments:

1. Make them **visible** (issue log, impediment board, stand-ups).
2. **Analyse** the root cause and impact.
3. **Resolve at the lowest appropriate level** — the team first, then the PM/scrum master.
4. **Escalate** through the agreed path when the resolution is outside the team's authority (functional manager, sponsor, steering committee) — with facts and options.
5. **Track** to closure and capture lessons learned; identify related risks.

> **Trap.** The servant leader removes impediments **the team cannot remove itself** — e.g., getting access to a test environment, resolving a dependency with another department. Solving every problem *for* a self-organising team (or ignoring impediments) are both wrong.

17.3 Benefits and value realisation

- **Benefits management plan** (Chapter 3): target benefits, owner, metrics, timeframe, assumptions, risks.
- **Value-based delivery (Process task 3)**: identify value components with stakeholders; **prioritise by value** (MoSCoW, WSJF, cost of delay); **deliver incrementally** (MVP, releases); **measure** benefits with KPIs and **OKRs** (objectives and key results); adjust based on feedback.
- **Leading vs lagging indicators**: leading indicators predict future results (e.g., feature adoption, pipeline of sales demos); lagging indicators confirm results after the fact (e.g., revenue, customer churn).
- **Transition to operations**: training, documentation, support model, service-level agreements, operational readiness reviews, hypercare period, handover of benefit tracking.
- Benefits are often realised **after** the project closes — the benefit owner continues measuring them.

17.4 Supporting organisational change (Business Environment task 7)

Projects create change; people must adopt it for benefits to occur. The PM supports **organisational change management (OCM)** — often with change managers and business leaders.

Change models

Model	Stages / elements
Lewin	**Unfreeze** (create readiness) → **Change** (move) → **Refreeze** (stabilise in new state)
Kotter's 8 steps	1 Create urgency · 2 Form a powerful guiding coalition · 3 Create a vision for change · 4 Communicate the vision · 5 Remove obstacles (empower action) · 6 Create short-term wins · 7 Build on the change (consolidate) · 8 Anchor the changes in corporate culture
ADKAR® (Prosci)	**A**wareness of the need for change · **D**esire to participate · **K**nowledge of how to change · **A**bility to implement · **R**einforcement to sustain
Bridges' transition model	**Ending, losing, letting go** → **Neutral zone** → **New beginning** (focuses on people's psychological transition, not the change itself)
Virginia Satir change model	Late status quo → Introduction of a foreign element → **Chaos** → Transforming idea → **Practice and integration** → New status quo (performance dips before improving)
Diffusion of innovations (Rogers)	Innovators, early adopters, early majority, late majority, laggards — target early adopters as champions

Practical actions

- **Assess readiness** (culture, capacity for change, past change experiences, leadership support).
- **Engage stakeholders** early; identify **champions/change agents**; involve users in design.
- **Communicate** the *why*, *what* and *what's in it for me* — repeatedly, in multiple channels.
- **Train and support** (coaching, super-users, help desks).
- Expect and **address resistance** — listen to concerns; resistance often signals legitimate issues.
- **Celebrate quick wins**; **reinforce** with metrics, recognition, updated processes and policies.
- Avoid **change saturation** — too many simultaneous changes for the same people.

17.5 Evaluating external business environment changes (Business Environment task 8)

External changes — **regulatory**, **economic** (inflation, interest rates, exchange rates, tariffs), **market** (competitors, customer preferences), **technological** (new platforms, AI), **social** and **environmental** — can alter the project's value, scope, risks and constraints.

Approach:

1. **Scan** the environment continuously (PESTLE, market research, regulatory watch, supplier news).
2. **Assess the impact** on the business case, scope, schedule, budget, risks, compliance.
3. **Recommend options** (continue, adapt scope, re-plan, pause, cancel) and involve the decision-makers (sponsor, steering committee, product owner).

4. **Process changes** through governance (change requests, backlog reprioritisation, re-baselining).
5. **Communicate** changes and update the risk register, assumption log and business case.

17.6 Continuous improvement (Business Environment task 6)

- **Lessons learned** — captured throughout and shared (Chapter 7).
- **Retrospectives** — every iteration; produce **actionable** improvements that are tracked (often added to the backlog).
- **PDCA / Kaizen** — small, frequent improvements; **Six Sigma DMAIC** for process problems; **Lean** waste elimination; **value stream mapping**.
- **Root-cause analysis** (5 Whys, fishbone) — fix causes, not symptoms.
- **Experiments** — try a change for a limited time, measure, keep or discard.
- **Update OPAs** — templates, checklists, processes — so the organisation improves, not just the project.
- **Process audits** and **maturity models** (e.g., OPM3, CMMI) at organisational level.

17.7 Sustainability and ESG

The principle *Integrate sustainability within all project areas* (Chapter 4) plays out in the business environment as:

- **Triple bottom line** — people, planet, prosperity.
- **ESG** — environmental (emissions, energy, waste, water, biodiversity), social (labour practices, health and safety, community impact, diversity and inclusion, human rights in the supply chain), governance (ethics, transparency, anti-corruption, board oversight).
- **UN Sustainable Development Goals (SDGs)** — 17 global goals many organisations align to.
- **Sustainable procurement** — supplier codes of conduct, environmental and social criteria in source selection, local sourcing.
- **Life-cycle thinking** — design for durability, repair, reuse, recycling (**circular economy**); **life cycle assessment (LCA)** of environmental impact.
- **Carbon footprint** — measure and reduce project and product emissions (scope 1, 2, 3 emissions in corporate reporting).
- **Regulatory sustainability reporting** — organisations may be legally required to report sustainability data; projects may need to provide it.
- **Social sustainability of the team** — sustainable pace, well-being, fair treatment.

17.8 Artificial intelligence in project management (Appendix X3)

The Eighth Edition adds an appendix on AI. Key ideas:

Common use cases

- **Planning and scheduling** — draft WBS/backlogs, generate schedule options, estimate from historical data, optimise resources.
- **Risk management** — identify risks from documents and history, predict delays or overruns, analyse sentiment in communications.

- **Reporting and communications** — summarise meetings, draft status reports, translate, tailor messages for audiences.
- **Analytics and forecasting** — predictive analytics on cost and schedule, anomaly detection, scenario simulation.
- **Knowledge management** — search lessons learned, answer questions from project documents.
- **Administrative automation** — meeting notes, action tracking, document drafting, procurement document analysis.

Responsible and ethical use

- **Human oversight and accountability** — AI supports decisions; **people remain accountable** for them (principle: *Be an accountable leader*). Validate outputs before use.
- **Transparency** — disclose when content or decisions are AI-assisted; explainability of models.
- **Data privacy and confidentiality** — don't feed confidential, personal or client data into tools that are not approved; follow data protection laws and company policy.
- **Bias and fairness** — AI can replicate biases in its training data (e.g., in resource selection); check for fairness.
- **Accuracy** — AI can generate plausible but wrong outputs ("hallucinations"); verify facts and calculations.
- **Intellectual property** — respect copyright and licensing; clarify ownership of AI-generated content.
- **Security** — approved tools, access controls.
- **Sustainability** — consider the energy use of AI.

Adopting AI

- Start with a clear **purpose and value** (use cases with measurable benefits); assess **data readiness and quality**; run **pilots**; establish **AI governance** (policies, approved tools, roles); **train** people (AI literacy); manage the **organisational change** (fears about jobs, trust); monitor results and **improve**.

AI projects themselves have particular characteristics: experimental, data-dependent, iterative model development, uncertain outcomes (adaptive approaches fit well), plus ethical and regulatory risks.

> **Exam tip.** In AI scenarios, the best answer usually **uses AI to augment** the team while keeping **human judgment, transparency, data protection and compliance** — e.g., "validate the AI-generated schedule with the team before baselining", "use only the organisation's approved AI tools for confidential data".

17.9 The Project Management Office (Appendix X2)

A **PMO** is an organisational structure that standardises project-related governance processes and facilitates the sharing of resources, methodologies, tools and techniques.

Type	Degree of control	Role
Supportive	**Low**	Consultative: templates, best practices, training, access to lessons learned
Controlling	**Moderate**	Support **and** require compliance (frameworks, methodologies, templates, governance reviews)

Type	Degree of control	Role
Directive	**High**	Directly **manages** the projects; project managers report to the PMO

Other forms: **enterprise PMO (EPMO)** aligning portfolio to strategy; **program management office**; **agile PMO / value delivery office (VDO)** that coaches teams, supports agile practices and focuses on value rather than compliance; **centre of excellence**.

PMO value proposition: better alignment with strategy, consistent governance, improved decision-making (portfolio visibility), capability development, resource optimisation, knowledge management, and **customer focus** — a PMO must demonstrate value to its customers (executives, PMs, teams) and evolve, otherwise it is seen as bureaucracy. PMO **maturity models** help assess and improve capabilities.

Chapter 17 Review Questions

Answer all questions first, then check the answer key for Chapter 17 in Appendix B.

17.1 During execution, a new data protection regulation takes effect that affects how the project's system stores customer data. What should the project manager do first?

- **A.** Cancel the project
- **B.** Ask the team to continue as planned, because the scope baseline has already been approved
- **C.** Ignore it until the next phase gate
- **D.** Assess the regulation's impact with compliance or legal experts

17.2 A team member reports that the test environment has been unavailable for three days and the team cannot resolve it. In an agile team, who is primarily expected to help remove this impediment?

- **A.** The product owner alone
- **B.** The customer
- **C.** Nobody; the team should wait for the environment to be repaired
- **D.** The scrum master or servant-leader project manager

17.3 Which change model includes the stages awareness, desire, knowledge, ability and reinforcement?

- **A.** Lewin's unfreeze-change-refreeze
- **B.** Kotter's 8 steps
- **C.** ADKAR
- **D.** Bridges' transition model

17.4 Which type of PMO provides templates, training and best practices but exercises a low degree of control over projects?

- **A.** Supportive
- **B.** Controlling
- **C.** Directive
- **D.** Enterprise

17.5 A project manager wants to use a public generative AI tool to summarise a confidential client contract. What is the most appropriate action?

- **A.** Upload the contract to any free AI tool to save time, since the summary is only for internal use
- **B.** Use only an approved tool for confidential data, follow policy and verify the summary
- **C.** Ask the client to summarise the contract instead
- **D.** Avoid all use of AI because it is never appropriate

17.6 An AI tool generates a project schedule in minutes. What should the project manager do before baselining it?

- **A.** Discard it because AI outputs cannot be used in projects
- **B.** Send it to the customer as the final commitment
- **C.** Baseline it immediately, because AI tools are more accurate than people at estimating work
- **D.** Review and validate it with the team, checking assumptions and estimates

17.7 Which situation is an issue rather than a risk?

- **A.** A supplier might go bankrupt next year
- **B.** The exchange rate may change before payment
- **C.** A key supplier has announced it will stop production next week
- **D.** A team member may resign after the bonus period

17.8 In Kotter's model, what comes right after creating a sense of urgency?

- **A.** Forming a powerful guiding coalition
- **B.** Anchoring the change in the culture
- **C.** Creating and celebrating short-term wins
- **D.** Removing obstacles

17.9 Which two practices support sustainable procurement? (Choose two.)

- **A.** Ignoring suppliers' carbon emissions because they are external
- **B.** Removing all evaluation criteria to speed up selection
- **C.** Requiring suppliers to follow a code of conduct on labour practices
- **D.** Including environmental and social criteria in the source selection criteria
- **E.** Always selecting the lowest price regardless of other factors

17.10 A retrospective identifies that code reviews are delaying delivery. What is the best way to ensure continuous improvement?

- **A.** Agree a specific improvement experiment with an owner, and review its effect
- **B.** Record the problem in the register and wait for the final lessons learned meeting to discuss it
- **C.** Remove code reviews entirely
- **D.** Ask the PMO to issue a new policy

17.11 Users of a new system resist the change and continue using old spreadsheets. What should the project manager do first?

- **A.** Disable the spreadsheets immediately
- **B.** Report the users to HR
- **C.** Understand the reasons for resistance by engaging the users

- **D.** Declare the project complete, because the system was delivered and works as specified

17.12 A sharp rise in import tariffs will increase the cost of key materials by 20%. What is the best first step?

- **A.** Absorb the increase silently using the management reserve, so that the sponsor is not worried
- **B.** Analyse the impact on the business case and present options to decision-makers
- **C.** Immediately switch to an unqualified local supplier
- **D.** Wait until the materials are invoiced

17.13 Which statement about leading and lagging indicators is correct?

- **A.** They are the same thing
- **B.** Leading indicators help predict results; lagging indicators measure them afterwards
- **C.** Lagging indicators are always better than leading indicators, because they are based on facts
- **D.** Leading indicators are only financial

Chapter 18 — Tailoring

After this chapter you should be able to:

- explain what tailoring is, why it matters and who does it;
- describe what can be tailored and the steps of the tailoring process;
- identify situational factors that drive tailoring decisions;
- recognise good and bad tailoring in scenarios.

18.1 What tailoring is

Tailoring is the deliberate adaptation of the approach, governance, processes, methods and artifacts to fit the project's context — its objectives, complexity, risks, stakeholders, team, organisation and environment. The PMBOK® Guide is a body of knowledge, not a methodology; **every project needs tailoring**.

Why tailor?

- Deliver **more value** by spending effort where it matters (and not on unnecessary documentation or meetings).
- **Increase commitment** from the team, who help shape the way of working.
- Improve **efficiency** and **responsiveness** to change.
- Meet **compliance** and **governance** needs proportionately (more rigour where risk is higher).
- Make the approach **customer- and context-oriented**.

Who tailors? The project manager collaborates with the **team**, **sponsor**, **PMO** and other governance bodies. Tailoring decisions are documented (often in the project management plan or the team's working agreement) and **revisited** as the project evolves. Tailoring must stay within **organisational policies and mandatory requirements** — you can't tailor away a legal obligation.

18.2 What can be tailored

Element	Examples
Life cycle and development approach	Predictive, adaptive or hybrid; number and type of phases; iteration length; release cadence
Processes	Add (e.g., extra safety reviews), remove (e.g., formal procurement when nothing is bought), modify (e.g., lighter change control for low-impact changes), blend (combine processes), align (with organisational standards)
Engagement	Who is involved and how (people, empowerment, integration of contractors and partners)
Tools	Scheduling software, collaboration tools, AI tools — chosen by the team within organisational constraints
Methods and artifacts	Which techniques and documents are used, and how formal they are

Element	Examples
Governance	Gate reviews, reporting frequency, approval thresholds
Performance domains	Depth of attention to each domain based on the project's needs

18.3 The tailoring process

A common four-step sequence:

1. **Select the initial development approach** — using the factors in Chapter 5 (product, project, organisation). Assess suitability (e.g., agile suitability filter).
2. **Tailor for the organisation** — align with the organisation's policies, PMO requirements, governance, approved tools and processes; the PMO or value delivery office may review and approve tailoring.
3. **Tailor for the project** — consider the **product/deliverable** (compliance, criticality, type, industry, uncertainty, complexity), the **project team** (size, location, experience, culture, access to the customer) and the **culture** (buy-in, trust, empowerment, organisational culture).
4. **Implement ongoing improvement** — review tailoring at phase gates, retrospectives and lessons learned; adjust as the project and team learn.

18.4 Situational factors that drive tailoring

Factor	Pushes toward more formality / predictive elements	Pushes toward lighter / adaptive elements
Regulatory and compliance requirements	High (medical devices, aviation, nuclear)	Low
Criticality / safety	Life-critical	Low impact
Requirements stability	Stable, well understood	Uncertain, emerging
Cost of change	High (physical construction)	Low (software)
Team size and distribution	Large, distributed, many vendors	Small, co-located
Team experience	Inexperienced with adaptive practices	Experienced, self-organising
Stakeholder availability	Limited, infrequent	Frequent, engaged product owner
Contract type	Fixed-price with defined scope	Collaborative, incremental contracts
Organisational culture	Hierarchical, low trust	Empowered, high trust
Project size and duration	Large, long	Small, short

18.5 Tailoring the performance domains — examples

Domain	Tailoring questions
Governance	How many gates? Who approves what? How formal is change control? What reporting does the steering committee need?
Scope	WBS, backlog or both? How detailed are requirements up front? How will acceptance work?
Schedule	Detailed CPM schedule, release plan, Kanban flow? Iteration length?
Finance	Fixed budget vs incremental funding? EVM required? Level of estimate accuracy?
Stakeholders	How many and how diverse? Formal communications plan or information radiators?
Resources	Internal team, vendors or blended? Co-located or virtual? Procurement rules?
Risk	Qualitative only, or quantitative analysis too? Risk-adjusted backlog? Size of reserves?

18.6 Good vs bad tailoring

Good tailoring is intentional, justified by the context, agreed with the right people, documented, compliant, and revisited.

Bad tailoring includes:

- Skipping essential controls (e.g., no change control on a fixed-price contract; no testing for a safety-critical product).
- Copying another project's approach without considering differences.
- "Agile" as an excuse for no planning, no documentation and no accountability.
- Heavy documentation and approvals on a small, low-risk, fast-moving initiative.
- Tailoring that breaches organisational policy or the law.

> **Exam tip.** When a question asks what to do at the start of a project in an unfamiliar context (new client, new industry, new team), the best answer often involves **assessing the context and tailoring** the approach with the team and stakeholders — not applying a standard methodology blindly, and not abandoning all structure.

Chapter 18 Review Questions

Answer all questions first, then check the answer key for Chapter 18 in Appendix B.

18.1 What is the primary purpose of tailoring?

- **A.** To adapt the approach, governance and processes to the project's context
- **B.** To remove all documentation from the project
- **C.** To apply every PMBOK Guide process to every project
- **D.** To allow the project manager to ignore any organisational policies that slow the project down

18.2 Which is the first step of the tailoring process?

- **A.** Tailor for the project team's culture

- **B.** Archive the lessons learned
- **C.** Implement ongoing improvement of the approach
- **D.** Select the initial development approach

18.3 A small internal project with low risk and no external suppliers is required by the PMO to complete a 40-page procurement plan template. What is the best course of action?

- **A.** Agree tailoring with the PMO, documenting that procurement is not applicable
- **B.** Complete the full template with invented content
- **C.** Ignore the PMO's requirement without telling anyone, since the project is small and low-risk
- **D.** Cancel the project

18.4 A medical device project is being planned. The team wants to use agile practices for software development. What is the best tailoring decision?

- **A.** Let each developer choose a different approach
- **B.** Ban agile practices because medical devices are regulated
- **C.** Use adaptive practices for software, keeping the required regulatory controls
- **D.** Drop regulatory documentation, because agile values working software over comprehensive documentation

18.5 Which two factors would typically push a project toward more formal, predictive tailoring? (Choose two.)

- **A.** High cost of change after construction starts
- **B.** Highly uncertain and emerging requirements
- **C.** Strict regulatory requirements for documented verification
- **D.** A small, co-located, experienced team
- **E.** Frequent availability of an engaged product owner

18.6 When should tailoring decisions be revisited?

- **A.** Never; once approved, tailoring decisions are fixed for the whole of the project
- **B.** Only after project closure
- **C.** Throughout the project, such as at phase gates and retrospectives
- **D.** Only if the sponsor changes

18.7 A project manager joins a new organisation and is assigned a project in an industry she does not know. What should she do first regarding the approach?

- **A.** Apply the methodology she used in her previous company unchanged, since it worked well there
- **B.** Assess the context, product, team and stakeholders, and tailor with them
- **C.** Avoid any structured approach until problems appear
- **D.** Ask the team to choose a framework by vote without analysis

18.8 Which statement describes bad tailoring?

- **A.** Adding an extra safety review for a high-risk construction activity
- **B.** Removing change control from a fixed-price contract project to "be more agile"
- **C.** Shortening status reports for a small, low-risk project with the sponsor's agreement
- **D.** Using a backlog for the software component of a hybrid project

Chapter 19 — Models, Methods and Artifacts: the Practitioner's Toolbox

After this chapter you should be able to:

- distinguish models, methods and artifacts;
- select an appropriate model, method or artifact for a given purpose;
- recognise the purpose of the artifacts most often named in exam questions.

The Seventh Edition introduced the idea of a toolbox of **models, methods and artifacts (MMAs)** and the Eighth Edition keeps it. You do not need to memorise the list — but you must know **what each item is for**, because exam options often name them.

- **Model** — a thinking strategy to explain a process, framework or phenomenon (e.g., Tuckman's ladder).
- **Method** — a means of achieving an outcome, output, result or deliverable (e.g., a SWOT analysis, planning poker).
- **Artifact** — a template, document, output or project deliverable (e.g., the risk register).

19.1 Commonly used models

Category	Models	Where explained
Leadership	Situational Leadership® II; OSCAR coaching model	Ch. 16
Communication	Cross-cultural communication; effectiveness of communication channels (richness vs distance); gulf of execution and evaluation	Ch. 12
Motivation	Hygiene and motivational factors (Herzberg); intrinsic vs extrinsic motivation (Pink); theory of needs (McClelland); Theory X, Y, Z	Ch. 16
Change	Managing change in organisations; ADKAR®; 8-step process for leading change (Kotter); Virginia Satir change model; transition model (Bridges)	Ch. 17
Complexity	Cynefin framework; Stacey matrix	Ch. 5
Team development	Tuckman ladder; Drexler/Sibbet team performance model	Ch. 16
Other	Conflict model (Thomas–Kilmann); negotiation (win-win, BATNA); planning; process groups/focus areas; salience model; Kano model	Ch. 8, 12, 16

19.2 Commonly used methods

Data gathering and analysis

Method	Purpose
Alternatives analysis	Evaluate options to choose the best way to do the work

Method	Purpose
Assumption and constraint analysis	Check that assumptions and constraints are valid and integrated
Benchmarking	Compare actual or planned practices with comparable organisations to find ideas and standards
Business justification analysis — payback, IRR, NPV, ROI, cost-benefit, BCR	Justify or select investments
Check sheet	Tally data during inspections
Cost of quality	Balance conformance vs non-conformance costs
Decision tree analysis	Evaluate alternatives under uncertainty (EMV)
Earned value analysis	Measure scope, schedule and cost performance
Expected monetary value	Quantify the value of uncertain outcomes
Forecast	Predict future performance (EAC, ETC, Monte Carlo)
Influence diagram	Graph the decisions, uncertainties and outcomes
Life cycle assessment	Evaluate total environmental impact of a product
Make-or-buy analysis	Decide whether to produce internally or purchase
Probability and impact matrix	Prioritise risks
Process analysis	Find process improvements
Regression analysis	Examine relationships among variables for forecasting
Reserve analysis	Evaluate remaining reserves vs remaining risk
Root-cause analysis	Find the underlying cause (5 Whys, fishbone)
Sensitivity analysis	Identify the variables with the most impact (tornado)
Simulation	Model combined effects of uncertainty (Monte Carlo)
Stakeholder analysis	Understand stakeholders' interests and influence
SWOT analysis	Strengths, weaknesses (internal), opportunities, threats (external)
Trend analysis	Is performance improving or deteriorating over time?
Value stream mapping	Visualise the flow of value and waste
Variance analysis	Determine cause and degree of difference between plan and actual
What-if scenario analysis	Assess the effect of scenarios on objectives

Estimating

Affinity grouping, analogous, function point (software size measurement), multipoint (three-point) estimating, parametric, relative estimating (story points), single-point estimating, story point estimating, wideband Delphi (planning poker), bottom-up.

Meetings and events

Event	Purpose
Backlog refinement	Clarify, split, estimate and order backlog items
Bidder conference	Give all prospective sellers the same information
Change control board meeting	Decide on change requests
Daily stand-up	Synchronise and identify impediments (15 min)
Iteration planning / review / retrospective	Plan the iteration; demonstrate the increment; improve the process
Kickoff	Launch the project/phase; align on objectives and roles
Lessons learned meeting	Capture knowledge to improve current and future projects
Planning meeting	Create, elaborate or review a plan and gain commitment
Project review / phase gate	Assess status and decide whether to continue
Release planning	Set the high-level plan for releases
Risk review	Analyse status of existing risks, identify new ones
Status meeting	Exchange and analyse progress information
Steering committee	Senior direction and decisions
Wrap-up / close-out	Obtain final acceptance, close the project

Other methods

Impact mapping (goals → actors → impacts → deliverables), modelling (prototypes, simulations), net promoter score (customer loyalty: % promoters − % detractors), prioritisation schema (MoSCoW, Kano, WSJF), timeboxing, design thinking (empathise, define, ideate, prototype, test), retrospectives formats (start-stop-continue, sailboat, 4Ls), customer journey mapping, personas.

19.3 Commonly used artifacts

Category	Artifacts
Strategy	Business case, project brief, project charter, project vision statement, roadmap, benefits management plan
Logs and registers	Assumption log, backlog, change log, issue log, lessons learned register, risk-adjusted backlog, risk register, stakeholder register, decision log
Plans	Change control plan, communications management plan, financial (cost) management plan, iteration plan, sourcing (procurement) strategy plan, project management plan, quality management plan, release plan, requirements management plan, resource management plan, risk management plan, scope management plan, schedule management plan, stakeholder engagement plan, test plan
Hierarchy charts	Organisational breakdown structure, product breakdown structure, resource breakdown structure, risk breakdown structure, work breakdown structure

Category	Artifacts
Baselines	Budget, milestone schedule, performance measurement baseline, project schedule, scope baseline
Visual data and information	Affinity diagram, burndown/burnup chart, cause-and-effect diagram, cumulative flow diagram, cycle time chart, dashboards, flowchart, Gantt chart, histogram, information radiator, lead time chart, prioritisation matrix, project schedule network diagram, requirements traceability matrix, RACI chart, scatter diagram, S-curve, stakeholder engagement assessment matrix, story map, throughput chart, use case, value stream map, velocity chart, control chart, heat map
Reports	Quality report, risk report, status report, final report
Agreements and contracts	Fixed-price, cost-reimbursable, T&M, indefinite delivery indefinite quantity (IDIQ), other agreements (MOU, MOA, SLA, BOA, letter of intent)
Other	Activity list, bid documents, metrics, project calendar, requirements documentation, project team charter, user story, definition of done, definition of ready, team agreements

19.4 Quick "which artifact?" guide

You need to...	Use...
Record a current problem and its owner	**Issue log**
Record an uncertain event	**Risk register**
Record something believed true for planning	**Assumption log**
Track all change requests and their status	**Change log**
Record decisions and their rationale	**Decision log**
See who does what on each activity	**RACI chart / RAM**
Link requirements to their source and tests	**Requirements traceability matrix**
See stakeholders' current vs desired engagement	**Stakeholder engagement assessment matrix**
Show the project's total scope as deliverables	**WBS** (predictive) / **product backlog, story map** (adaptive)
Describe each WBS element in detail	**WBS dictionary**
Show key dates to executives	**Milestone chart / roadmap**
Show remaining work in a sprint	**Burndown chart**
Show scope changes and progress together	**Burnup chart**
Detect bottlenecks in flow	**Cumulative flow diagram**
Show cumulative planned spending over time	**S-curve** (cost baseline)
Show the logical sequence and critical path	**Project schedule network diagram**
Formalise the team's values and agreements	**Team charter**
Prioritise threats by probability and impact	**Probability and impact matrix / heat map**

Chapter 19 Review Questions

Answer all questions first, then check the answer key for Chapter 19 in Appendix B.

19.1 Which of the following is a model rather than a method or an artifact?

- **A.** Planning poker
- **B.** Burnup chart
- **C.** Tuckman ladder
- **D.** Risk register

19.2 A project manager needs a document that records decisions made, who made them and why, so the reasoning is not lost. Which artifact is most appropriate?

- **A.** Change log
- **B.** Decision log
- **C.** Issue log
- **D.** Assumption log

19.3 Which method would best help a team decide whether to develop a component in-house or purchase it?

- **A.** Affinity grouping
- **B.** Net promoter score
- **C.** Value stream mapping
- **D.** Make-or-buy analysis

19.4 Which visual artifact plots cumulative planned cost over time and typically has a characteristic shape?

- **A.** S-curve
- **B.** Tornado diagram
- **C.** Pareto chart
- **D.** Histogram

19.5 A team wants to visualise each step of its delivery process, including waiting times between steps, to eliminate waste. Which method fits best?

- **A.** Value stream mapping
- **B.** Bidder conference
- **C.** SWOT analysis
- **D.** Regression analysis

19.6 Match each need with the most appropriate artifact.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Record a current problem with an owner and due date	**A.** Issue log
2. Record a condition believed to be true for planning purposes	**B.** Risk register
3. Record an uncertain event with probability and impact	**C.** Assumption log

Item	Option
4. Track submitted changes and their approval status	**D.** Change log

19.7 Net promoter score is best used to measure:

- **A.** The number of defects found in each release of the product
- **B.** Team velocity
- **C.** Customer loyalty and willingness to recommend
- **D.** The schedule performance index

19.8 Which artifact shows a prioritised arrangement of user stories along the user's journey, often sliced into releases?

- **A.** Resource breakdown structure
- **B.** WBS dictionary
- **C.** Story map
- **D.** RACI chart

19.9 Which method helps an analyst identify the most likely future value of a variable based on its relationship with other variables?

- **A.** Bidder conference
- **B.** Regression analysis
- **C.** Check sheet
- **D.** Retrospective

19.10 A steering committee wants a single page showing status of milestones, budget, risks and key metrics. Which artifact is most suitable?

- **A.** The team charter
- **B.** The WBS dictionary
- **C.** The activity list
- **D.** A dashboard

Chapter 20 — Formulas and Calculations Master Sheet

After this chapter you should be able to: perform every calculation that commonly appears on the PMP exam, quickly and accurately, and interpret the result.

> **Exam tip.** The exam has relatively few pure calculation questions, but it often gives you a number (e.g., CPI = 0.8) and asks what it means or what to do. Interpretation matters as much as arithmetic. You will have an on-screen calculator.

20.1 Earned value management

Measure	Formula	Good if...
Cost variance	**CV = EV − AC**	≥ 0
Schedule variance	**SV = EV − PV**	≥ 0
Cost performance index	**CPI = EV ÷ AC**	≥ 1
Schedule performance index	**SPI = EV ÷ PV**	≥ 1
Estimate at completion (typical: current CPI continues)	**EAC = BAC ÷ CPI**	≤ BAC
EAC (atypical: rest at budgeted rate)	**EAC = AC + (BAC − EV)**	
EAC (CPI and SPI both affect remaining work)	**EAC = AC + (BAC − EV) ÷ (CPI × SPI)**	
EAC (re-estimate)	**EAC = AC + bottom-up ETC**	
Estimate to complete	**ETC = EAC − AC**	
Variance at completion	**VAC = BAC − EAC**	≥ 0
To-complete performance index (to meet BAC)	**TCPI = (BAC − EV) ÷ (BAC − AC)**	≤ 1 (easier)
TCPI (to meet EAC)	**TCPI = (BAC − EV) ÷ (EAC − AC)**	
Percent complete	**EV ÷ BAC**	
Percent spent	**AC ÷ BAC**	
Planned value at a point (linear plan)	**PV = BAC × (elapsed time ÷ total time)**	

Interpretation quick table

CPI	SPI	Status
> 1	> 1	Under budget, ahead of schedule
> 1	< 1	Under budget, behind schedule
< 1	> 1	Over budget, ahead of schedule
< 1	< 1	Over budget, behind schedule

Drill: BAC $100,000; PV $50,000; EV $40,000; AC $45,000. CV = −5,000 · SV = −10,000 · CPI = 0.889 · SPI = 0.80 · EAC = 100,000 ÷ 0.889 ≈ 112,500 · ETC ≈ 67,500 · VAC ≈ −12,500 · TCPI = 60,000 ÷ 55,000 ≈ 1.09.

20.2 Estimating

Measure	Formula
Triangular (simple average)	**E = (O + M + P) ÷ 3**
Beta / PERT	**E = (O + 4M + P) ÷ 6**
PERT standard deviation	**σ = (P − O) ÷ 6**
PERT variance	**$\sigma^2 = [(P - O) \div 6]^2$**
Path standard deviation	**√(sum of activity variances)**
Confidence ranges	±1σ ≈ 68.3% · ±2σ ≈ 95.5% · ±3σ ≈ 99.7%
Parametric	**Quantity × unit rate**
ROM / budget / definitive ranges	−25% to +75% · −10% to +25% · −5% to +10%

Drill: O = 10, M = 16, P = 28 → PERT E = (10 + 64 + 28) ÷ 6 = **17**; σ = 18 ÷ 6 = **3**; 95% range ≈ 17 ± 6 = **11 to 23**.

20.3 Schedule network

Measure	Formula
Early finish	**EF = ES + duration** (with ES of first activity = 0)
Late start	**LS = LF − duration**
Total float	**TF = LS − ES = LF − EF**
Free float	**FF = ES (earliest successor) − EF (activity)**
Critical path	Longest path; usually zero total float

(If your course uses "day 1" conventions — EF = ES + duration − 1 — the float results are the same. Be consistent.)

20.4 Agile and flow

Measure	Formula
Velocity	Story points of **done** items per iteration
Iterations remaining	**Remaining story points ÷ average velocity** (round up)
Little's Law	**Cycle time = WIP ÷ throughput**
Throughput	Items completed ÷ time period

Measure	Formula
Lead time	Request → delivery
Cycle time	Start of work → completion

Drill: 180 points remaining, velocity 24 → 7.5 → **8 iterations**.

20.5 Communication

Measure	Formula
Communication channels	**n (n − 1) ÷ 2**

Drill: 6 people → 15 channels; adding 3 people (9) → 36 channels (+21).

20.6 Risk

Measure	Formula
Expected monetary value	**EMV = probability × impact** (threats negative, opportunities positive)
Risk score (qualitative)	**probability × impact** (on the scales defined in the risk management plan)
Decision tree	Expected value of each branch = cost/value + Σ(probability × outcome); choose the best expected value

Drill: 10% chance of $200,000 loss and 30% chance of $50,000 gain → EMV = −20,000 + 15,000 = **−5,000**.

20.7 Finance and selection

Measure	Formula / rule
Present value	**$PV = FV \div (1 + r)^n$**
Future value	**$FV = PV \times (1 + r)^n$**
NPV	Σ PV of inflows − Σ PV of outflows; choose the **highest** (> 0)
IRR	Rate where NPV = 0; choose the **highest**
ROI	**(Benefit − cost) ÷ cost × 100%**
Payback period	**Investment ÷ annual net cash inflow** (uneven flows: accumulate until recovered)
Benefit–cost ratio	**Benefits ÷ costs**; > 1 is favourable
Opportunity cost	Value of the **best alternative not chosen**
Straight-line depreciation	**(Cost − salvage value) ÷ useful life**

Drill: invest $50,000 in a tool that saves $20,000 per year for 4 years → ROI over 4 years = (80,000 − 50,000) ÷ 50,000 = **60%**; payback = **2.5 years**.

20.8 Procurement

Measure	Formula
Make-or-buy break-even (rent vs buy)	Purchase cost + (daily operating cost × d) = daily rental × d → **d = purchase cost ÷ (rental − operating cost)**
CPIF / FPIF fee adjustment	**Fee = target fee + seller share × (target cost − actual cost)**
Total price (CPIF)	**Actual cost + fee**
FPIF price	**Actual cost + adjusted profit**, capped at the **ceiling price**
Point of total assumption	**PTA = (ceiling price − target price) ÷ buyer's share ratio + target cost**

20.9 Quality

Measure	Formula / rule
Control limits	Mean ± 3σ (process data)
Rule of seven	7 consecutive points on one side of the mean → investigate
Cost of quality	Conformance (prevention + appraisal) + non-conformance (internal + external failure)
Six Sigma	3.4 defects per million opportunities

20.10 Mixed drills (answers below)

1. EV = 120, AC = 100, PV = 150. What are CPI and SPI?
2. BAC = 500, CPI = 1.25. EAC (typical)?
3. A team of 12 adds 4 people. How many channels are added?
4. O = 2, M = 5, P = 14. PERT estimate and σ?
5. Project A: NPV $80k over 4 years; Project B: NPV $95k over 6 years. Which one, and opportunity cost?
6. 25 items in progress, throughput 5 per week. Cycle time?
7. Option 1 costs $40k with a 50% chance of $10k extra; option 2 costs $44k with no risk. Better option by expected cost?
8. FPIF: target cost $80k, target profit $8k, ceiling $96k, share 60/40, actual cost $90k. Price paid?

Answers: 1) CPI 1.2, SPI 0.8 (under budget, behind schedule). 2) 400. 3) 12 people → 66 channels; 16 people → 120 channels; **54 added**. 4) E = (2 + 20 + 14) ÷ 6 = **6**; σ = 12 ÷ 6 = **2**. 5) **Project B** (higher NPV — duration already accounted for); opportunity cost **80*k* * * .6)25 ÷ 5 = * * 5*weeks* * * .7)*Option*1 : 40 + 5 = * *45k**; option 2: $44k → **option 2**. 8) Overrun $10k; seller absorbs 40% = $4k; profit = 8 − 4 = 4*k*; *price* = 90 + 4 = * *94k** (below the ceiling of $96k).

Chapter 20 Review Questions

Answer all questions first, then check the answer key for Chapter 20 in Appendix B.

20.1 A project has PV = $60,000, EV = $54,000 and AC = $60,000. What is the SPI? (Give one decimal.)

(Fill in the blank.)

Answer: _____

20.2 BAC = $800,000, EV = $300,000, AC = $400,000. If the remaining work will be performed at the budgeted rate, what is the ETC?

- **A.** $400,000
- **B.** $500,000
- **C.** $666,667
- **D.** $1,066,667

20.3 A project with 7 stakeholders adds 5 more. How many additional communication channels are created?

- **A.** 5
- **B.** 21
- **C.** 45
- **D.** 66

20.4 Which EAC formula should be used when the original estimate is considered fundamentally flawed?

- **A.** AC + (BAC − EV)
- **B.** AC + (BAC − EV) / (CPI × SPI)
- **C.** BAC / CPI
- **D.** AC + bottom-up ETC

20.5 An activity has a PERT estimate of 20 days and a standard deviation of 2 days. What is the approximate range for 95% confidence?

- **A.** 16 to 24 days
- **B.** 14 to 26 days
- **C.** 20 to 24 days
- **D.** 18 to 22 days

20.6 A tool costs $60,000 and saves $15,000 per year. What is the payback period?

- **A.** 2 years
- **B.** 4 years
- **C.** 6 years
- **D.** 15 years

20.7 The TCPI to meet the BAC is calculated as 1.35. What does this mean?

- **A.** The remaining work must be done 35% more efficiently than planned to meet the budget
- **B.** The project is 35% under budget
- **C.** The project has a 35% chance of success if it continues at its current rate of performance

- **D.** The project will finish 35% early

20.8 Straight-line depreciation: equipment costs $25,000, has a salvage value of $5,000 and a useful life of 4 years. What is the annual depreciation?

- **A.** $4,000
- **B.** $5,000
- **C.** $6,250
- **D.** $7,500

20.9 Rental of a crane costs $900 per day. Buying costs $54,000 plus $300 per day in operating cost. After how many days is buying cheaper?

- **A.** More than 60 days
- **B.** More than 180 days
- **C.** More than 45 days
- **D.** More than 90 days

20.10 A team has 160 story points remaining and an average velocity of 30 points per sprint. How many sprints are needed to complete the work? (Whole number.)

(Fill in the blank.)

Answer: _____

Chapter 21 — Exam Strategy and the PMI Mindset

After this chapter you should be able to:

- apply the "PMI mindset" to choose the best answer among several plausible options;
- use a repeatable technique to read and answer situational questions;
- manage time, breaks and stress on exam day;
- avoid the most common traps.

21.1 The PMI mindset — 25 rules of thumb

These are not laws, but they describe how the "best" answer is usually chosen.

Leadership and people

1. **Be a servant leader.** Support, coach, facilitate and remove impediments; don't command and control unless there is an emergency.
2. **Empower the team.** Let self-organising teams decide *how* to do their work; involve the team in estimates and plans.
3. **Talk to people directly.** Address problems face-to-face (or the closest equivalent) and privately, before escalating.
4. **Collaborate to resolve conflict.** Problem-solving is the default; forcing only for emergencies or ethics.
5. **Recognise and develop people.** Training, mentoring and recognition are good investments.
6. **Respect culture and diversity.** Adapt communication; include everyone.

Analysis and action 7. **Assess before acting.** When something happens, first understand it: gather facts, review the plan, analyse impact and root cause. 8. **But don't just analyse — act.** Among options, prefer the one that *does something* appropriate after analysis, rather than waiting or ignoring. 9. **Fix root causes, not symptoms.** 10. **Be proactive.** Anticipate risks, engage stakeholders early, plan communication, prevent problems. 11. **Prevention over inspection; quality is built in.**

Process and governance 12. **Follow the agreed process.** Changes to baselines go through change control; contracts through contract change control; agile changes through the backlog and product owner. 13. **Know who decides.** The product owner decides priorities; the sponsor/steering committee decides on business case and major changes; the CCB on changes per thresholds; the team on how to build. The PM facilitates and recommends. 14. **Escalate only when needed** — when the issue exceeds your authority or the team's ability, and after trying to resolve it yourself — and escalate with **options**, not just problems. 15. **Keep documents current.** Update the risk register, issue log, lessons learned, stakeholder register as things happen. 16. **Never implement unapproved changes**, and don't gold-plate. 17. **Close properly.** Formal acceptance, lessons learned, contract closure, archiving, then release resources.

Value, ethics and environment 18. **Focus on value and outcomes**, not just outputs or the triple constraint. 19. **Be honest and transparent.** Report bad news early with options; never hide problems or falsify data. 20. **Follow laws, regulations, policies and the Code of Ethics.** Disclose conflicts of interest;

refuse bribes; report illegal conduct. 21. **Consider sustainability and the wider impact** of decisions. 22. **Use AI responsibly** — as an aid, with human validation, transparency and data protection.

Context 23. **Identify the approach** (predictive, agile, hybrid) and answer in that context. 24. **Tailor** — the right amount of process for the situation. 25. **Assume you are in a well-run organisation** with the usual artifacts (charter, plans, registers) unless the question says otherwise — and that the PM has the authority the question describes.

21.2 A repeatable technique for situational questions

1. **Read the last sentence first** — what is being asked? (*first*, *next*, *best*, *most likely*, *least*, *EXCEPT*, *what should have been done to prevent…*)
2. **Read the scenario and underline the key facts**: the approach (sprints? baseline?), the phase (planning, executing, closing?), who is involved, what has already been done, the real problem (often not the one stated).
3. **Predict the answer** before reading the options.
4. **Eliminate** clearly wrong options — usually two:
   - options that ignore the problem, delay without reason, or are passive;
   - options that escalate prematurely or hand the problem to someone else;
   - options that break rules (unapproved changes, ethics violations, bypassing the product owner);
   - options that use the wrong approach for the context (formal CCB for a backlog item in Scrum; "just start coding" in a regulated predictive project).
5. **Choose between the remaining two** by asking: *Which one happens first? Which one addresses the root cause? Which one is more aligned with servant leadership, value and the agreed process?*
6. **Watch for "first" vs "next".** "First" usually means analysis or understanding (review, assess, discuss). "Next" assumes something has already been done — read carefully what that is.
7. **Don't add facts.** Answer the question as written, not the situation you imagine.
8. **Flag and move on** if you're stuck for more than ~2 minutes. Never leave a question blank — there is no penalty for guessing.

21.3 Typical question patterns and their usual answers

Pattern	Usually best
A stakeholder asks for a change in a predictive project	Assess impact → change request → CCB decision → update plan → communicate
A stakeholder asks for a change in an agile project	Add to the backlog; product owner prioritises
A team member is underperforming	Private conversation to understand causes; coaching/training; then involve functional manager if needed
Two team members are in conflict	Let them try first; then facilitate a collaborative discussion
The sponsor wants a delivery date the team believes is impossible	Analyse with the team; present data and options (scope, resources, phasing); negotiate
A risk occurs that was identified	Implement the planned response (risk owner); update the register

Pattern	Usually best
An unidentified problem occurs	Treat it as an issue: assess, workaround, issue log, then identify new risks
A new stakeholder appears	Add to the stakeholder register, analyse, plan engagement
The team is missing sprint commitments	Discuss in the retrospective; look at capacity, WIP, impediments, estimation
A vendor is late	Review the contract and performance; communicate with the vendor through the agreed channel
The project is cancelled	Close the project: document, transfer, close contracts, lessons learned, release resources
A team member asks the PM to decide a technical approach (agile)	Coach the team to decide
The product owner is unavailable	Coach on the role; agree availability; escalate if needed
A regulatory change appears mid-project	Assess impact with experts; process changes; compliance is mandatory
A request to hide bad news	Refuse; report transparently with options

21.4 Time management and exam day

- **Pacing:** ~80 seconds per question. Checkpoints: Q60 at ~80 min, Q120 at ~160 min, Q180 at ~235 min.
- **Breaks:** take both. Before each break, review flagged questions in that section — you can't return after the break.
- **Calculations:** write the formula, then plug in numbers. Re-check signs (negative CV = over budget).
- **Case-study sets:** read the case once carefully, then refer back for each question; the questions may test different domains.
- **Multiple-response:** the question tells you how many to choose; partial credit is not given.
- **Matching/drag-and-drop:** start with the pairs you are sure of; use elimination for the rest.
- **Online proctored exams:** test your system in advance; clear your desk; have ID ready; no notes, phones or watches.
- **Physical readiness:** sleep, hydration, a light meal. Arrive early (or log in early).
- **Mental readiness:** expect hard questions — the exam feels difficult for almost everyone. Many unscored pretest items are deliberately difficult; don't let them shake you.

21.5 The final two weeks

- Take at least **three full-length timed simulations** (this package has ten).
- Review **every** explanation, especially where you chose the second-best answer — those reveal mindset gaps.
- Keep an **error log**: question, your answer, correct answer, why, which principle or process it tests.
- Re-read Chapters 4 (principles), 7 (governance/change), 15 (agile), 16 (people) and this chapter.
- Memorise the formula sheet (Chapter 20) and practise ten calculations per day.

- Stop heavy studying the day before the exam; review the mindset rules and rest.

Chapter 21 Review Questions

Answer all questions first, then check the answer key for Chapter 21 in Appendix B.

21.1 A question asks what the project manager should do FIRST when a team member reports that a critical supplier may deliver late. Which type of answer is most likely to be correct?

- **A.** Ignoring the report until the delay has been confirmed in writing by the supplier's management
- **B.** An action that gathers information and assesses the situation first
- **C.** Terminating the supplier's contract
- **D.** Escalating immediately to the sponsor

21.2 In an agile project, a senior manager asks the project manager to add a feature to the current sprint. What is the best response?

- **A.** Submit the request to the change control board, since all scope changes need formal approval
- **B.** Refuse the request because the sprint is locked
- **C.** Add the feature to the current sprint immediately
- **D.** Explain that the product owner prioritises new work, and connect them

21.3 Which answer pattern is usually WRONG on the PMP exam?

- **A.** Escalating to the sponsor before trying to understand or resolve the problem
- **B.** Meeting privately with a team member to discuss the concerns about their recent performance
- **C.** Updating the risk register when new risks are identified
- **D.** Facilitating a collaborative discussion to resolve conflict

21.4 Before taking the first optional break, what should a candidate do?

- **A.** Skip remaining questions to save time
- **B.** Nothing; breaks have no effect on navigation
- **C.** Review the flagged questions in the first section, since they cannot be revisited
- **D.** Leave the flagged questions to review at the end of the exam, when there is more time

21.5 The sponsor asks the project manager to report the project as "green" to the board although it is significantly behind schedule. What should the project manager do?

- **A.** Report the true status, with recovery options, and discuss the concern with the sponsor
- **B.** Report green as requested
- **C.** Avoid sending any report this month
- **D.** Report green as requested, but tell the team the truth so that they can recover the delay

21.6 A scenario describes a project that uses a WBS, a schedule baseline and a change control board. Which mechanism is most appropriate for handling a requested scope change?

- **A.** Add the request to the product backlog for the product owner to prioritise in the next sprint
- **B.** Integrated change control: assess the impact and submit a change request
- **C.** Implement it if it takes less than one day
- **D.** Ask the team to vote on it

21.7 Which two techniques help you choose between two remaining plausible answers? (Choose two.)

- **A.** Ask which option addresses the root cause rather than a symptom
- **B.** Choose the option that involves the most senior person
- **C.** Ask which option would logically happen first in the sequence
- **D.** Choose the longest option
- **E.** Choose the option with the most technical terms

21.8 What is a good pacing target for the PMP exam (180 questions, 240 minutes)?

- **A.** Pacing does not matter, because the time allowed for the PMP exam is effectively unlimited
- **B.** About 2 minutes per question
- **C.** About 80 seconds per question, reaching question 60 around the 80-minute mark
- **D.** About 30 seconds per question

Appendix A — Glossary of Key Terms

Definitions are written for exam preparation (in plain language). Terms are listed alphabetically.

Term	Definition
Acceptance criteria	Conditions that must be met before a deliverable or user story is accepted.
Accepted deliverables	Deliverables formally accepted by the customer or sponsor (output of Validate Scope).
Accuracy	How close a measured value is to the true value.
Activity	A distinct, scheduled portion of work performed during a project; decomposed from work packages.
Actual cost (AC)	The realised cost incurred for the work performed during a specific period.
Adaptive approach	A development approach in which requirements are elaborated and validated frequently during delivery (agile).
Affinity diagram	A technique that groups many ideas into themes.
Agreement	Any document or communication defining the initial intentions of a project, e.g., contract, MOU, SLA.
Analogous estimating	Estimating using historical data from a similar activity or project.
Assumption	A factor considered true, real or certain without proof, for planning purposes.
Assumption log	Record of assumptions and constraints throughout the project.
Backlog refinement	Ongoing clarification, splitting, estimation and ordering of backlog items.
BAC (budget at completion)	The sum of all budgets established for the work to be performed; equals the cost baseline total.
Baseline	The approved version of a work product, changed only through formal change control.
Benefits management plan	Describes how and when benefits will be delivered and measured, with a benefit owner.
Bidder conference	Meeting with prospective sellers before proposals are prepared, so all receive the same information.
Bottom-up estimating	Estimating by aggregating estimates of lower-level components.
Burndown / burnup chart	Graphs of remaining work / completed work over time.
Business case	Documented economic feasibility study used to justify a project and assess its continuing validity.
Change control board (CCB)	A group responsible for reviewing, evaluating, approving, deferring or rejecting changes.
Change log	A list of changes submitted and their status.

Term	Definition
Change request	A formal proposal to modify a document, deliverable or baseline (corrective, preventive, defect repair, update).
Charter (project)	Document issued by the sponsor that authorises the project and gives the PM authority.
Code of accounts	Numbering system that uniquely identifies each WBS element.
Configuration management	Managing the specifications of product and documents: identification, status accounting, verification and audit.
Constraint	A limiting factor that affects the execution of a project.
Contingency reserve	Time or money for identified risks (known-unknowns); part of the baseline.
Control account	A management control point where scope, budget and schedule are integrated for performance measurement.
Control chart	Graph of process data over time against control limits, used to determine stability.
Cost baseline	Approved time-phased budget, excluding management reserve.
Cost of quality (COQ)	Costs of conformance (prevention, appraisal) plus costs of non-conformance (internal, external failure).
Crashing	Shortening the schedule by adding resources to critical-path activities at the least cost.
Critical path	The longest path through a schedule network; determines the shortest project duration.
Cumulative flow diagram (CFD)	Chart showing work in each state over time; reveals bottlenecks.
Cycle time	Time from starting work on an item to completing it.
Decomposition	Dividing scope and deliverables into smaller, more manageable parts.
Definition of done (DoD)	Team checklist of criteria every increment must meet to be considered done.
Definition of ready (DoR)	Criteria a backlog item must meet before the team can start it.
Deliverable	Any unique, verifiable product, result or capability produced to complete a process, phase or project.
Development approach	Method used to create and evolve the product: predictive, iterative, incremental, adaptive, hybrid.
Earned value (EV)	The measure of work performed expressed in terms of the budget authorised for that work.
Emotional intelligence	Ability to identify, assess and manage one's own emotions and those of others.
Enterprise environmental factors (EEFs)	Conditions, not under the team's immediate control, that influence the project.
Escalate (risk strategy)	Passing a risk outside the project's scope or authority to the appropriate owner.

Term	Definition
Estimate at completion (EAC)	Expected total cost of completing all work.
Expected monetary value (EMV)	Probability × impact; sum across outcomes.
Fast tracking	Performing activities in parallel that were planned in sequence.
Float (slack)	Time an activity can be delayed: total float (without delaying the project), free float (without delaying successors).
Focus area	Eighth-Edition name for a process group: Initiating, Planning, Executing, Monitoring and Controlling, Closing.
Gold plating	Adding features or quality beyond requirements, not requested by the customer.
Governance	Framework of authority, accountability, decision rights and oversight directing the project.
Grade	Category assigned to products with the same use but different technical characteristics.
Hybrid approach	A combination of predictive and adaptive elements.
Increment	A usable, potentially releasable piece of the product.
Information radiator	A large, visible display of project information (task board, burn chart).
Issue	A current condition or situation that may affect objectives and must be resolved.
Issue log	Document recording issues, owners, actions and status.
Iteration	A timeboxed cycle of development (sprint).
Kanban	Method for managing flow by visualising work and limiting WIP.
Lag / Lead	Required delay / allowed acceleration between a predecessor and a successor.
Lead time	Time from a request to its delivery.
Lessons learned register	Document recording knowledge gained during the project, updated throughout.
Management reserve	Budget withheld for unidentified risks (unknown-unknowns); outside the cost baseline.
Milestone	A significant point or event; zero duration.
Minimum viable product (MVP)	Smallest product that allows learning whether the solution delivers value.
Monte Carlo simulation	Computer simulation that uses probability distributions to model possible outcomes.
Net present value (NPV)	Present value of inflows minus present value of outflows.
Organisational process assets (OPAs)	Plans, processes, policies, procedures and knowledge bases of the performing organisation.
Parametric estimating	Estimating using a statistical relationship between historical data and other variables.

Term	Definition
Performance domain	A group of related activities critical for the effective delivery of outcomes (seven in the Eighth Edition).
Performance measurement baseline (PMB)	Integrated scope, schedule and cost baselines used for earned value comparisons.
Planned value (PV)	The authorised budget assigned to scheduled work.
Portfolio	Projects, programs and operations managed as a group to achieve strategic objectives.
Precision	How close repeated measurements are to each other.
Predictive approach	A development approach in which scope, time and cost are determined early in the life cycle.
Principle	A fundamental norm, truth or value guiding behaviour (six in the Eighth Edition).
Product backlog	Ordered list of everything that might be needed in the product; owned by the product owner.
Product owner	Person accountable for maximising the value of the product and ordering the backlog.
Program	Related projects and activities managed in a coordinated way to obtain benefits.
Progressive elaboration	Iteratively increasing the level of detail as more information becomes available.
Project	A temporary endeavour undertaken to create a unique product, service or result.
Project management plan	Document describing how the project will be executed, monitored and controlled, and closed.
Quality	Degree to which inherent characteristics fulfil requirements.
RACI chart	Responsibility matrix: Responsible, Accountable, Consulted, Informed.
Requirements traceability matrix	Grid linking requirements from their origin to the deliverables and tests that satisfy them.
Residual risk	Risk that remains after responses are implemented.
Resource levelling / smoothing	Adjusting dates to balance resource demand (may change the end date) / within float (does not).
Risk	An uncertain event or condition that, if it occurs, has a positive or negative effect on objectives.
Risk appetite / threshold	Willingness to accept uncertainty / measurable level of acceptable variation.
Risk register	Repository of identified risks, their analysis and responses.
Rolling wave planning	Planning near-term work in detail and future work at a higher level.
Scope baseline	Approved scope statement, WBS and WBS dictionary.

Term	Definition
Scope creep	Uncontrolled expansion of scope without adjustments to time, cost and resources.
Secondary risk	Risk arising as a direct result of implementing a risk response.
Servant leadership	Leadership that focuses on serving the team: removing impediments, coaching, empowering.
Sourcing strategy plan	Eighth-Edition plan describing what will be acquired externally, how and from whom.
Sponsor	Person or group providing resources and support for the project and accountable for enabling success.
Sprint	Scrum's timeboxed iteration of one month or less.
Stakeholder	Anyone who may affect, be affected by, or perceive itself to be affected by the project.
Stakeholder register	Document with identification, assessment and classification of stakeholders.
Story points	Relative units estimating the size and complexity of backlog items.
Sustainability	Meeting present needs without compromising the future — economic, environmental and social.
Tailoring	Deliberate adaptation of approach, governance and processes to the context.
Team charter	Document of team values, agreements and operating guidelines.
Throughput	Number of items completed per unit of time.
Timebox	A fixed period of time for an activity.
To-complete performance index (TCPI)	Cost efficiency needed on remaining work to meet a target (BAC or EAC).
Trigger condition	Event or situation indicating a risk is about to occur.
User story	Short description of value for a user: "As a... I want... so that...".
Validate Scope	Process of formalising acceptance of completed deliverables.
Value	The worth, importance or usefulness of something to stakeholders.
Velocity	Amount of work (story points) a team completes per iteration.
Work breakdown structure (WBS)	Hierarchical decomposition of the total scope of work.
Work package	Lowest level of the WBS, for which cost and duration are estimated and managed.
Work performance data / information / reports	Raw observations / analysed data compared with plans / compiled documents for decisions.
Workaround	An unplanned response to a risk that occurred or an issue.

Appendix B — Answer Keys for the Chapter Review Questions

Each answer includes a short explanation. If you missed a question, re-read the related section before moving on.

Chapter 1 — The PMP® Exam (2026) and How the PMBOK® Guide Fits

1.1 — Answer: A

Correct: A. 170

The exam contains 180 questions, of which 170 are scored and 10 are unscored pretest items. The figures 175 scored and 230 minutes belong to the earlier format.

1.2 — Answer: B

Correct: B. 26%

Business Environment rose from 8% to 26%. People is 33% and Process is 41%.

1.3 — Answer: A

Correct: A. Business Environment

In the 2026 outline, risk planning and management is a Business Environment task, together with governance, compliance, change control, issues and impediments, continuous improvement, organisational change and the external environment.

1.4 — Answer: B

Correct: B. 36 months

With a four-year degree the requirement is 36 months of experience leading projects within the last eight years. 24 months applies to GAC-accredited degrees and 60 months to a secondary diploma or associate's degree.

1.5 — Answer: B

Correct: B. 40%

About 40% of items reflect predictive approaches; the remaining ~60% reflect agile or hybrid approaches.

1.6 — Answer: D, E

Correct: D. Align stakeholder expectations / E. Help ensure knowledge transfer

People contains: develop a common vision, manage conflicts, lead the team, engage stakeholders, align stakeholder expectations, manage stakeholder expectations, help ensure knowledge transfer, and plan and manage communication. Evaluate project status and procurement are Process tasks; organisational change is a Business Environment task.

1.7 — Answer: C

Correct: C. The exam follows the content outline; the PMBOK Guide is a core reference for questions

The PMP is based on the ECO. The PMBOK Guide is a key reference among several, and questions test the application of its concepts in realistic situations rather than memorisation of text.

1.8 — Answer: C

Correct: C. A performance rating, such as Above Target or Needs Improvement

PMI reports an overall Pass/Fail result plus a performance rating per domain. It does not publish raw scores or a fixed pass mark.

1.9 — Answer: D

Correct: D. They can no longer be reviewed or changed

Once you start a break, the previous block of questions is locked. Review flagged questions before taking the break.

1.10 — Answer: D

Correct: D. The best action may differ by context, e.g. backlog reprioritisation instead of a change request

The values behind the answers are the same, but the mechanisms differ: in an agile context new work usually goes to the product backlog for the product owner to prioritise, while in a predictive context a change to a baseline goes through integrated change control.

1.11 — Answer: 1→A, 2→B, 3→D, 4→C

Correct pairs: 1. Scored questions → 170; 2. Exam duration → 240 minutes; 3. Weight of the People domain → 33%; 4. Weight of the Process domain → 41%

180 questions are delivered (170 scored + 10 pretest) in 240 minutes. People is 33%, Process 41% and Business Environment 26%.

Chapter 2 — PMBOK® Guide 8 at a Glance: Structure, Evolution and Core Vocabulary

2.1 — Answer: C

Correct: C. 6 principles, 7 performance domains, 5 focus areas and 40 processes

The Eighth Edition is organised around 6 principles, 7 performance domains, 5 focus areas and 40 processes. Twelve principles and eight domains describe the Seventh Edition; ten knowledge areas and 49 processes describe the Sixth.

2.2 — Answer: C

Correct: C. Quality

The seven domains are Governance, Scope, Schedule, Finance, Stakeholders, Resources and Risk. Quality is embedded across domains: planning in Define Scope, assurance in Governance and control in Monitor and Control Scope.

2.3 — Answer: C

Correct: C. Higher customer retention and lower processing costs

The output is the system, the outcome is the faster processing, and the benefit is the gain the organisation realises from that outcome, such as retention and cost savings.

2.4 — Answer: B

Correct: B. A program coordinates related components for added benefits; a portfolio groups components to meet strategy

Programs are about coordination of related work for benefits; portfolios are about selecting and prioritising investments for strategy. Portfolio components need not be related.

2.5 — Answer: A

Correct: A. The lessons learned repository from previous projects

OPAs are plans, processes, policies, procedures and knowledge bases owned by the performing organisation, such as historical information and lessons learned. Regulations, culture and market conditions are EEFs.

2.6 — Answer: D

Correct: D. Stakeholders performance domain

Plan Communications Management, Manage Communications and Monitor Communications are processes of the Stakeholders domain, reflecting the close link between communication and engagement.

2.7 — Answer: A

Correct: A. Temporary means a definite beginning and end, not a short duration

Temporary refers to a definite start and end. A project can run for years, and the product or outcome it creates can be long-lasting.

2.8 — Answer: D

Correct: D. Principles alone were too abstract, so processes returned as non-prescriptive guidance

The Eighth Edition keeps the principles and domains of the Seventh Edition but reconnects them with practical, non-prescriptive processes that apply to predictive, adaptive and hybrid approaches.

2.9 — Answer: B, C

Correct: B. Artificial Intelligence / C. Procurement

The Eighth Edition appendices include the PMO (X2), Artificial Intelligence (X3) and Procurement (X4). Earned value, the Agile Manifesto and business analysis are covered elsewhere, not as these appendices.

2.10 — Answer: B

Correct: B. Replacing the software is a project; running monthly payroll is operations

The replacement is temporary and unique, so it is a project. Monthly payroll is ongoing and repetitive, so it is operations, which the project changed.

2.11 — Answer: 1→C, 2→A, 3→B, 4→D

Correct pairs: 1. Principles → Why — which beliefs and behaviours should guide decisions?; 2. Performance domains → What — which areas of activity must be managed effectively?; 3. Processes and focus areas → How — which work is performed, in what order, with which tools?; 4. Tailoring → Fit — how should the approach be adapted to this context?

Principles give the why, domains the what, and processes the how; tailoring adapts all of them to the specific project, organisation and environment.

2.12 — Answer: B

Correct: B. An enterprise environmental factor

External regulations are conditions not under the immediate control of the team that constrain or influence the project, which defines an enterprise environmental factor.

Chapter 3 — The Standard: A System for Value Delivery, Governance and the Project Environment

3.1 — Answer: C

Correct: C. Project charter

The charter, issued by the sponsor or initiator, formally authorises the project and documents the PM's authority. The business case justifies the investment and the benefits plan describes how benefits will be realised.

3.2 — Answer: A

Correct: A. The business case

The business case contains the justification and cost-benefit analysis for the investment. Phase gates are the natural point to confirm it is still valid and decide to continue, change or stop the project.

3.3 — Answer: D

Correct: D. The sponsor or the designated benefit owner

The benefits management plan names a benefit owner, often the sponsor or an operational manager, who is accountable for realising benefits after transition. The PM's accountability normally ends at project closure.

3.4 — Answer: C

Correct: C. Functional

In a functional structure the functional manager controls resources and budget; the PM role is part-time with little or no authority.

3.5 — Answer: C

Correct: C. Negotiate with the functional manager for the engineer's time

In a matrix the functional manager controls the resource, so the PM must negotiate and use influence. Escalation is reserved for when negotiation fails; bypassing the manager breaks governance.

3.6 — Answer: D

Correct: D. The organisation's culture and appetite for risk

Culture and risk appetite influence the project but are not under the team's immediate control, so they are EEFs. Templates, repositories and procedures are organisational process assets.

3.7 — Answer: B, C

Correct: B. The benefit owner responsible for realisation / C. The timeframe for realising target benefits

A benefits management plan describes target benefits, strategic alignment, timeframe, benefit owner, metrics, assumptions and risks. Critical path, ground rules and WBS details belong to other artifacts.

3.8 — Answer: A

Correct: A. Strategy guides the work, and performance and benefit information lets leaders adjust strategy

Leadership sets direction downward; projects and operations feed back performance, lessons and benefit data upward. Without honest upward feedback, the organisation cannot rebalance its portfolio.

3.9 — Answer: B

Correct: B. A product-oriented approach that delivers value continuously

Long-lived product teams that deliver value continuously reflect product management considerations described in the Standard, distinct from temporary project teams.

3.10 — Answer: D

Correct: D. Power Skills

Power Skills are the interpersonal skills; Ways of Working covers delivery approaches and techniques; Business Acumen covers strategy, finance and market understanding.

3.11 — Answer: A

Correct: A. Develop it with the sponsor, who issues and signs it

The PM often helps develop the charter, but it is issued by the sponsor or initiator at a level appropriate to fund and authorise the project. The PM cannot authorise their own project.

3.12 — Answer: B

Correct: B. Operations

Once deliverables are transitioned, operations run and sustain them, enabling benefits over time. Projects create the outputs; operations keep them working.

Chapter 4 — The Six Principles of Project Management

4.1 — Answer: B

Correct: B. Adopt a holistic view

Considering the ripple effects of one decision on several areas and on the wider environment is systems thinking, the core of adopting a holistic view.

4.2 — Answer: D

Correct: D. Re-evaluate the business case with the sponsor and consider changing direction

Value, not completion, is the measure of success. Continuing only because of money already spent is sunk-cost thinking; the right action is to revisit the business case with the decision-maker.

4.3 — Answer: C

Correct: C. Agreeing on a definition of done and automated tests before development starts

Embedding quality means building it in from the start through clear criteria and preventive practices. Final inspection alone is detection, gold plating adds risk without value, and quality is everyone's responsibility.

4.4 — Answer: B

Correct: B. Inform the sponsor promptly, correct the report and explain the planned actions

Accountable leaders are transparent and own their mistakes. Correcting the record promptly preserves trust; hiding or shifting blame violates the Code of Ethics.

4.5 — Answer: C

Correct: C. Integrate sustainability within all project areas

Sustainability requires considering environmental and social impacts in sourcing decisions. Value is broader than price, and a holistic view would also consider reputational and regulatory risk.

4.6 — Answer: A

Correct: A. Encourage the team to decide the breakdown and offer support

Self-organising teams decide how to do their work. The PM supports and coaches rather than dictating, which builds ownership and capability.

4.7 — Answer: B, C

Correct: B. Reporting a project as green although critical milestones are slipping / C. Accepting a valuable gift from a bidder during an open tender

Misleading status reports and conflicts of interest breach transparency and integrity. Escalating through the agreed path, admitting mistakes and seeking input are accountable behaviours.

4.8 — Answer: D

Correct: D. Because embedding quality makes it part of every domain, especially Governance and Scope

Quality is integrated rather than isolated: it is planned with scope, assured through governance and controlled with scope, reflecting that it is everyone's responsibility.

4.9 — Answer: A

Correct: A. Integrate sustainability, including the team's social sustainability and a sustainable pace

Social sustainability includes people's well-being and a sustainable pace. Persistent overtime also damages quality and engagement, so it threatens several principles, but sustainability is the most direct.

4.10 — Answer: B

Correct: B. Optimise risk responses

Optimise risk responses was absorbed into the Risk performance domain. The other three map to the principles Focus on value, Be an accountable leader and Build an empowered culture.

4.11 — Answer: C

Correct: C. Lower call-centre volume and higher customer satisfaction after launch

Value-oriented metrics measure outcomes and benefits for the organisation and customers. Code volume, spend and meeting counts measure activity, not value.

4.12 — Answer: D

Correct: D. They are non-prescriptive and guide behaviour in every type of project

Principles are fundamental norms that guide behaviour; they do not prescribe specific procedures and they are consistent with, not a replacement for, the Code of Ethics.

Chapter 5 — Life Cycles and Development Approaches

5.1 — Answer: A

Correct: A. Iterative

Repeated cycles that improve the same solution through feedback describe an iterative approach. Incremental delivers usable pieces over time; predictive plans once and builds once.

5.2 — Answer: D

Correct: D. Incremental delivery

Delivering usable portions of the product over time is incremental. Agile approaches combine incremental delivery with iterative refinement.

5.3 — Answer: B

Correct: B. A phase gate

A phase gate (stage gate, kill point) is a decision point at the end of a phase. The decision can be to continue, continue with modifications, repeat the phase or end the project.

5.4 — Answer: A

Correct: A. Well-understood requirements, and changes after construction starts are very expensive

Stable, clear requirements and a high cost of change favour planning up front. The other characteristics favour adaptive approaches.

5.5 — Answer: A

Correct: A. Hybrid

Combining adaptive delivery for the uncertain UI with a predictive approach for the regulated, sequential integration is a hybrid approach.

5.6 — Answer: B

Correct: B. Probe, sense, respond — run safe-to-fail experiments and adapt

Complex situations have emergent cause and effect, so experimentation and feedback (as in agile) are appropriate. Best practice suits clear situations, expert analysis suits complicated ones, and acting first suits chaos.

5.7 — Answer: A, C

Correct: A. The likelihood of change in the project / C. The team's access to the customer or business representative

The filter assesses culture (buy-in, trust, decision-making), team (size, experience, access to the business) and project factors (likelihood of change, criticality, incremental delivery).

5.8 — Answer: D

Correct: D. Changes are possible but go through formal change control against the baselines

Predictive approaches constrain change through baselines and change control, but changes still occur and are evaluated formally.

5.9 — Answer: D

Correct: D. Rolling wave planning

Rolling wave planning is a form of progressive elaboration: near-term work is planned in detail, future work at a higher level.

5.10 — Answer: C

Correct: C. Continuous delivery

Releasing features as soon as they are ready through automation is continuous delivery, typical of DevOps environments.

5.11 — Answer: C

Correct: C. Predictive

High agreement and high certainty represent a simple situation in which detailed up-front planning works well.

5.12 — Answer: B

Correct: B. Release plan or product roadmap

Roadmaps and release plans show features over time at a high level; iteration plans then add detail for each sprint.

Chapter 6 — Focus Areas and the 40 Processes

6.1 — Answer: C

Correct: C. 19

The distribution is Initiating 2, Planning 19, Executing 8, Monitoring and Controlling 10, Closing 1, totalling 40 processes.

6.2 — Answer: C

Correct: C. Initiate Project or Phase and Identify Stakeholders

Initiating contains only Initiate Project or Phase (the charter) and Identify Stakeholders.

6.3 — Answer: B

Correct: B. Assess and Implement Changes

Assess and Implement Changes, in the Governance domain, is the integrated change control process.

6.4 — Answer: D

Correct: D. Develop Schedule

The Schedule domain has three processes. Develop Schedule absorbs activity definition, sequencing and duration estimating together with building the schedule model.

6.5 — Answer: B

Correct: B. None; the risk register is a project document updated as needed

Project documents such as the risk register, issue log and lessons learned register are updated as the project progresses. Changes to baselines and the project management plan require formal change control.

6.6 — Answer: A

Correct: A. The WBS dictionary

The scope baseline consists of the approved project scope statement, the WBS (scope structure) and the WBS dictionary.

6.7 — Answer: B

Correct: B. Work performance data

Raw observations from execution are work performance data. Once compared with the plan and analysed, they become information; compiled for stakeholders, they become reports.

6.8 — Answer: D

Correct: D. Corrective action

Corrective action realigns current performance with the plan. Preventive action addresses future performance, defect repair fixes non-conforming products, and an update changes controlled documents.

6.9 — Answer: B, E

Correct: B. In adaptive approaches the focus areas occur within each iteration / E. Monitoring and Controlling activities occur throughout the project

Focus areas group work by its nature; they repeat in every phase or iteration and monitoring runs throughout. Closing applies to phases and contracts as well, and everything can be tailored.

6.10 — Answer: D

Correct: D. Governance

Governance has Initiate Project or Phase (initiating), two planning processes, three executing, two monitoring and controlling, and Close Project or Phase (closing).

6.11 — Answer: A

Correct: A. It includes contingency reserves but excludes management reserves

The cost baseline is the approved time-phased budget including contingency reserves for identified risks. Management reserves, for unknown-unknowns, are part of the project budget but not the baseline.

6.12 — Answer: 1→C, 2→A, 3→D, 4→B

Correct pairs: 1. Create WBS → Develop Scope Structure; 2. Develop Team and Manage Team → Lead the Team; 3. Plan Cost Management → Plan Financial Management; 4. Collect Requirements → Elicit and Analyze Requirements

The Eighth Edition renames and merges several processes: Create WBS became Develop Scope Structure, the two team processes became Lead the Team, Plan Cost Management became Plan Financial Management and Collect Requirements became Elicit and Analyze Requirements.

6.13 — Answer: A

Correct: A. Monitor and Control Project Performance

Monitor and Control Project Performance consolidates work performance information from the control processes into work performance reports. Manage Communications then distributes them.

Chapter 7 — Governance Performance Domain

7.1 — Answer: B

Correct: B. Time and materials with a not-to-exceed limit

When scope cannot yet be defined but work must start, T&M (ideally capped) or cost-reimbursable contracts are appropriate. Fixed-price contracts require a well-defined scope.

7.2 — Answer: A

Correct: A. Firm fixed price

In an FFP contract the seller must deliver for the agreed price regardless of actual costs. Cost-reimbursable contracts shift most risk to the buyer, and T&M sits in between.

7.3 — Answer: B

Correct: B. Scope creep; it should go through integrated change control

Uncontrolled additions requested by the customer are scope creep. Gold plating is when the team adds extras on its own initiative. The request should be captured as a change request and evaluated before implementation.

7.4 — Answer: D

Correct: D. Evaluate its impact on scope, schedule, cost, quality, risk and value

Before a decision, the PM analyses the impact and options so the decision-maker can make an informed choice. Implementing before approval and rejecting without analysis are both wrong.

7.5 — Answer: C

Correct: C. Explicit knowledge can be codified; tacit knowledge is personal and shared through interaction

Explicit knowledge is easily written down; tacit knowledge such as experience and judgment is transferred through conversation, observation and collaboration.

7.6 — Answer: D

Correct: D. Throughout the project, then transferred to the organisation's repository at closure

Lessons learned are captured continuously so they can improve the current project as well as future ones; at closure they are transferred to the organisational lessons learned repository.

7.7 — Answer: C

Correct: C. Obtain formal acceptance before completing closure

Closure requires formal acceptance, administrative closure, contract closure and lessons learned. Releasing resources before formal acceptance risks being unable to address remaining issues.

7.8 — Answer: C

Correct: C. Manage Quality Assurance

Manage Quality Assurance in the Governance domain focuses on processes, using audits and process analysis. Inspecting deliverables is quality control within Monitor and Control Scope; formal acceptance is Validate Scope.

7.9 — Answer: A

Correct: A. Fixed price incentive fee

FPIF combines a fixed-price structure with a target cost, target profit, share ratio and ceiling price, rewarding efficient performance.

7.10 — Answer: B

Correct: B. Request for quotation

An RFQ is used when the item is well defined and price is the main criterion. RFPs suit complex solutions; RFIs gather market information early.

7.11 — Answer: D, E

Correct: D. Settling open claims and making final payments / E. Providing formal written notice that the contract is complete

Contract closure includes verifying deliverables, settling claims, final payments, formal notice of completion and archiving. RFPs, SOWs and bidder conferences belong to earlier procurement activities.

7.12 — Answer: B

Correct: B. Perform closure: document the status, transfer completed work, close contracts and capture lessons

Even when a project is terminated early, closure is needed to protect the organisation, settle obligations and preserve knowledge.

7.13 — Answer: D

Correct: D. Add it to the product backlog for the product owner to prioritise

In adaptive work, change is welcome but flows through the backlog, which the product owner orders by value. The sprint goal is normally protected from mid-sprint changes.

7.14 — Answer: A

Correct: A. Configuration verification and audit

Identification defines controlled items, status accounting records and reports their state, and verification and audit confirm correctness and implementation of approved changes.

7.15 — Answer: C

Correct: C. 40 days

Break-even: 12,000 + 100d = 400d, so 300d = 12,000 and d = 40. For more than 40 days, buying is cheaper.

Chapter 8 — Scope Performance Domain

8.1 — Answer: A

Correct: A. The WBS includes all of the project's work, and only that work

The 100% rule means the total of the work at the lowest levels rolls up to the total scope of the project, with nothing missing and nothing extra.

8.2 — Answer: C

Correct: C. Work package

Work packages are the lowest level of the WBS. Activities are decomposed from work packages in schedule development and are not part of the WBS itself.

8.3 — Answer: D

Correct: D. Validate Scope

Validate Scope takes verified deliverables (which passed quality control) and obtains formal acceptance from the customer or sponsor.

8.4 — Answer: C

Correct: C. Gold plating

Gold plating is the team adding unrequested features on its own initiative. Scope creep is uncontrolled expansion, often from customer requests, without adjusting time and cost.

8.5 — Answer: A

Correct: A. Requirements traceability matrix

The RTM traces requirements backward to their origin and forward to design, deliverables and tests, helping to ensure value and assess change impact.

8.6 — Answer: A

Correct: A. Facilitate a workshop with both stakeholders to reach agreement

Facilitated workshops reconcile differences and build consensus. Choosing by seniority or doing both ignores value and creates waste.

8.7 — Answer: B

Correct: B. Acceptance criteria are story-specific; the definition of done applies to every item

Acceptance criteria describe what a particular story must do to be accepted. The DoD is a shared checklist of quality conditions that every increment must meet.

8.8 — Answer: A

Correct: A. Project exclusions

Explicitly stating what is out of scope manages expectations and prevents misunderstandings later.

8.9 — Answer: B, E

Correct: B. Negotiable / E. Testable

INVEST stands for Independent, Negotiable, Valuable, Estimable, Small and Testable. Value must be to the user or customer.

8.10 — Answer: B

Correct: B. Non-functional requirement

Performance, capacity, security and usability qualities are non-functional requirements. Functional requirements describe behaviours of the product.

8.11 — Answer: C

Correct: C. They become expected basic features as customers get used to them

Delighters lose their novelty and become performance or basic expectations; yesterday's innovation becomes today's standard.

8.12 — Answer: B

Correct: B. The product owner

The product owner is accountable for maximising value and ordering the backlog, informed by stakeholders and the developers' input on risks and dependencies.

8.13 — Answer: D

Correct: D. Within Monitor and Control Scope

Quality control merged into Monitor and Control Scope, producing verified deliverables. Assurance of processes is Manage Quality Assurance in Governance, and formal acceptance is Validate Scope.

8.14 — Answer: D

Correct: D. A minimum viable product

An MVP is the smallest product that validates the key assumption with real users, supporting the principle of focusing on value.

Chapter 9 — Quality: Embedded Across Governance and Scope

9.1 — Answer: A

Correct: A. The first is low grade but high quality; the second is high grade but low quality

Grade refers to the category of features; quality refers to how well requirements are met. Low grade can be acceptable; low quality is always a problem.

9.2 — Answer: A

Correct: A. External failure costs

Failures discovered by the customer, such as warranty work and liability, are external failure costs and are usually the most expensive.

9.3 — Answer: C

Correct: C. The process may be out of control and the cause should be investigated

Under the rule of seven, seven consecutive points on one side of the mean indicate a non-random pattern, so the process should be investigated even though no point is outside the limits.

9.4 — Answer: C

Correct: C. Pareto diagram

A Pareto diagram ranks causes by frequency, highlighting the vital few that produce the majority of problems.

9.5 — Answer: D

Correct: D. Auditing whether the team follows the agreed testing process

Assurance focuses on processes, for example quality audits. Inspecting and testing deliverables are quality control activities.

9.6 — Answer: B

Correct: B. Precise but not accurate

Tightly clustered results are precise; being consistently off the true value means they are not accurate.

9.7 — Answer: B

Correct: B. The customer or the requirements

Specification limits reflect customer requirements. Control limits are calculated from process data, typically at plus or minus three sigma.

9.8 — Answer: D

Correct: D. Keep the definition of done and adjust the sprint scope with the product owner

Quality is not negotiable within the definition of done. When time is short, the scope of the sprint is negotiated with the product owner rather than lowering quality.

9.9 — Answer: C, D

Correct: C. Training the team in a new coding standard / D. Design reviews before construction

Prevention costs are incurred to avoid defects, such as training and design reviews. Rework is internal failure, recalls are external failure and inspection is appraisal.

9.10 — Answer: C

Correct: C. Scatter diagram

A scatter diagram plots two variables to reveal correlation. It does not by itself prove causation.

9.11 — Answer: B

Correct: B. As part of Define Scope

Quality planning has been incorporated into Define Scope, where quality requirements, metrics and the approach to verification are defined alongside the scope.

9.12 — Answer: D

Correct: D. Philip Crosby

Crosby emphasised conformance to requirements, zero defects and the idea that the cost of non-conformance makes investment in quality pay for itself.

Chapter 10 — Schedule Performance Domain

10.1 — Answer: 10

PERT expected duration = (O + 4M + P) / 6 = (6 + 36 + 18) / 6 = 60 / 6 = 10 days.

10.2 — Answer: B

Correct: B. 2 days

Standard deviation = (P − O) / 6 = (18 − 6) / 6 = 2 days.

10.3 — Answer: D

Correct: D. Fast tracking critical-path activities

Fast tracking overlaps activities that were planned in sequence. It adds risk and potential rework but little or no cost. Crashing adds cost; smoothing does not shorten the project.

10.4 — Answer: B

Correct: B. Resource smoothing

Resource smoothing works only within float, so the end date does not change. Resource levelling may extend the schedule and change the critical path.

10.5 — Answer: C

Correct: C. 14 days

Paths: A-B-E = 4 + 6 + 2 = 12; A-C-D-E = 4 + 3 + 5 + 2 = 14. The longest path, A-C-D-E, is critical at 14 days.

10.6 — Answer: A

Correct: A. 2 days

B can start at day 4 and finish at day 10. E cannot start before day 12 (after D). B's late finish is 12, late start 6, so total float = 6 − 4 = 2 days.

10.7 — Answer: D

Correct: D. Mandatory

A mandatory (hard logic) dependency is inherent in the nature of the work. Discretionary dependencies reflect preferences and are the usual targets of fast tracking.

10.8 — Answer: C

Correct: C. 4 weeks

Cycle time = WIP / throughput = 12 / 3 = 4 weeks. Reducing WIP is the most direct way to reduce cycle time.

10.9 — Answer: C

Correct: C. 24 weeks

300 / 25 = 12 sprints; 12 sprints × 2 weeks = 24 weeks, assuming velocity and scope remain stable.

10.10 — Answer: D

Correct: D. Burnup chart with a total-scope line

A burnup chart shows completed work against a separate total-scope line, so additions to scope are visible as the scope line rises.

10.11 — Answer: A

Correct: A. Velocity is team-specific and should not be used to compare teams

Story points are relative to each team, so velocities are not comparable. Using velocity for comparison encourages point inflation and undermines trust.

10.12 — Answer: D, E

Correct: D. It is the longest path through the schedule network / E. Activities on it usually have zero total float

The critical path is the longest path and determines the shortest project duration; its activities normally have zero float. It can change during the project as activities are delayed or completed.

10.13 — Answer: A

Correct: A. A bottleneck in testing

A widening band means work is accumulating in that state faster than it leaves it, indicating a bottleneck that limits flow.

10.14 — Answer: B

Correct: B. Start-to-finish

Finish-to-start is the most common; start-to-finish is rarely used, for example in shift handovers.

Chapter 11 — Finance Performance Domain

11.1 — Answer: 0.90

CPI = EV / AC = 45,000 / 50,000 = 0.90. The project gets 90 cents of work for every dollar spent, so it is over budget.

11.2 — Answer: D

Correct: D. $625,000

CPI = 200,000 / 250,000 = 0.8. EAC = BAC / CPI = 500,000 / 0.8 = $625,000.

11.3 — Answer: A

Correct: A. −$20,000

SV = EV − PV = 200,000 − 220,000 = −$20,000, so the project is behind schedule.

11.4 — Answer: C

Correct: C. 1.20

TCPI = (BAC − EV) / (BAC − AC) = 300,000 / 250,000 = 1.20. The team would need to be 20% more efficient than planned on the remaining work.

11.5 — Answer: B

Correct: B. Contingency reserve

The cost baseline includes contingency reserves for identified risks. Management reserves for unknown-unknowns are added to obtain the project budget but are excluded from the baseline.

11.6 — Answer: A

Correct: A. Ahead of schedule and over budget

SPI above 1 means more work was done than planned (ahead). CPI below 1 means the work cost more than budgeted (over budget).

11.7 — Answer: D

Correct: D. −25% to +75%

Early rough order of magnitude estimates are typically −25% to +75%. Definitive estimates later in planning are −5% to +10%.

11.8 — Answer: A

Correct: A. Project A; opportunity cost $95,000

NPV already accounts for timing, so choose the higher NPV. The opportunity cost is the value of the option not chosen: $95,000.

11.9 — Answer: C

Correct: C. $310,000

For atypical variance, EAC = AC + (BAC − EV) = 80,000 + 230,000 = $310,000.

11.10 — Answer: C

Correct: C. Sunk costs should not drive decisions about future investment

Money already spent cannot be recovered regardless of the decision. The decision should be based on future costs, benefits and value.

11.11 — Answer: C, E

Correct: C. It addresses unknown-unknowns / E. Using it normally requires approval from management and a change to the cost baseline

Management reserve covers unidentified risks; it is outside the cost baseline and its use requires management approval and a baseline change.

11.12 — Answer: B

Correct: B. 4 years

Payback = initial investment / annual net inflow = 400,000 / 100,000 = 4 years.

11.13 — Answer: B

Correct: B. $200,000

PV = FV / (1 + r)^n = 242,000 / 1.21 = $200,000.

11.14 — Answer: D

Correct: D. Perform funding limit reconciliation and adjust the schedule of work if needed

Funding limit reconciliation compares planned expenditure with funding constraints and may require rescheduling work so spending does not exceed available funds.

Chapter 12 — Stakeholders Performance Domain (including Communications)

12.1 — Answer: C

Correct: C. 38

Channels = n(n − 1)/2. With 8 people: 28. With 12 people: 66. The increase is 66 − 28 = 38.

12.2 — Answer: C

Correct: C. Keep satisfied

High power, low interest stakeholders should be kept satisfied so they do not become obstacles. High–high is manage closely; low power–high interest is keep informed.

12.3 — Answer: A

Correct: A. Power, urgency and legitimacy

The salience model uses power, urgency and legitimacy. Grids use pairs such as power/interest or impact/influence.

12.4 — Answer: B

Correct: B. Plan engagement actions to understand and address the manager's concerns

The gap between current and desired engagement drives targeted engagement actions. Understanding the concerns is the first step to moving the stakeholder toward support.

12.5 — Answer: D

Correct: D. Pull communication via a shared repository

Pull communication suits large volumes of information and large audiences who access it at their discretion.

12.6 — Answer: A

Correct: A. Interactive communication, such as a facilitated meeting

Complex or sensitive topics that require mutual understanding and decisions are best handled through real-time, multidirectional interactive communication.

12.7 — Answer: C

Correct: C. Add her to the stakeholder register and analyse her interests

Stakeholder identification is continuous. A newly discovered stakeholder must be recorded and analysed so that engagement can be planned.

12.8 — Answer: D

Correct: D. The sender, by checking for feedback

The sender encodes the message clearly and confirms it was understood; the receiver decodes it, acknowledges receipt and provides feedback.

12.9 — Answer: B, D

Correct: B. The frequency and format of each communication / D. The person responsible for communicating each type of information

The plan defines what, when, how often, in what format, by whom, to whom and through which methods information is communicated, plus escalation and constraints.

12.10 — Answer: B

Correct: B. Discuss her information needs and adjust the communication approach

Monitoring communications means checking whether information needs are met and adapting the approach, for example an executive summary or a dashboard.

12.11 — Answer: B

Correct: B. Information radiators such as a visible task board and burndown chart

Information radiators make progress, impediments and work in progress highly visible and support transparency in adaptive teams.

12.12 — Answer: D

Correct: D. Outward

Outward refers to stakeholders outside the project team and organisation. Upward is senior management, downward is the team and sideward is peers.

12.13 — Answer: A

Correct: A. It may contain sensitive assessments of stakeholders' attitudes

Notes on resistance and strategies to change attitudes can be politically sensitive, so the PM uses judgment about how widely to share them.

Chapter 13 — Resources Performance Domain (including Procurement)

13.1 — Answer: D

Correct: D. Exactly one

There should be only one Accountable person per activity to avoid confusion about ownership. Several people can be Responsible, Consulted or Informed.

13.2 — Answer: B

Correct: B. The team members together, facilitated by the project manager

The team charter captures values, agreements and operating guidelines. It is most effective when the team develops it collaboratively and revisits it.

13.3 — Answer: B

Correct: B. Answer through the formal process so that all sellers receive the same information

Fairness requires that all bidders receive the same information at the same time. Clarifications are shared with all prospective sellers through the formal channel.

13.4 — Answer: C

Correct: C. Alternative dispute resolution, such as mediation or arbitration

Claims are resolved preferably through negotiation, then alternative dispute resolution; litigation is the last resort.

13.5 — Answer: $199,000

Underrun = $20,000; seller's share = 20% × 20,000 = $4,000; fee = 15,000 + 4,000 = $19,000; total = 180,000 + 19,000 = $199,000.

13.6 — Answer: C

Correct: C. Discuss it privately with the team member to understand the causes

Performance issues are addressed early, privately and respectfully, focusing on causes and support. Escalation or replacement come later if needed.

13.7 — Answer: A

Correct: A. Monitor and Control Resourcing

Monitor and Control Resourcing covers physical and virtual resources and externally sourced resources. People performance is managed through Lead the Team.

13.8 — Answer: A

Correct: A. Raise the concern through the contract channels with the seller's management

The buyer manages the seller through the contract and the seller's management, not by directing the seller's employees. Concerns are raised formally and documented.

13.9 — Answer: A

Correct: A. To give all prospective sellers the same, clear understanding of the procurement

Bidder conferences are held before proposals are submitted so all prospective sellers receive the same information, which supports fairness.

13.10 — Answer: A, D

Correct: A. Conflict resolution process / D. Decision-making process

Team charters include values, communication guidelines, decision-making and conflict resolution processes, meeting guidelines and team agreements.

13.11 — Answer: D

Correct: D. The halo effect

The halo effect is assuming someone will be good at everything because they excel at one thing.

13.12 — Answer: C

Correct: C. $120,000

PTA = ((ceiling price – target price) / buyer share) + target cost = (15,000 / 0.75) + 100,000 = $120,000.

13.13 — Answer: B

Correct: B. An amount withheld from each payment until the work is completed satisfactorily

Retainage protects the buyer by holding back part of each payment until completion. Pre-agreed compensation for late delivery is liquidated damages.

Chapter 14 — Risk Performance Domain

14.1 — Answer: D

Correct: D. −$8,000

EMV of the threat = 0.25 × −80,000 = −20,000. EMV of the opportunity = 0.40 × 30,000 = +12,000. Net = −$8,000.

14.2 — Answer: A

Correct: A. Transfer

Insurance transfers the financial impact to a third party in exchange for a premium. The risk itself is not eliminated.

14.3 — Answer: B

Correct: B. Exploit

Exploiting an opportunity aims to make it certain to happen. Enhancing only increases probability or impact.

14.4 — Answer: D

Correct: D. Ensure the risk owner implements the planned contingency response

When a trigger occurs, the planned contingency response is executed by the risk owner. The register and reports are then updated.

14.5 — Answer: C

Correct: C. A workaround

A workaround is an unplanned response to an event or issue that occurs. Contingency and fallback plans are prepared in advance.

14.6 — Answer: A

Correct: A. Sensitivity analysis displayed as a tornado diagram

Sensitivity analysis shows the relative impact of each uncertain variable; the tornado diagram displays the largest effects at the top.

14.7 — Answer: C

Correct: C. Escalate the risk to the appropriate owner in the organisation

Risks outside the project's scope or authority should be escalated to the program, portfolio or relevant business owner, who takes ownership.

14.8 — Answer: B

Correct: B. Residual risk

Residual risk remains after responses have been implemented. Secondary risk is a new risk created by the response itself.

14.9 — Answer: D

Correct: D. Option Y, expected cost $114,000

X: 100,000 + 0.3 × 50,000 = $115,000. Y: 110,000 + 0.2 × 20,000 = $114,000. Y has the lower expected cost.

14.10 — Answer: B, D

Correct: B. Including risk response activities in the backlog and prioritising them with features / D. Using timeboxed spikes to reduce technical uncertainty

A risk-adjusted backlog and spikes reduce uncertainty early. Freezing the backlog, postponing risk discussion and eliminating demos increase risk.

14.11 — Answer: B

Correct: B. To evaluate how effective the risk process and responses are

Risk audits assess whether risk processes and responses are working. Identification and reassessment happen in other activities.

14.12 — Answer: A

Correct: A. Appetite is the uncertainty accepted for reward; thresholds are measurable limits reflecting it

Appetite is a general willingness to take risk, and thresholds translate it into measurable limits, such as "no more than 5% cost variance".

14.13 — Answer: D

Correct: D. A target near 20 May gives 85% confidence; 1 May leaves a 50% chance of being late

Simulation results give confidence levels for dates. Communicating the probability helps stakeholders choose a target that matches their risk appetite.

14.14 — Answer: C

Correct: C. Perform Risk Analysis

The Sixth Edition's two analysis processes were merged into Perform Risk Analysis.

Chapter 15 — Agile and Hybrid Delivery in Depth

15.1 — Answer: D

Correct: D. Documentation still has value, but delivering a working product is valued more

The manifesto values items on the left more than those on the right; items on the right still have value. Teams produce documentation that is useful, not documentation for its own sake.

15.2 — Answer: D

Correct: D. The product owner

The product owner orders the backlog and maximises value. The scrum master serves the team and the organisation; developers decide how to build the increment.

15.3 — Answer: B

Correct: B. Suggest they continue after the daily scrum so the event stays focused

The daily scrum is a 15-minute inspection of progress toward the sprint goal. Detailed problem-solving happens afterwards with the people involved.

15.4 — Answer: A

Correct: A. The product owner

Only the product owner has the authority to cancel a sprint, usually when the sprint goal becomes obsolete.

15.5 — Answer: C

Correct: C. To create a pull system that improves flow and exposes bottlenecks

Limiting WIP reduces multitasking, shortens cycle time and reveals bottlenecks. Keeping everyone busy is not the goal; flow is.

15.6 — Answer: A

Correct: A. Test-driven development

TDD follows a red-green-refactor cycle: write a failing test, write code to pass it, then improve the design.

15.7 — Answer: B

Correct: B. LeSS

Large-Scale Scrum (LeSS) scales Scrum with one product owner and one product backlog for many teams and minimal additional roles.

15.8 — Answer: A, C

Correct: A. Removing impediments that block the team / C. Shielding the team from interruptions so they can focus

Servant leaders remove impediments, protect focus, facilitate and coach. Assigning tasks, deciding designs and reporting individual metrics undermine self-organisation.

15.9 — Answer: A

Correct: A. Task switching

Task switching causes time loss from context changes. Limiting WIP and dedicating people to teams reduce this waste.

15.10 — Answer: C

Correct: C. Time, cost and quality

DSDM fixes time, cost and quality and flexes the scope using MoSCoW prioritisation.

15.11 — Answer: B

Correct: B. A working session where stakeholders inspect the increment and adapt the backlog

The sprint review inspects the product increment with stakeholders and adapts the backlog. The retrospective focuses on process improvement.

15.12 — Answer: D

Correct: D. The lowest and highest estimators explain their reasoning, then the team re-estimates

Discussing outliers surfaces different assumptions and knowledge; the team then votes again to converge.

15.13 — Answer: C

Correct: C. Aligning release plans with hardware milestones at defined integration points

Hybrid delivery requires explicit synchronisation of milestones and releases, managed dependencies and consistent governance and reporting.

15.14 — Answer: C

Correct: C. A timeboxed investigation to reduce uncertainty before committing

Spikes are short, timeboxed investigations that answer technical or design questions and reduce risk.

Chapter 16 — Leading People: Vision, Teams, Conflict, Motivation, Influence and Ethics

16.1 — Answer: A

Correct: A. Meet with both to understand their positions and help them find a solution

Collaborating or problem-solving addresses the root cause and builds commitment. Forcing may be needed in emergencies; withdrawing leaves the problem unresolved; escalation is premature.

16.2 — Answer: D

Correct: D. Force/direct

In emergencies, safety, legal or ethical situations, a direct decision may be necessary. Collaboration is preferred when time and the situation allow.

16.3 — Answer: C

Correct: C. Salary and working conditions

Hygiene factors prevent dissatisfaction but do not motivate on their own. Recognition, growth and responsibility are motivators.

16.4 — Answer: B

Correct: B. Forming; clarify goals, roles and ground rules

In forming, people are getting oriented. The leader provides direction, clarifies objectives and roles, and helps the team establish agreements such as a team charter.

16.5 — Answer: D

Correct: D. Delegating

High competence and high commitment call for a delegating style with low direction and low support.

16.6 — Answer: A

Correct: A. Expert and referent power

Expert power (knowledge) and referent power (respect, credibility) build lasting influence. Coercive power tends to damage relationships.

16.7 — Answer: D

Correct: D. Disclose the conflict and stay out of the selection decision

The Code of Ethics requires disclosure of real or potential conflicts of interest and refraining from influencing the decision unless the stakeholders approve after full disclosure.

16.8 — Answer: C

Correct: C. Empathy

Empathy is the ability to understand others' emotions. Self-awareness and self-regulation concern one's own emotions.

16.9 — Answer: B

Correct: B. The team believes the target is unachievable, so expectancy is low

Motivation depends on expectancy, instrumentality and valence. If the team does not believe effort can achieve the target, a valuable reward will not motivate.

16.10 — Answer: B, E

Correct: B. Pairing operations staff with developers during the final iterations / E. Job shadowing and joint handover workshops before go-live

Tacit knowledge transfers best through interaction such as shadowing, pairing and workshops, which should start before closure. Sending documents at the end and single points of knowledge are weak practices.

16.11 — Answer: B

Correct: B. Facilitate a session with the product owner and team to clarify the vision

A shared vision aligns priorities. Revisiting the vision and objectives with the product owner builds a common understanding without taking over the PO's decisions.

16.12 — Answer: A

Correct: A. The best alternative to a negotiated agreement, i.e. the walk-away option

BATNA is your best option if negotiation fails. Knowing it prevents accepting a worse deal. ZOPA is the overlap between parties' acceptable ranges.

16.13 — Answer: B

Correct: B. Offer ways to contribute that suit the person, such as one-to-one or written input

Cultural awareness and inclusion mean adapting how input is gathered so all voices are heard, without embarrassing anyone.

16.14 — Answer: C

Correct: C. Coaching helps a person find their own solutions; mentoring shares the mentor's experience

Coaching is performance-focused and question-driven; mentoring is relationship- and experience-based, often career-oriented.

Chapter 17 — The Business Environment: Compliance, Issues, Change, Improvement, Sustainability, AI and PMOs

17.1 — Answer: D

Correct: D. Assess the regulation's impact with compliance or legal experts

Compliance is mandatory, so the PM first assesses the impact with experts, then processes the necessary changes through governance and updates plans and risks.

17.2 — Answer: D

Correct: D. The scrum master or servant-leader project manager

Servant leaders remove impediments the team cannot remove itself, such as access to environments or cross-department dependencies.

17.3 — Answer: C

Correct: C. ADKAR

ADKAR is Prosci's individual change model: awareness, desire, knowledge, ability and reinforcement.

17.4 — Answer: A

Correct: A. Supportive

Supportive PMOs are consultative. Controlling PMOs require compliance, and directive PMOs manage projects directly.

17.5 — Answer: B

Correct: B. Use only an approved tool for confidential data, follow policy and verify the summary

Responsible AI use protects confidentiality and complies with policy while keeping human verification of outputs.

17.6 — Answer: D

Correct: D. Review and validate it with the team, checking assumptions and estimates

AI augments planning, but people remain accountable. Validating outputs with those who will do the work ensures accuracy and buy-in.

17.7 — Answer: C

Correct: C. A key supplier has announced it will stop production next week

An issue is a current condition that must be resolved now. The other options describe uncertain future events, which are risks.

17.8 — Answer: A

Correct: A. Forming a powerful guiding coalition

The order is urgency, coalition, vision, communicate the vision, remove obstacles, short-term wins, build on the change, anchor it in the culture.

17.9 — Answer: C, D

Correct: C. Requiring suppliers to follow a code of conduct on labour practices / D. Including environmental and social criteria in the source selection criteria

Sustainable procurement considers environmental and social performance in selection and contracts, not price alone.

17.10 — Answer: A

Correct: A. Agree a specific improvement experiment with an owner, and review its effect

Effective continuous improvement turns retrospective findings into concrete, owned actions that are tried, measured and reviewed.

17.11 — Answer: C

Correct: C. Understand the reasons for resistance by engaging the users

Resistance often signals real concerns (training, usability, workload). Understanding them enables targeted actions such as training, communication or design changes.

17.12 — Answer: B

Correct: B. Analyse the impact on the business case and present options to decision-makers

External environment changes should be assessed for impact and handled through governance with options for the sponsor or steering committee.

17.13 — Answer: B

Correct: B. Leading indicators help predict results; lagging indicators measure them afterwards

For example, adoption rate during a pilot is a leading indicator of future revenue, which is a lagging indicator.

Chapter 18 — Tailoring

18.1 — Answer: A

Correct: A. To adapt the approach, governance and processes to the project's context

Tailoring adapts the way of working to the context. It neither applies everything blindly nor removes necessary controls or policies.

18.2 — Answer: D

Correct: D. Select the initial development approach

The usual sequence is: select the initial development approach, tailor for the organisation, tailor for the project, then implement ongoing improvement.

18.3 — Answer: A

Correct: A. Agree tailoring with the PMO, documenting that procurement is not applicable

Tailoring for the organisation means working with the PMO to adapt requirements to the context, documenting the rationale, rather than ignoring policy or producing waste.

18.4 — Answer: C

Correct: C. Use adaptive practices for software, keeping the required regulatory controls

Tailoring can combine adaptive delivery with the rigour required by regulation. Compliance obligations cannot be tailored away.

18.5 — Answer: A, C

Correct: A. High cost of change after construction starts / C. Strict regulatory requirements for documented verification

High cost of change and strict regulation favour planning and formal controls. Engaged product owners, uncertain requirements and small experienced teams favour adaptive approaches.

18.6 — Answer: C

Correct: C. Throughout the project, such as at phase gates and retrospectives

Implementing ongoing improvement means tailoring is reviewed and adjusted as the team and project learn.

18.7 — Answer: B

Correct: B. Assess the context, product, team and stakeholders, and tailor with them

Tailoring starts with understanding the context. Copying an approach without considering differences is poor tailoring.

18.8 — Answer: B

Correct: B. Removing change control from a fixed-price contract project to "be more agile"

Removing essential controls in a context that needs them, such as contract change control, is poor tailoring. The other examples adapt the approach to the context sensibly.

Chapter 19 — Models, Methods and Artifacts: the Practitioner's Toolbox

19.1 — Answer: C

Correct: C. Tuckman ladder

A model is a thinking strategy that explains a phenomenon, such as team development. Planning poker is a method; the risk register and burnup chart are artifacts.

19.2 — Answer: B

Correct: B. Decision log

A decision log captures decisions and their rationale. Issue, assumption and change logs serve other purposes.

19.3 — Answer: D

Correct: D. Make-or-buy analysis

Make-or-buy analysis compares the costs, risks and benefits of internal development versus purchase.

19.4 — Answer: A

Correct: A. S-curve

The time-phased cost baseline, plotted cumulatively, typically forms an S-curve.

19.5 — Answer: A

Correct: A. Value stream mapping

Value stream mapping shows value-adding and non-value-adding time across the whole flow, highlighting waste such as waiting.

19.6 — Answer: 1→A, 2→C, 3→B, 4→D

Correct pairs: 1. Record a current problem with an owner and due date → Issue log; 2. Record a condition believed to be true for planning purposes → Assumption log; 3. Record an uncertain event with probability and impact → Risk register; 4. Track submitted changes and their approval status → Change log

Issues are current problems, assumptions are beliefs used for planning, risks are uncertain events, and the change log tracks change requests and decisions.

19.7 — Answer: C

Correct: C. Customer loyalty and willingness to recommend

NPS subtracts the percentage of detractors from the percentage of promoters, measuring customer loyalty.

19.8 — Answer: C

Correct: C. Story map

Story mapping arranges stories by user activities horizontally and by priority vertically, helping plan releases.

19.9 — Answer: B

Correct: B. Regression analysis

Regression analysis examines relationships among variables to support forecasting.

19.10 — Answer: D

Correct: D. A dashboard

Dashboards summarise key performance information visually for decision-makers.

Chapter 20 — Formulas and Calculations Master Sheet

20.1 — Answer: 0.9

SPI = EV / PV = 54,000 / 60,000 = 0.9. The project has completed 90% of the work planned to date.

20.2 — Answer: B

Correct: B. $500,000

With atypical variance, EAC = AC + (BAC − EV) = 400,000 + 500,000 = 900,000. ETC = EAC − AC = $500,000.

20.3 — Answer: C

Correct: C. 45

7 people: 21 channels. 12 people: 66 channels. The increase is 45.

20.4 — Answer: D

Correct: D. AC + bottom-up ETC

When the original estimating assumptions are no longer valid, the team re-estimates the remaining work bottom-up and adds it to the actual cost.

20.5 — Answer: A

Correct: A. 16 to 24 days

About 95% of outcomes fall within ±2 standard deviations: 20 ± 4 = 16 to 24 days.

20.6 — Answer: B

Correct: B. 4 years

Payback = 60,000 / 15,000 = 4 years.

20.7 — Answer: A

Correct: A. The remaining work must be done 35% more efficiently than planned to meet the budget

A TCPI above 1 means future performance must exceed planned efficiency. A value such as 1.35 often signals that the BAC is no longer achievable without changes.

20.8 — Answer: B

Correct: B. $5,000

(Cost − salvage) / life = (25,000 − 5,000) / 4 = $5,000 per year.

20.9 — Answer: D

Correct: D. More than 90 days

Break-even d = 54,000 / (900 − 300) = 90 days. Beyond 90 days, buying is cheaper.

20.10 — Answer: 6

160 / 30 = 5.33, rounded up to 6 sprints, because a partial sprint still has to be scheduled.

Chapter 21 — Exam Strategy and the PMI Mindset

21.1 — Answer: B

Correct: B. An action that gathers information and assesses the situation first

"First" usually points to understanding the situation and its impact before taking larger actions such as escalation or termination.

21.2 — Answer: D

Correct: D. Explain that the product owner prioritises new work, and connect them

In agile contexts, the product owner prioritises backlog items. The PM facilitates the connection and protects the sprint goal rather than refusing outright.

21.3 — Answer: A

Correct: A. Escalating to the sponsor before trying to understand or resolve the problem

Premature escalation is a common wrong answer. Escalation is appropriate when an issue exceeds the PM's authority or the team's ability, ideally with options.

21.4 — Answer: C

Correct: C. Review the flagged questions in the first section, since they cannot be revisited

Once a break begins, the previous section is locked. Reviewing flagged questions beforehand prevents losing the chance to change answers.

21.5 — Answer: A

Correct: A. Report the true status, with recovery options, and discuss the concern with the sponsor

Honesty is a mandatory standard of the Code of Ethics. The PM reports accurately and provides options, while working with the sponsor.

21.6 — Answer: B

Correct: B. Integrated change control: assess the impact and submit a change request

The scenario indicates a predictive context, so changes to the baseline go through integrated change control.

21.7 — Answer: A, C

Correct: A. Ask which option addresses the root cause rather than a symptom / C. Ask which option would logically happen first in the sequence

Root cause and sequence are reliable tie-breakers. Length, jargon and seniority are not indicators of the best answer.

21.8 — Answer: C

Correct: C. About 80 seconds per question, reaching question 60 around the 80-minute mark

240 minutes for 180 questions is about 1.33 minutes per question, so 60 questions take about 80 minutes.